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College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

Avianna's Inland Resort Online Booking System

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Abstract

Resorts increasingly depend on web applications to take guest reservations, avoid overbooking, communicate with guests, and monitor business performance. Avianna's Inland Resort in Sta. Barbara, Iloilo, Philippines, still handled reservations through walk-ins, calls, texts, and paper logbooks. This paper presents the design and implementation of a web-based online booking system for the resort, covering the customer portal, the administrator panel, the front-desk receptionist workflow, the database, and the server-side logic. The system was built with PHP, MySQL (MariaDB), HTML5, CSS3 (Bootstrap 5), JavaScript, and Chart.js. It provides live pricing, e-mail transaction notifications, booking status management (Pending, Approved, Checked In, Checked Out), express guest registration, incidentals computation, safe cancellation archival, an analytics dashboard, an event announcer, and receipt generation. The paper describes the architecture, database design, development methodology, and testing, and reports the findings of functional and user acceptance evaluation.

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The resort industry in the Philippines had been growing continuously due to the increasing demand of its services. Many individuals had come to value leisure, recreation, and family-oriented places as people engaged in activities such as holidays, weekend outings, and special events. Resorts had become the preferred destinations for family meets, birthday celebrations, team-building, excursions, and vacations. Consequently, resort owners had striven to provide quality services that appeal to the diverse needs of their clients and stay ahead of the competition. Additionally, customers had increasingly used digital technology to search, connect, inquire, book, and pay for their preferred resort services, thus creating a need for enhanced online services.

In this regard, Avianna's Inland Resort in Sta. Barbara, Iloilo, had become one of the recreational facilities that had been in operation for a relatively short time. The resort had been owned and managed by Mr. JJ S. Layson for about seven months. It had a stretch of land covering 3500 square meters. The resort had already provided a wide range of facilities and services to cater to the needs of different customers. Although it was relatively new, Avianna's Inland resort had been able to provide cottages, overnight rooms, pavilions, and event spaces together with recreational facilities for families, groups, and individuals.

Among the facilities provided at Avianna's Inland Resort, there were five cottages that had been available for day and overnight use. The day entrance and cottage fees for the day varied from ₱600 to ₱800, while the overnight cottage fee could cost up to ₱2000. Each cottage could accommodate a maximum of twelve persons. Additionally, the resort had provided private or soundproof rooms with air conditioning, television, mini sound system, and electric fans. The rooms had been provided to ensure that guests had a pleasant experience. Moreover, the resort had also provided pavilions for birthdays, family meets, outings, and meetings. The main pavilion could host up to 80 persons for ₱4000 while the poolside pavilion could accommodate 40 persons at ₱2500. Additionally, there was a semi-open pavilion for smaller open-air gatherings at ₱2000. The overnight rates for all types of pavilions ranged from ₱6000 to ₱12000 depending on the number of guests. The wide range of facilities and services had enabled the resort to appeal to different customer needs and offer quality recreational services.

In addition to the range of recreational facilities and services provided at the resort, various other attractions had been made available to enhance guest experience. These included a mini zoo, gaming area, and a retail store. These had complemented the resort's recreational services and attracted more customers. However, despite the wide range of services and facilities offered by the resort, it had been using manual processes to manage reservations, information of its customers, and availability of its facilities. In this regard, customers walked in or contacted the resort through telephone and made their reservations. Additionally, all the information about reservations was recorded manually, which made it difficult to organize and record customer information.

The absence of an official website and an online booking platform has also limited the resort's ability to provide customers with convenient access to its services. The customers have needed to contact the resort to inquire about available accommodations, rates, schedules, and the reservation procedure. Not only has this required additional time and effort on the customer's end, but it also placed additional work on the resort personnel. In addition, the resort management had to manually manage the reservations. This resulted in the possibility for double bookings, scheduling conflicts, inaccurate customer records, and issues with the monitoring of the availability of the cottages, rooms, and pavilions. The resort's operations had become less efficient and less capable of responding quickly to

inquiries and reservation requests.

The increasing reliance on digital technology in the tourism and hospitality industry had made these challenges even more pressing. The customers had expected the resorts to provide the establishment with an online platform in which they can access information, compare available accommodations, check schedules, and submit their requests without having to physically visit the resort. The resorts that had implemented the digital booking platform had gained the opportunity to enhance their accessibility to customers and simplify the reservation process. In contrast, the lack of such an online booking system at Avianna's Inland Resort has highlighted the problem in terms of the ability to reach the potential customers who prefer digital and remote transactions. As such, there was a need to implement such an online booking system that can help facilitate both the customers' convenience and the improved resort management.

With the aforementioned problems in mind, the development of a web-based Online Booking System for Avianna's Inland Resort has been proposed. This system has been intended to provide customers with an accessible platform where they can access the resorts' information, facilities, accommodation rates, and available services. The system has also been intended to allow customers to browse the cottages, rooms, and pavilions, check their availability, and submit their reservation requests online. With the system in place, the reservation process has been expected to become faster, more organized, and less dependent on manual transactions. Confirmations and notifications have been incorporated as well to help facilitate the communication between customers and resort personnel.

In the case of resort management, the proposed system has been intended to provide a centralized platform for managing the customer information, reservation records, facility availability, and transaction records. The system has been expected to provide the necessary monitoring of the reservations, which would help prevent double bookings and scheduling conflicts. In addition, the automation of the reservation-related processes had been expected to reduce the manual work, paperwork, and the possibility for human errors. Lastly, the system's automated reservation reports shall be leveraged to help the management monitor the bookings, records, and pertinent information. Features related to authentication and account security had also been considered to assure that the management and customers could secure the necessary information.

Overall the development of the Online Booking System has been deemed appropriate to respond to the operational and technological problems experienced by Avianna's Inland Resort. By leveraging the online reservations, records management, facility monitoring, notifications, and automated reservation reports, the proposed system has been expected to improve the efficiency of the resort's operations, while also enhancing the service's accessibility and convenience for the resort customers. The development of such a system will also help enhance the resort's online visibility and competitive standing within the tourism and hospitality industry. Ultimately, the adoption of the proposed online booking system shall be leveraged to improve the resort's operational efficiency, customer satisfaction, long-term growth, and sustainability as an emerging recreational destination in the Municipality of Sta. Barbara, Iloilo.

1.2 Statement of the Problem

The continuous advancement of information and communication technology had significantly transformed the operations of businesses across various industries, particularly within the tourism and hospitality sector. Technological innovations have enabled organizations to improve the quality of their services, streamline operational processes, and provide customers with faster and more convenient access to information and services. The widespread availability of internet connectivity and digital platforms had also influenced customer behavior, making online transactions an essential component of business operations. As a result, resorts and other accommodation establishments had increasingly adopted web-based systems to automate reservations, manage customer information, and improve operational efficiency.

The tourism and hospitality industry had become highly competitive as customers increasingly sought establishments that offered convenience, accessibility, and quality service. Rather than relying on traditional

reservation methods, customers had preferred online platforms that enabled them to browse accommodation options, view resort facilities, compare prices, check availability, and confirm reservations instantly. Online booking systems had become an important business tool because they provided customers with twenty-four-hour access to reservation services while helping business owners manage reservations more accurately and efficiently. Consequently, establishments that had continued to rely on manual processes had experienced increasing difficulty in meeting the changing expectations of customers and remaining competitive within the industry.

Avianna's Inland Resort in Sta. Barbara, Iloilo, had relied primarily on manual procedures in managing reservations, customer inquiries, and record keeping. Reservation transactions had been conducted mostly through walk-in customers, phone calls, text messages, and social media inquiries, while customer information and reservation details had been manually recorded in logbooks and paper-based documents. Although these methods had enabled the resort to operate during its initial stages, they had become increasingly inefficient as the number of customers continued to grow. The increasing volume of reservations had required additional time and effort from resort personnel, making reservation management more difficult and reducing the overall efficiency of daily operations.

The manual reservation process had often resulted in delays in confirming bookings because every reservation had to be verified manually before being finalized. Customers had sometimes waited for extended periods before receiving confirmation regarding the availability of cottages, rooms, pavilions, and other resort facilities. During weekends, holidays, and peak vacation seasons, when reservation requests had significantly increased, resort personnel had experienced greater difficulty in responding promptly to customer inquiries. These delays had affected the quality of customer service and had reduced the convenience expected by customers seeking fast and reliable reservation services.

The use of manual record keeping had increased the possibility of inaccurate customer information, incomplete reservation records, and misplaced booking documents. Since reservation information had been recorded manually, typographical errors and omissions had occasionally occurred during the recording process. Retrieving customer records had also required resort personnel to manually search through reservation logbooks, resulting in slower transaction processing and increased administrative workload. These challenges had reduced the accuracy and reliability of reservation records and had made customer information more difficult to manage.

Another significant concern had been the possibility of scheduling conflicts and double bookings. Because reservation schedules had been monitored manually, resort personnel had sometimes encountered difficulties in determining the real-time availability of accommodations and resort facilities. Multiple customers requesting the same cottage, room, or pavilion on identical dates had increased the possibility of reservation conflicts. Without an automated reservation management system capable of updating booking information immediately, ensuring the accuracy of reservation schedules had become more difficult. Such situations had affected customer satisfaction, reduced confidence in the resort's reservation process, and had the potential to negatively influence the reputation of the establishment.

The resort had also encountered challenges in monitoring the utilization of its cottages, rooms, pavilions, and other recreational facilities. Since reservation records had been maintained manually, management had experienced difficulty in determining which facilities were available, reserved, or occupied at a particular time. This limitation had affected the efficient allocation of accommodations and had sometimes resulted in underutilization of available facilities or scheduling conflicts. Likewise, monitoring reservation trends and occupancy rates had become difficult because reservation information had not been stored within a centralized and organized database.

To reservation management concerns, the resort had experienced difficulties in preparing reports and maintaining organized customer records. Daily, weekly, monthly, and annual reservation reports had been prepared manually by reviewing reservation logbooks and compiling booking information. This process had consumed considerable time and effort while increasing the possibility of computational errors, duplicate entries, and incomplete reports. As a result, management had encountered delays in obtaining accurate business information needed for evaluating operational performance, monitoring customer demand, and making informed management decisions.

The absence of a dedicated website and online booking platform had limited the resort's ability to promote its

services and reach a wider customer market. Customers had often needed to contact the resort through phone calls or social media platforms simply to inquire about accommodation rates, entrance fees, available amenities, operating schedules, and reservation availability. Since responses had depended on the availability of resort personnel, customers had sometimes experienced delays in obtaining the information they needed before making reservation decisions. This situation had reduced customer convenience and had limited the resort's ability to provide efficient and accessible customer service.

The lack of an official online booking platform had also affected the resort's competitiveness within the tourism and hospitality industry. Many competing resorts had already implemented web-based reservation systems that enabled customers to make reservations anytime and anywhere. These establishments had benefited from increased online visibility, improved customer engagement, and more efficient business operations. In contrast, Avianna's Inland Resort had continued to rely on traditional reservation procedures, which had limited its ability to attract new customers and maximize business opportunities in an increasingly digital environment.

Recognizing these operational challenges, this study proposed the development of a Web-Based Online Booking System for Avianna's Inland Resort. The proposed system had been designed to automate reservation management, centralize customer records, provide real-time monitoring of accommodation availability, and simplify administrative processes. It had also been intended to minimize reservation errors, reduce manual workloads, improve report generation, and enhance communication between customers and resort management. Furthermore, the system had been expected to provide customers with convenient online access to resort information, accommodation availability, pricing details, and reservation services regardless of time and location.

The implementation of the proposed system had also been envisioned to strengthen the resort's online presence by providing an official digital platform capable of showcasing its facilities, services, amenities, and promotional offers. Through the integration of online reservation services, the resort had been expected to improve customer satisfaction, increase operational efficiency, enhance management decision-making through accurate reports, and strengthen its competitiveness within the tourism and hospitality industry. Ultimately, the proposed Online Booking System had been anticipated to contribute to the long-term sustainability, continued growth, and overall success of Avianna's Inland Resort.

The study sought to determine how Avianna’s Inland Resort could improve its existing manual reservation and booking processes through the development and implementation of a Web-Based Online Booking System. Since the resort had primarily relied on walk-in transactions, phone calls, text messages, and social media inquiries to accommodate reservation requests, the study aimed to determine how the existing process could be improved to make reservation transactions more organized, efficient, accurate, and convenient for both customers and resort personnel. It specifically examined how the proposed system could simplify the process of checking accommodation availability, recording customer information, processing reservation requests, confirming bookings, and updating reservation records. By replacing or minimizing manual procedures, the study sought to determine whether the system could reduce the time and effort required to process reservations and provide a more systematic approach to managing booking transactions.

The study also sought to determine how the proposed online booking system could minimize common reservation problems, particularly double bookings, inaccurate customer records, incomplete reservation information, and lost or misplaced booking records. Under the existing manual process, reservation information had been recorded and monitored through paper-based records and other communication channels, which could result in inconsistencies and difficulties in retrieving updated information. The study therefore examined how a centralized database could help store and organize customer information, reservation details, booking dates, selected facilities, number of guests, and reservation status. It also considered how real-time updates of booking information could help prevent scheduling conflicts by allowing resort personnel to determine whether cottages, rooms, pavilions, and other facilities were available, reserved, or occupied. Through these functions, the study aimed to determine whether the proposed system could improve the accuracy, reliability, and security of reservation records.

Another concern of the study was how the proposed system could provide customers with more convenient and accessible access to resort information and reservation services. The study examined how customers could use the

online platform to view information about Avianna's Inland Resort, including its facilities, cottages, rooms, pavilions, accommodation rates, available services, and other relevant details without having to personally visit or contact the resort. It also sought to determine how the system could allow customers to check accommodation availability and submit reservation requests online regardless of their location and time of inquiry. Through an accessible online platform, the study aimed to provide customers with a more convenient way of planning their visits, selecting suitable accommodations, and completing reservation transactions while reducing the time required for direct communication with resort personnel.

The study further aimed to determine how the proposed system could assist resort management in efficiently monitoring cottages, rooms, pavilions, and other resort facilities. Since facility availability had previously been monitored through manual records and direct communication, management had experienced difficulties in obtaining updated information regarding the status of accommodations. The proposed system was therefore intended to provide a centralized monitoring mechanism that could display current reservation schedules and facility availability. The study examined how such functionality could help management identify available and reserved facilities, organize schedules, prevent overlapping bookings, and maximize the utilization of the resort's resources. It also considered how centralized facility monitoring could support better planning and allocation of accommodations during regular operating days, weekends, holidays, and peak seasons.

To improve customer satisfaction by providing faster, more convenient, and accessible services. The study recognized that customers had increasingly expected establishments in the tourism and hospitality industry to provide efficient digital services that reduced unnecessary waiting time and simplified transactions. Through online access to resort information, availability checking, reservation submission, booking confirmation, and notifications, the system was expected to improve the overall customer experience. The study therefore examined whether the implementation of these features could reduce delays in responding to inquiries and confirming reservations while providing customers with greater convenience and confidence when making bookings. Improved accessibility and faster service were also considered important factors in increasing customer satisfaction and encouraging customers to choose the resort for future visits.

The study also sought to determine how the proposed system could automate report generation, reservation tracking, and customer record management. Under the existing manual process, resort personnel had needed to review logbooks and manually compile information when preparing reservation reports and monitoring booking activities. This process had consumed time and had increased the possibility of errors, duplicate entries, and incomplete information. The proposed system was intended to automate the generation of reservation reports based on stored booking data, allowing management to obtain organized information regarding reservations, customer records, facility utilization, and booking status. The study examined how automated reporting could reduce administrative workload, improve record accuracy, and provide management with timely information that could be used for monitoring operations and supporting business decisions.

The study aimed to determine how the implementation of the online booking system could strengthen Avianna's Inland Resort's online presence and improve its competitiveness within the tourism and hospitality industry. The absence of a dedicated online booking platform had limited the resort's ability to reach customers who preferred digital services and online transactions. Through the proposed system, the resort could have a centralized digital platform where its facilities, services, accommodation options, rates, and other relevant information could be presented to potential customers. The study examined how this digital presence could help increase the resort's visibility, make its services more accessible to a wider market, and allow it to compete more effectively with other resorts that had already adopted online booking technologies. The system was also expected to provide opportunities for the resort to communicate more effectively with customers and promote its available facilities and services through a reliable online platform.

The study also sought to identify the features and functionalities that should be included in the proposed system to effectively meet the needs of both customers and resort management. For customers, the system needed to provide functions that supported convenient browsing of resort information, viewing of facilities and accommodation rates, checking availability, submitting reservations, receiving confirmations, and communicating with resort personnel. For management and authorized staff, the system needed to provide functions for managing customer accounts,

maintaining reservation records, monitoring facility availability, updating accommodation information, managing booking statuses, generating reports, and maintaining organized records. Security and authentication features were also considered necessary to protect customer information and ensure that only authorized users could access administrative functions. By identifying these requirements, the study aimed to ensure that the proposed system would be functional, user-friendly, reliable, and appropriate for the operational needs of the resort.

The study sought to determine how the implementation of the Web-Based Online Booking System could contribute to the overall operational efficiency and long-term growth of Avianna's Inland Resort. The proposed system was expected to reduce dependence on manual procedures, minimize reservation-related errors, improve record management, simplify facility monitoring, and provide management with more accurate and timely information. At the same time, it was expected to provide customers with faster, more convenient, and accessible services that could improve their overall experience. By integrating reservation management, customer record management, facility monitoring, automated reporting, communication, and online information access into a centralized platform, the system was expected to support more efficient resort operations. Ultimately, the study aimed to determine whether the adoption of the proposed online booking system could help Avianna's Inland Resort improve its service delivery, strengthen its digital presence, increase customer satisfaction, enhance competitiveness, and establish a more efficient operational foundation that could support its continued growth and long-term sustainability in the tourism and hospitality industry.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to design, develop, and implement a web-based Online Booking System for Avianna's Inland Resort located in Santa Barbara that will improve reservation management, enhance customer convenience, automate resort operations, and strengthen the resort's online presence. The proposed system aims to provide a modern, efficient, and user-friendly platform that enables customers to access resort information, check accommodation availability, and make reservations online anytime and anywhere through internet-enabled devices such as smartphones, tablets, laptops, and desktop computers.

The study is motivated by the need to address the limitations of the current manual reservation system, which relies heavily on walk-in transactions, phone calls, and informal communication channels. These traditional methods often result in inefficiencies such as delayed booking confirmations, difficulty in tracking reservations, and inconsistencies in customer records. By introducing a computerized and centralized booking system, the study aims to modernize resort operations and improve overall service delivery.

Specifically, the study seeks to replace the existing manual reservation process with an automated system that can efficiently handle booking transactions in a structured and organized manner. This includes streamlining the entire reservation workflow from customer inquiry, account registration, accommodation selection, booking submission, approval, and confirmation. Through automation, the system is expected to reduce dependency on manual encoding and minimize administrative errors that may negatively affect customer experience.

Another key objective of the study is to improve the accuracy, reliability, and integrity of resort data. By utilizing a centralized database, all customer records, reservation details, and accommodation information will be securely stored and systematically organized. This ensures that data can be easily retrieved, updated, and managed in real time, reducing the risks of data loss, duplication, and inconsistency that are commonly associated with manual record-keeping systems.

The system also aims to provide real-time monitoring of accommodation availability, allowing both customers and administrators to view updated information regarding cottages, rooms, and pavilions. This feature plays a crucial role in preventing double bookings and scheduling conflicts, ensuring that all reservations are properly managed and accurately reflected in the system. Real-time updates also enhance transparency and improve decision-making for both users and management.

Furthermore, the proposed Online Booking System seeks to enhance customer satisfaction by offering a convenient and accessible platform for making reservations. Customers will be able to easily browse available facilities, view detailed descriptions, check pricing, and select preferred dates without physically visiting the resort or waiting for manual responses. This level of convenience is expected to significantly improve user experience and encourage more bookings through the digital platform.

The study also aims to strengthen communication between customers and resort management through automated system notifications and messaging features. Customers will receive booking confirmations, reservation updates, and important announcements in a timely manner. At the same time, administrators will be able to respond to inquiries and manage customer interactions more efficiently, reducing communication gaps and misunderstandings.

In addition, the system is designed to support the administrative and operational functions of Avianna's Inland Resort. Through the administrative dashboard, resort personnel will be able to manage user accounts, monitor reservations, update facility information, and generate various reports. These reports may include booking summaries, occupancy rates, and transaction records, which are essential for operational analysis and business planning.

Another important objective of the study is to enhance operational efficiency by reducing the workload of resort staff. With the automation of repetitive tasks such as recording reservations, checking availability, and preparing reports, employees can focus more on improving customer service and managing resort activities. This leads to better productivity and more efficient use of human resources.

The study also seeks to integrate essential security features to ensure that the system is safe, reliable, and protected from unauthorized access. Security measures such as user authentication, password protection, and data validation will be implemented to safeguard sensitive customer and administrative information. This helps maintain trust and confidence among system users.

Moreover, the system aims to support the digital transformation of Avianna's Inland Resort by establishing a strong online presence in the tourism and hospitality industry. As more customers rely on online platforms for travel planning and accommodation booking, having a web-based reservation system will allow the resort to remain competitive and accessible to a wider audience.

Finally, the study intends to develop a system that adheres to important software quality standards such as functionality, usability, reliability, efficiency, and security. These quality attributes ensure that the system is not only operational but also effective, user-friendly, and capable of sustaining long-term use.

This study aims to create a comprehensive and efficient Online Booking System that will modernize reservation processes, enhance customer experience, improve administrative efficiency, support data-driven decision-making, and contribute to the long-term growth, sustainability, and competitiveness of Avianna's Inland Resort in the tourism and hospitality industry.

1.3.2 Specific Objectives

The study specifically aimed to develop an online platform that enabled customers to view resort information, facilities, accommodation rates, and available services conveniently. It also aimed to provide an online reservation system that allowed customers to book cottages, rooms, and pavilions anytime and anywhere. Furthermore, the study sought to implement a database management system that efficiently stored, organized, and managed customer information, reservation records, and transaction details. The system also provided real-time booking monitoring, which helped the management track accommodation availability and prevent double bookings and scheduling conflicts. Authentication and account security features were implemented to protect user and customer information, safeguard sensitive data, and ensure secure transactions.

The study aimed to automate the reservation process to reduce manual workload, paperwork, and human errors in booking management. It generated reservation reports automatically to support easier monitoring, record keeping, and business decision-making. The system also improved communication between customers and resort staff through online inquiries, booking confirmations, and notification features. It helped the resort management efficiently monitor cottages, rooms, pavilions, and other resort facilities while enhancing customer satisfaction by providing faster, more convenient, and accessible services through digital technology. Finally, the study aimed to strengthen Avianna's Inland Resort's online presence, improve its competitiveness in the tourism and hospitality industry, and promote operational efficiency while supporting the resort's long-term growth and sustainability through the use of modern information systems and digital solutions.

General Objective

The general objective of this study was to design, develop, and implement a web-based Online Booking System for Avianna's Inland Resort that would enhance the efficiency of reservation management, improve customer service, streamline business operations, and provide customers with a convenient and accessible platform for viewing resort information and making reservations online. The system was developed as a centralized and automated solution for managing resort transactions and addressing the limitations of the existing manual reservation processes, which had relied heavily on walk-in inquiries, phone calls, text messages, social media communications, and handwritten or spreadsheet-based record keeping. Through the development of the proposed system, the study sought to provide the resort with a more organized, reliable, and efficient method of handling its reservation and customer management activities.

The proposed system was intended to automate the reservation process, beginning with customer registration and account management and continuing through accommodation browsing, availability checking, booking submission,

reservation verification, approval, and confirmation. By integrating these processes into a single web-based platform, the system was expected to reduce the resort's dependence on manual encoding and paper-based documentation. The automation of reservation activities was also intended to minimize delays, inconsistencies, duplicate entries, and human errors that could occur during manual processing. Through a centralized database, reservation information could be stored, organized, retrieved, and updated more efficiently, thereby improving the accuracy and reliability of the resort's booking records.

Another important objective of the study was to provide real-time monitoring of accommodation availability. The system was intended to allow customers and authorized resort personnel to view updated information regarding the availability and reservation status of cottages, rooms, pavilions, and other facilities. This functionality was designed to help prevent double bookings, scheduling conflicts, and misunderstandings regarding facility availability. By ensuring that reservation information was consistently updated within a centralized database, the system was expected to provide management with a more reliable method of monitoring bookings and allocating available facilities. This feature was particularly important during weekends, holidays, and peak seasons when the volume of reservation requests could increase significantly.

The study also aimed to enhance the customer experience by providing a convenient, accessible, and user-friendly online booking environment. Through the proposed system, customers could access resort information using internet-enabled devices such as smartphones, tablets, laptops, and desktop computers. The platform was intended to allow customers to browse the available facilities, view accommodation descriptions and rates, check availability, and submit reservation requests without the need to personally visit the resort or repeatedly contact resort personnel. By providing access to reservation services regardless of time and location, the system was expected to reduce waiting time, improve convenience, and make the booking process more accessible to potential customers.

The system was intended to strengthen communication between customers and resort management. Features such as booking confirmations, reservation updates, notifications, and online inquiries were incorporated to provide customers with timely information regarding their transactions. These communication functions were expected to reduce misunderstandings, improve transparency, and allow resort personnel to respond more efficiently to customer concerns. By providing customers with clear and timely updates, the system was intended to establish a more organized communication process between the resort and its guests.

The study also sought to support the administrative operations of Avianna's Inland Resort by providing management with tools for efficient customer and reservation data management. Authorized administrators were intended to manage customer accounts, maintain reservation records, update accommodation information, monitor booking statuses, and manage the availability of resort facilities through a centralized administrative platform. The system was also intended to generate automated reports containing relevant reservation and operational information, such as booking summaries, reservation status, customer records, occupancy information, and transaction records. These reports were expected to provide management with accurate and organized information that could support operational planning, monitoring, evaluation, and business decision-making.

This study emphasized the integration of modern information technology into the daily operations of Avianna's Inland Resort as part of its digital transformation. The implementation of a web-based Online Booking System was intended to strengthen the resort's online presence and provide it with a dedicated digital platform for presenting its facilities, accommodations, services, rates, and other relevant information. Through improved online accessibility, the resort was expected to reach a wider audience and provide potential customers with easier access to its services. The adoption of digital technology was also intended to help the resort respond to changing customer expectations and remain competitive within the continuously developing tourism and hospitality industry.

In terms of operational efficiency, the system was intended to minimize repetitive manual tasks, reduce paperwork, improve record organization, and streamline reservation-related workflows. By automating routine activities such as reservation recording, availability monitoring, customer record management, confirmation, and report generation, the system was expected to reduce the administrative workload of resort personnel. This would allow staff to devote more time and attention to customer service, facility management, and other operational responsibilities. As a result, the proposed system was expected to contribute to better utilization of human resources and more efficient delivery of resort services.

The study also considered important software quality attributes to ensure that the developed system would meet the operational and user requirements of the resort. The system was evaluated in terms of functionality, usability, reliability, efficiency, and security. Functionality ensured that the system could perform its intended reservation and management tasks correctly. Usability focused on providing customers and administrators with an understandable and user-friendly interface. Reliability emphasized the system's ability to consistently perform its functions and maintain accurate reservation information. Efficiency considered the system's ability to process transactions and retrieve information within reasonable response times, while security focused on protecting customer information, reservation records, and other sensitive data from unauthorized access or misuse.

The general objective of the study was to develop a functional, secure, reliable, and efficient Web-Based Online Booking System that would modernize the reservation processes of Avianna's Inland Resort. The system was intended to improve the overall customer experience by providing convenient and accessible online reservation services while helping management organize customer records, monitor facility availability, process bookings, generate reports, and make more informed operational decisions. Through the integration of automated reservation management, centralized data storage, real-time availability monitoring, online communication, and administrative reporting, the proposed system was expected to improve the resort's operational efficiency and service delivery. In the long term, the implementation of the system was intended to strengthen Avianna's Inland Resort's digital presence, enhance its competitiveness, support its continued growth, and contribute to its sustainability within the tourism and hospitality industry.

1.4 Scope and Delimitation

This study focused on the design, development, and implementation of a web-based Online Booking System for Avianna's Inland Resort located in Sta. Barbara, Iloilo. The proposed system was developed to improve the resort's existing reservation management process by replacing manual booking procedures with a computerized and internet-based solution. The study focused on providing customers with a convenient platform where they could access resort information, view amenities and accommodation rates, check the availability of cottages, rooms, and pavilions, and submit reservation requests online. At the same time, the system provided resort management with a centralized platform for monitoring and managing customer information, reservation records, accommodation schedules, booking status, and reservation-related reports. The development of the system was intended to reduce the limitations associated with manual reservation procedures and provide a more organized, accessible, accurate, and efficient method of managing booking transactions.

The system was developed as a web-based application that could be accessed through various internet-enabled devices, including desktop computers, laptops, tablets, and smartphones. Customers, administrators, and authorized resort personnel could access the system using standard web browsers without the need to install additional software applications. The web-based platform allowed users to access reservation-related services from different locations, provided that a stable internet connection was available. This approach was intended to make the system more accessible to customers who preferred online transactions while also allowing resort personnel to manage reservation activities through a centralized platform.

The study covered the development of several major modules and functionalities necessary for the efficient operation of the resort's reservation and booking processes. On the customer side, the system included user registration and login functions that allowed customers to create accounts and securely access their personal information and reservation records. Customers could view information about Avianna's Inland Resort, including its facilities, amenities, cottages, rooms, pavilions, accommodation rates, and available services. The system also allowed customers to make online reservations for cottages, rooms, and pavilions by providing the necessary booking information and selecting their preferred dates and accommodations. Customers could view available schedules and accommodation availability before submitting reservation requests, which helped reduce the possibility of selecting already reserved facilities.

The customer module also provided reservation confirmation and booking status monitoring functions. Through these features, customers could determine whether their reservation requests had been received, processed,

approved, or otherwise updated by the resort management. The system also included online inquiry and communication features that allowed customers to send questions or concerns regarding reservations and resort services. In addition, customers could access their reservation history and review relevant booking details through their accounts. These functions were intended to provide customers with greater convenience, transparency, and accessibility throughout the reservation process while reducing their dependence on walk-in inquiries, phone calls, and other manual forms of communication.

For resort management and authorized administrators, the system included functions for managing customer accounts and reservation records. Administrators could view and organize customer information, monitor submitted reservations, and update booking records when necessary. The system also provided real-time monitoring of room, cottage, and pavilion availability, allowing authorized personnel to determine which facilities had been available, reserved, or occupied. This function was intended to assist management in preventing double bookings, identifying scheduling conflicts, and improving the allocation of available facilities. Administrators could also approve, update, and manage reservations based on the actual availability and policies of the resort.

The administrator module further allowed authorized personnel to update accommodation details, rates, descriptions, and other relevant resort information displayed on the system. This ensured that customers could access updated information when browsing the resort's facilities and services. The system also included a report generation function that allowed administrators to generate reservation and transaction reports based on information stored in the database. These reports were intended to assist management in monitoring booking activities, maintaining organized records, and supporting operational planning and decision-making. Database management and record organization were also included to provide centralized storage of customer information, reservation details, accommodation records, and other relevant system data. Furthermore, administrators could monitor customer inquiries and booking activities to provide appropriate responses and maintain effective communication with customers.

The study also covered the implementation of database management to ensure that customer information, reservation records, accommodation details, and transaction data were stored in an organized and centralized manner. The database was intended to improve the retrieval, updating, and management of records while reducing the problems associated with handwritten and scattered reservation documents. Security features, including user authentication and login verification, were also incorporated into the system to help protect customer and administrative information from unauthorized access. Access to administrative functions was restricted to authorized users to help maintain the confidentiality and integrity of reservation and customer records.

Another major focus of the study was the automation of manual reservation processes. The system was designed to reduce paperwork, minimize repetitive encoding tasks, lessen the possibility of booking conflicts, and improve the overall efficiency of resort operations. By allowing customers to submit reservations and inquiries through the online platform, the system reduced the need for customers to personally visit the resort or repeatedly communicate with resort personnel to obtain basic reservation information. The automation of booking records, availability monitoring, confirmation, and report generation was also intended to reduce the administrative workload of resort personnel and allow them to focus more on customer service and other operational responsibilities.

Target Users

The primary target users of the proposed Online Booking System were the customers of Avianna's Inland Resort, the resort administrator or management, and the resort receptionist. Resort customers were considered the primary end users of the customer-facing portion of the system. They could use the platform to view resort information, amenities, accommodation rates, and available facilities, as well as check accommodation schedules and make online reservations for cottages, rooms, and pavilions. Customers could also monitor the status of their reservations, review their booking history, and communicate with the resort through the available inquiry functions. The system was intended to provide customers with a more convenient and accessible alternative to traditional reservation methods.

The resort administrator or management served as the primary users of the administrative functions of the system. The resort owner and authorized staff were responsible for managing customer accounts, reviewing and approving reservations, monitoring accommodation availability, updating resort and accommodation information, maintaining records, generating reports, and overseeing the overall operation of the booking system. The administrator functions

were intended to provide management with centralized access to the information needed for effective reservation monitoring and operational decision-making.

The receptionist was also identified as a target user because reception personnel were responsible for handling customer inquiries, monitoring reservation activities, and updating booking information. Through the system, receptionists could access relevant reservation records, check facility availability, respond to customer inquiries, and assist in maintaining updated booking information. Their access to the system was intended to reduce the amount of manual record checking and improve the speed and accuracy of reservation-related services.

Platform of the System

The proposed Online Booking System was developed as a web-based platform that could be accessed using desktop computers, laptops, tablets, and smartphones with internet connectivity. The system could be accessed through standard web browsers, allowing customers and authorized resort personnel to use the platform without installing separate software applications. The web-based design was selected to provide greater accessibility and flexibility for users while allowing the resort to maintain a centralized reservation and information management platform. However, the system's accessibility and performance depended on the availability of a stable internet connection and the proper maintenance of the hosting and system infrastructure.

Delimitation of the Study

The study was limited to the development and implementation of an online booking and reservation management system specifically for Avianna's Inland Resort. The system focused primarily on reservation management, customer accessibility, accommodation availability monitoring, customer record management, communication, and administrative reporting. Certain advanced features and business functions were deliberately excluded from the study to maintain a manageable development scope and to focus on the primary problems identified in the resort's existing reservation process.

The study did not include the development of a dedicated mobile application for Android or iOS devices. Although the web-based system could be accessed through smartphones and tablets using web browsers, the development of separate native or hybrid mobile applications was outside the scope of the study. The system also did not include advanced online payment gateways or direct integration with banking systems and other financial institutions. Payment-related processes that required external banking or payment service integration were therefore not included in the system's primary functionality.

The study also excluded GPS tracking and location-based services because these functions were not considered essential to the resort's primary reservation and booking management requirements. Similarly, artificial intelligence chatbots and advanced customer analytics were not included in the proposed system. The system focused on basic online inquiries and communication rather than automated AI-based customer support or advanced analysis of customer behavior and preferences.

Accounting, payroll, and employee salary management systems were also excluded from the scope of the study. Although the system generated reservation and transaction-related reports, it was not intended to replace a complete accounting or financial management system. Likewise, inventory management for retail store products, food supplies, cleaning materials, and other resort supplies was not included because the primary focus of the study was reservation and accommodation management rather than comprehensive inventory control.

The system also did not include integration with third-party travel agencies, hotel booking platforms, or external reservation marketplaces. Reservations were intended to be managed directly through the official web-based platform developed for Avianna's Inland Resort. Biometric authentication and advanced cybersecurity systems were also excluded from the scope of the study. The system incorporated basic authentication and access-control mechanisms appropriate for its intended users, but highly specialized security technologies were beyond the limitations of the project.

The system did not support offline reservation processing. Since the platform was web-based, customers and authorized personnel needed an internet connection to access and use its reservation and management functions. In

situations where internet connectivity was unavailable, the resort would still need to rely on its existing manual procedures until the system became accessible again. Therefore, the effectiveness and availability of the proposed system depended on stable internet connectivity, proper system maintenance, and the availability of the necessary technical infrastructure.

1.5 Significance of the Study

The study was considered significant because it addressed the challenges associated with the manual reservation procedures and traditional customer service processes of Avianna's Inland Resort in Sta. Barbara, Iloilo. As information and communication technology had continued to influence the tourism and hospitality industry, resorts and recreational establishments had increasingly adopted digital solutions to improve operational efficiency, enhance customer satisfaction, organize business information, and remain competitive. The development and implementation of the Web-Based Online Booking System for Avianna's Inland Resort had provided an opportunity to apply modern information technology to the resort's existing reservation and management processes.

The proposed system had modernized the reservation procedures of the resort by providing a centralized online platform through which customers could access resort information, view accommodation options, check facility availability, and submit reservation requests. It had also provided management with tools for organizing customer records, monitoring reservations, managing accommodation schedules, and generating reservation-related reports. Through the automation of these activities, the system has helped reduce reliance on manual record keeping, minimize repetitive administrative tasks, lessen the possibility of human errors, and improve the overall efficiency of reservation management. The system had also contributed to strengthening the resort's online presence by providing customers with a dedicated digital platform for accessing its services.

Furthermore, the study had contributed to the growing body of knowledge regarding the application of information systems in the tourism and hospitality sector. It demonstrated how a web-based reservation system could be used to address common operational challenges experienced by small and developing resort establishments. The study also provided an example of how digital technology could support customer service, reservation management, data organization, facility monitoring, and business decision-making. The developed system could serve as a useful reference for other resorts, recreational establishments, and small tourism-related businesses that intended to adopt similar digital solutions to improve their operations.

The study was particularly significant to the following beneficiaries:

Administrators and Resort Management

The administrators and management personnel of Avianna's Inland Resort were among the primary beneficiaries of the proposed Online Booking System. The system provided management with a more organized, efficient, and reliable method of handling reservation transactions and monitoring resort operations. Before the implementation of the system, reservation activities had primarily depended on manual logs, phone inquiries, paper-based records, text messages, and other forms of direct communication. These procedures had required considerable time and effort and had created the possibility of lost records, inaccurate information, scheduling conflicts, and double bookings.

Through the implementation of the system, administrators had been able to manage customer reservations and accommodation activities in a more systematic manner. The centralized platform allowed authorized personnel to monitor current, upcoming, and completed reservations while maintaining organized customer and booking records. Reservation information could be retrieved and updated more efficiently, reducing the time required to search through physical documents or manually review reservation logbooks. This improved accessibility to information had allowed administrators to respond more accurately and promptly to customer inquiries regarding reservation status, accommodation availability, booking dates, and other relevant details.

The system had also assisted administrators in monitoring the availability of cottages, rooms, pavilions, and other bookable facilities. By maintaining updated reservation information, the system helped management determine which accommodations had been available, reserved, or occupied. This functionality reduced the possibility of

scheduling conflicts and double bookings that could occur when facility availability was monitored manually. It also provided management with a more reliable method of allocating available accommodations and organizing reservation schedules.

Another important benefit to administrators had been the automated generation of reservation and operational reports. Instead of manually compiling information from various reservation records, administrators could obtain organized reports based on the data stored in the system. These reports provided useful information regarding reservations, occupancy, facility utilization, customer records, and other booking-related activities. The availability of organized reports had supported management in monitoring operational performance, identifying reservation patterns, evaluating facility utilization, and making more informed decisions regarding resort operations.

The system had also reduced the administrative burden associated with repetitive reservation and record-management tasks. Activities that previously required extensive manual encoding, checking, and documentation had become more manageable through the centralized digital platform. This allowed resort personnel to allocate more time to customer service and other important operational responsibilities. Overall, the system had improved the productivity and workflow of administrators by providing accurate, organized, and accessible information while reducing the time and effort required for routine reservation management.

End Users, Customers, and Guests

The customers and guests of Avianna's Inland Resort were also significant beneficiaries of the proposed Online Booking System. The system had been developed to provide customers with a more convenient, accessible, and efficient method of obtaining resort information and making reservations. As customers had increasingly preferred digital services for travel and accommodation transactions, the availability of an online reservation platform had provided a more modern alternative to traditional booking procedures.

Through the system, customers had been able to browse information about the resort's facilities, amenities, cottages, rooms, pavilions, accommodation rates, and available services. This allowed them to review relevant information before making a reservation and reduced the need to personally visit the resort or repeatedly contact resort personnel to obtain basic information. Customers could also check accommodation availability and select their preferred facility based on their intended reservation date and requirements.

The online reservation function had made the booking process more convenient because customers could submit reservation requests without personally visiting the resort. The system could be accessed through internet-enabled devices, including computers, tablets, and smartphones, allowing customers to make inquiries and reservations from different locations. This accessibility had reduced the time and effort required to complete reservation transactions and had provided customers with greater flexibility when planning their visits.

The system had also improved the speed and convenience of the reservation process by providing customers with booking status information and reservation confirmations. Customers could monitor the progress of their reservations and review important booking details through their accounts. The availability-monitoring feature also helped customers determine whether their preferred accommodations had been available before submitting a reservation request. This reduced the possibility of selecting already reserved facilities and allowed customers to make more informed decisions.

In addition, the system had improved communication between customers and resort management by providing an organized channel for reservation-related inquiries and updates. Customers had greater access to information regarding accommodation rates, available facilities, reservation schedules, and booking details. The availability of clear and organized information had improved transparency and reduced potential misunderstandings between customers and resort personnel.

The system had therefore contributed to a more convenient and reliable customer experience. The reduced need for physical visits, faster reservation processing, improved access to resort information, availability monitoring, and organized booking records had provided customers with greater control over their reservations. These improvements had contributed to increased convenience and satisfaction by making the overall reservation process easier, faster, and more accessible.

Avianna's Inland Resort as an Organization

Avianna's Inland Resort as a business organization had also benefited from the development and implementation of the proposed Online Booking System. The system had supported the resort's transition from traditional manual procedures toward a more organized and technology-driven method of managing reservations and customer services. This transition had contributed to improvements in operational efficiency, service delivery, record management, and customer engagement.

By replacing or reducing manual reservation procedures, the organization had been able to streamline several booking-related activities. The centralized system had provided management with organized access to customer records, reservation information, facility availability, and operational reports. This improved organization of information had reduced the administrative effort associated with maintaining paper-based records and manually tracking reservations. It had also helped minimize errors and improve the consistency of booking information.

The system had also strengthened the resort's online presence by providing a dedicated digital platform through which potential customers could access information about its facilities, accommodations, rates, and services. The online platform had therefore served not only as a reservation tool but also as an additional means of promoting the resort and increasing its visibility to potential customers. By making its services more accessible online, the resort had gained greater opportunities to reach customers who preferred digital transactions and online reservation services.

The availability of computerized reservation reports had also benefited the organization by providing management with organized information for evaluating business performance. Reservation data could be used to monitor booking activities, facility utilization, and customer demand. Such information could support management in identifying trends, planning operational activities, improving resource allocation, and making informed business decisions. The system had therefore provided management with a stronger information base for planning and evaluating the resort's operations.

The implementation of the system had also been expected to contribute to reduced administrative costs by minimizing paperwork, repetitive manual tasks, and the time required to process reservations and prepare reports. Improved operational efficiency could allow resort personnel to focus more on customer service and other activities that directly contributed to the quality of resort operations. In the long term, improved service delivery and customer accessibility could contribute to increased customer satisfaction, stronger customer relationships, and greater opportunities for business growth.

Online Booking System has provided Avianna's Inland Resort with an opportunity to improve its reservation management, strengthen its digital presence, enhance service quality, and increase its competitiveness within the tourism and hospitality industry. The system had supported the organization's efforts to adapt to changing customer expectations and the increasing use of digital technology in tourism-related services.

Future Researchers

The study was also significant to future researchers who intended to conduct related studies involving online booking systems, reservation management, tourism information systems, or digital transformation among small and medium-sized hospitality establishments. The study provided information regarding the development and implementation of a web-based reservation system designed to address actual operational challenges encountered by a local resort.

The findings, system features, development processes, and identified operational requirements could serve as references for future studies involving similar systems. Future researchers could use the study as a foundation for improving existing reservation technologies, developing additional features, evaluating system usability and performance, or integrating more advanced technologies into resort management systems. They could also explore areas that were outside the scope of the present study, such as mobile applications, online payment gateways, artificial intelligence-based customer support, advanced analytics, inventory management, and integration with third-party booking platforms.

The study could therefore contribute to future research and development efforts by demonstrating how information technology could be applied to improve reservation processes, customer service, and operational management within the tourism and hospitality industry.

Overall Significance

The development and implementation of the Web-Based Online Booking System had been significant because it addressed the reservation and customer service challenges experienced by Avianna's Inland Resort while providing practical benefits to its customers, administrators, and the organization as a whole. The system had improved access to resort information, simplified reservation procedures, organized customer and booking records, assisted in monitoring accommodation availability, supported automated reporting, and improved communication between customers and resort personnel. Through these improvements, the system had contributed to more efficient operations and a more convenient customer experience.

The importance of adopting digital technology as part of the modernization and development of local tourism and hospitality establishments. The implementation of the system had provided Avianna's Inland Resort with a foundation for improving its service delivery, strengthening its online presence, responding to changing customer expectations, and competing more effectively in an increasingly digital tourism environment. In the long term, the system had the potential to support continued operational improvement, customer satisfaction, business growth, and sustainability of Avianna's Inland Resort.

The researchers were among the significant beneficiaries of the study because the development and implementation of the Web-Based Online Booking System provided them with valuable academic, technical, and professional experiences. The project allowed them to apply the theoretical concepts, principles, and skills they had acquired throughout their Bachelor of Science in Information Technology program to an actual business environment. Instead of learning information technology concepts solely through classroom activities, the researchers were able to apply their knowledge to a real-world problem involving reservation management, customer service, data organization, and business operations. This experience provided them with a deeper understanding of how information technology could be used to develop practical solutions to organizational challenges.

Through the development of the system, the researchers strengthened their programming and software development skills by applying various programming concepts and techniques in the creation of the web-based application. They gained practical experience in web development, including interface design, system functionality, database integration, system testing, debugging, and continuous improvement. The development process required them to analyze the identified problems of Avianna's Inland Resort, determine the system requirements, design appropriate functionalities, and implement features that addressed the needs of both customers and resort management. These activities enhanced their ability to translate theoretical knowledge into functional software solutions.

The researchers also improved their knowledge and skills in database design and management. They learned how to identify the information required by the system, organize data into appropriate structures, establish relationships among different records, and implement methods for storing, retrieving, updating, and managing customer and reservation information. Through this process, they developed a better understanding of the importance of accurate and organized data in the operation of an information system. They also gained practical experience in maintaining data integrity and ensuring that information could be accessed efficiently when needed by authorized users.

Throughout the development process, the researchers encountered various technical, operational, and design challenges. These challenges helped strengthen their analytical and problem-solving abilities because they were required to identify the causes of problems, evaluate possible solutions, test alternatives, and make appropriate decisions. The process of troubleshooting system errors and improving system functions allowed them to develop greater patience, adaptability, and critical thinking skills. These experiences were particularly valuable because they demonstrated that software development involved continuous evaluation and improvement rather than simply completing the initial programming of the system.

The project also strengthened the researchers' research, documentation, and communication skills. They gathered information from the resort, analyzed existing reservation procedures, identified operational problems, documented system requirements, and prepared the necessary research and system development materials. Through these

activities, they gained a better understanding of how research findings could be translated into system requirements and how proper documentation could support the development and evaluation of an information system. They were also able to apply software engineering principles involving planning, analysis, design, development, testing, evaluation, implementation, and maintenance.

Furthermore, the study provided the researchers with practical experience in project planning and management. They were required to organize development activities, assign responsibilities, monitor progress, meet project objectives, manage time, and address challenges encountered throughout the project. Working collaboratively on the system also strengthened their teamwork, communication, coordination, and leadership skills. These experiences helped them understand the importance of cooperation and effective communication when developing information systems as part of a team.

The project also gave the researchers a deeper understanding of the role of information technology in improving business operations. By developing a system that addressed actual reservation and administrative problems, they were able to observe how digital solutions could reduce manual work, improve information organization, increase operational efficiency, minimize errors, and support management decision-making. The successful completion of the project therefore contributed to their academic growth and helped develop professional competencies that were relevant to the information technology industry. The knowledge, skills, and experiences gained throughout the study provided a strong foundation for their future academic activities, employment opportunities, and professional careers in the field of information technology.

Future Developers

Future developers could benefit from the study because it provided a practical reference for the design, development, implementation, and evaluation of web-based booking and reservation systems. The methodologies, system requirements, database structure, system architecture, development procedures, and evaluation results presented in the study could serve as useful guides for individuals or development teams planning to create similar systems for resorts, hotels, recreational facilities, and other service-oriented establishments.

The study could provide future developers with a better understanding of how a reservation management system could be designed to address common problems such as manual record keeping, double bookings, scheduling conflicts, inefficient facility monitoring, and delayed reservation processing. The system development process could also serve as a reference for applying software development methodologies to an actual business environment. Developers could examine the approaches used in analyzing requirements, designing system functions, implementing database structures, conducting testing procedures, and evaluating system performance and usability.

The study could also serve as a basis for future system enhancements. Although the developed system addressed the primary reservation and management needs identified during the study, future developers could expand its functionality based on changing business requirements and technological advancements. Possible enhancements could include the development of dedicated Android or iOS applications, integration of online payment gateways, cloud-based data storage, advanced customer analytics, artificial intelligence-based customer support, automated notifications, and other modern technologies. Future developers could therefore use the existing system as a foundation for developing more advanced and comprehensive tourism and hospitality information systems.

The study could also encourage future developers to consider user-centered design when developing business applications. By considering the requirements of both customers and resort management, future systems could be designed to provide interfaces and functions that were easier to understand, more accessible, and more suitable for actual users. In this way, the study could contribute to the continuous improvement and innovation of information systems within the tourism, hospitality, and service industries.

Future Researchers

Future researchers could benefit from the study as a source of information and reference material for conducting related research involving online reservation systems, hospitality management systems, tourism information technologies, and digital transformation. The methodologies, data-gathering techniques, findings, system requirements, development processes, and recommendations presented in the study could provide a foundation for

future investigations involving similar research topics.

The study could support future researchers who intended to conduct comparative studies involving manual and computerized reservation systems or evaluate the effectiveness of online booking platforms in different tourism and hospitality establishments. The findings could also provide useful information for researchers examining the relationship between digital technology, customer satisfaction, operational efficiency, and business competitiveness. Researchers could build upon the present study by examining how the adoption of information systems affected reservation volume, customer engagement, facility utilization, service quality, and management decision-making.

Furthermore, future researchers could investigate areas that were not included in the scope of the present study. These could include online payment integration, mobile booking applications, artificial intelligence, predictive analytics, cloud computing, automated customer support, advanced cybersecurity, inventory management, and integration with third-party tourism platforms. By exploring these areas, future research could contribute to the continued development of innovative technologies for the tourism and hospitality sector.

The study could also contribute to academic literature in the field of information technology by demonstrating how a web-based system could be developed to address specific operational problems within a local resort. Future researchers could use the study as a reference when developing research instruments, identifying system requirements, selecting appropriate development methodologies, or designing evaluation procedures. In this manner, the study could serve as a starting point for further research and innovation in tourism information systems.

Tourism Industry

The tourism and hospitality industry could also benefit from the development and implementation of the Online Booking System because the study demonstrated how digital technology could be applied to improve the management and delivery of tourism-related services. As customers had increasingly relied on digital platforms when searching for accommodations, checking availability, and making reservations, the adoption of online systems had become an important component of modern tourism operations.

The study demonstrated how a web-based booking system could help tourism establishments improve reservation management, customer accessibility, service delivery, and operational efficiency. By providing customers with convenient access to resort information and reservation services, digital platforms could reduce barriers to booking and improve customer engagement. At the same time, centralized reservation management could help tourism businesses organize customer information, monitor facility availability, reduce booking errors, and generate useful operational reports.

The implementation of such systems could also encourage small and medium-sized tourism enterprises to participate in digital transformation. Many local tourism establishments had traditionally depended on manual procedures because of limited resources, technical knowledge, or lack of appropriate systems. The study demonstrated that a web-based information system could provide practical solutions to common operational challenges and could support the modernization of tourism services.

The adoption of digital reservation technologies could further contribute to improved service standards and competitiveness within the industry. Tourism businesses that provided accessible and convenient online services could respond more effectively to changing customer expectations and compete with establishments that had already adopted digital platforms. The study therefore highlighted the importance of technological innovation in improving customer engagement, operational performance, and long-term business sustainability within the tourism and hospitality sector.

Local Community

The local community of Sta. Barbara, Iloilo, could also indirectly benefit from the successful implementation and continued operation of the proposed system. As Avianna’s Inland Resort improved its reservation processes, online accessibility, and overall service delivery, the resort could become more accessible to visitors from nearby communities and other locations. Increased accessibility and improved online visibility could contribute to greater interest in the resort and potentially increase the number of visitors who traveled to the area.

An increase in resort visitors could contribute to increased economic activity within the surrounding community. Customers visiting the resort could also patronize nearby restaurants, transportation services, stores, suppliers, and other local businesses. In this way, the growth of the resort could create opportunities for other businesses operating within or near the community. Increased visitor activity could also contribute to the promotion of Sta. Barbara as a recreational and tourism destination.

The continued development of the resort could potentially create additional employment opportunities as the demand for resort services increased. More visitors could require additional personnel for customer service, maintenance, food services, facility management, and other operational activities. These opportunities could provide additional sources of income for members of the local community.

The improved online presence of Avianna's Inland Resort could also contribute to greater visibility of local tourism attractions. By promoting the resort through a dedicated digital platform, potential visitors could become more aware of recreational opportunities available in Sta. Barbara. The study therefore demonstrated how the digital development of a local tourism establishment could have benefits that extended beyond the organization itself and could contribute indirectly to local economic activity, tourism promotion, and community development.

Bachelor of Science in Information Technology Students

Students pursuing the Bachelor of Science in Information Technology program could benefit from the study by using it as an academic and practical learning resource for future system development projects. The study provided an example of how concepts learned in information technology courses could be applied to an actual business problem. Through the system development process, students could gain a better understanding of how systems analysis, database design, web application development, software engineering, research methodology, testing, documentation, and project management were applied in a real-world context.

The study could also provide BSIT students with practical examples of the different stages involved in developing an information system. Students could examine how system requirements were identified, how user needs were translated into system functions, how databases were designed, how web-based interfaces were developed, and how system functionality was tested and evaluated. These examples could help students understand the relationship between theoretical concepts and actual software development activities.

It could provide ideas regarding system architecture, database management, user authentication, reservation management, reporting functions, and user-centered design. Students could also identify potential improvements and additional features that could be incorporated into future projects based on the limitations of the developed system.

The study could also encourage BSIT students to develop solutions that addressed actual problems within businesses and communities. Rather than focusing solely on technical functionality, the project demonstrated the importance of understanding user requirements, operational processes, and organizational needs when developing an information system. Through this perspective, students could gain a better appreciation of the role of information technology in solving practical problems and improving business operations.

1.6 Definition of Terms

The following terms were conceptually and operationally defined to provide a clear understanding of the important concepts, technologies, and terminologies used in this study. Since the research involved the design, development, and implementation of a web-based Online Booking System for Avianna's Inland Resort, the researchers included terms related to information technology, database management, web development, tourism, hospitality, and reservation management. The conceptual definitions described the generally accepted meanings of the terms, while the operational definitions explained how the terms were specifically applied within the context of the study.

The terms were arranged alphabetically to facilitate easy reference and understanding. The definitions were presented according to their relevance to the development and implementation of the Online Booking System and were intended to establish a consistent interpretation of the terminology used throughout the study.

Administrator - The administrator referred to the authorized resort personnel who managed customer accounts, reservations, accommodation availability, resort information, inquiries, and reports through the Online Booking System.

Authentication - refers to the login process used by customers and authorized resort personnel to verify their identity and access the appropriate functions of the Online Booking System.

Availability Monitoring - referred to the system function that allowed administrators and customers to determine whether cottages, rooms, pavilions, and other reservable facilities were available, reserved, or occupied for specific dates.

Booking - Operationally, booking referred to the process through which customers submitted requests to reserve cottages, rooms, pavilions, or other available facilities of Avianna's Inland Resort.

Booking System - Operationally, the booking system referred to the web-based platform developed in this study to allow customers to submit reservations while enabling resort personnel to manage booking records and accommodation availability.

Customer - Operationally, a customer referred to a person who accessed the Avianna's Inland Resort Online Booking System to view resort information, check accommodation availability, submit reservations, monitor booking status, or make inquiries.

Database - the database referred to the centralized storage component of the Online Booking System used to store customer information, user accounts, reservation records, accommodation details, booking status, and other relevant system data.

Database Management - Operationally, database management refers to the activities performed within the Online Booking System to organize and maintain customer records, reservation information, accommodation data, and other system-related information.

Digital Transformation - Operationally, digital transformation referred to the transition of Avianna's Inland Resort from manual reservation and record-keeping procedures toward a computerized and web-based reservation management process.

End User - Operationally, end users referred to customers, administrators, and authorized resort personnel who accessed and used the functions of the Online Booking System according to their assigned roles.

Hospitality Industry - Operationally, the hospitality industry referred to the business sector in which Avianna's Inland Resort operated, particularly in providing accommodation, recreational facilities, and leisure services to customers and guests.

Information System - Operationally, the information system referred to the web-based Online Booking System

developed to collect, process, store, retrieve, and manage customer and reservation information for Avianna's Inland Resort.

Internet - Operationally, the internet referred to the network connection required by customers and authorized resort personnel to access and use the web-based Online Booking System.

Online Booking System - Operationally, the Online Booking System referred to the web-based application developed specifically for Avianna's Inland Resort to provide online resort information, accommodation availability, reservation services, customer record management, booking monitoring, and administrative functions.

Online Reservation - Operationally, online reservation refers to the process through which customers use the Online Booking System to select available resort facilities, provide required booking information, choose reservation dates, and submit a reservation request electronically.

Password - Operationally, a password refers to the security credential created by users to access their registered accounts in the Online Booking System and protect their account information from unauthorized access.

Pavilion - Operationally, a pavilion referred to one of the event and recreational facilities available at Avianna's Inland Resort that customers could view and reserve through the Online Booking System based on availability and applicable rates.

Real-Time Monitoring - real-time monitoring referred to the Online Booking System's capability to display updated reservation and accommodation availability information, allowing authorized users to determine whether specific cottages, rooms, or pavilions had been available or already reserved.

Record Management - record management refers to the process of organizing and maintaining customer information, reservation records, booking histories, and other relevant data within the Online Booking System.

Report Generation - referred to the system function that automatically produced reservation, customer, occupancy, and transaction-related reports based on information stored in the system database.

Reservation - Operationally, a reservation refers to a customer's confirmed or submitted request to use a cottage, room, pavilion, or other reservable facility of Avianna's Inland Resort on a specified date.

Resort - Operationally, the resort referred to Avianna's Inland Resort in Sta. Barbara, Iloilo, which served as the subject establishment of the study and the intended organization for the implementation of the Online Booking System.

Security - Operationally, security referred to the authentication, account protection, access control, and other security measures incorporated into the Online Booking System to help protect customer and administrative information.

System - Operationally, the system referred to the Web-Based Online Booking System developed to manage reservations, customer information, accommodation availability, inquiries, and administrative activities of Avianna's Inland Resort.

Tourism - Operationally, tourism referred to the sector of activity associated with visitors and travelers that Avianna's Inland Resort served by providing recreational facilities, accommodations, and leisure services.

User Account - Operationally, a user account refers to the registered profile created by customers or authorized personnel to access specific functions of the Online Booking System according to their assigned user role.

User Interface - Operationally, the user interface referred to the screens, menus, buttons, forms, navigation elements, and other visual components of the Online Booking System through which customers and administrators accessed and performed system functions.

Web-Based System - Operationally, the web-based system referred to the Online Booking System of Avianna's

Inland Resort, which could be accessed through standard web browsers using internet-enabled computers, tablets, and smartphones.

Website - Operationally, the website referred to the online platform developed for Avianna's Inland Resort that presented information about the resort's facilities, accommodations, rates, services, and reservation options while providing customers with access to the Online Booking System.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Over the last ten years, the use of online reservation systems for accommodation has rapidly increased worldwide. This trend has become especially important for rural lodging businesses because it allows even small establishments to connect with a large number of international customers in an affordable and convenient way. However, only a few major reservation platforms now control most of the market, affecting locally operated national and regional booking systems. Features such as price comparison, customer ratings, and online reviews influence competition among businesses, quality standards, marketing approaches, and revenue distribution.

The study investigated these effects on rural accommodation providers in Western Norway through semi-structured interviews. The research explored how and why businesses started using Booking.com, currently the largest online reservation platform, and how it changed their operations. Findings revealed that Booking.com plays a major role in rural tourism reservations in Norway and has both positive and negative impacts on businesses and destinations. Although many accommodation providers benefit from increased visibility and bookings, national and regional organizations lose revenue that once supported training, marketing, lobbying, and tourism development activities. The paper also suggests possible new functions for these organizations in the future.

Another study discussed the development of rural tourism in developed countries since the 1970s. Rural tourism became popular as an alternative tourism form and as a tool for rural development and environmental conservation. The paper compared the growth of rural tourism with Butler's Tourism Area Life Cycle, which includes stages such as emergence, expansion, increasing complexity, and eventual challenges caused by competition, weak governance, social changes, and technological advancements.

The researchers analyzed 1,848 articles published since 2000 using the Scopus database to examine how research has responded to changes in rural tourism. The findings showed that tourism studies have shifted from traditional sightseeing toward experiential tourism activities. Researchers also focused on topics such as governance, leadership, networking, product innovation, and marketing. The paper concluded by recommending the creation of a "New Generation Rural Tourism" that emphasizes research-based planning, sustainable management, improved marketing strategies, and stronger destination development. It also proposed establishing an international research group dedicated to studying and supporting sustainable rural tourism development.

As the hospitality industry seeks to provide superior service and ensure guest satisfaction, several factors affect companies. Hospitality firms are characterised by a wide range of services, including accommodation, food and beverage operations, and additional services such as transportation, entertainment, and recreation. Companies within the hospitality sector, despite providing similar services, are not homogeneous. However, there are a set of similar traits that are present in hotels in general: companies operate 24 h a day, seven days a week; are both a production and services industry, with production and sales combined in the same place; combine a wide range of operations that require a high degree of coordination; require many different skills from staff, but have also a high number of unskilled staff; services are provided directly to guests, but they leave the premise without any tangible product; and have a high fixed cost, a fixed rate, but depend on a fluctuating demand, many times seasonal and unpredictable , as crises such as the COVID-19 pandemic showed.

The hospitality sector is a major contributor to the global economy. It generates revenue, creates jobs, and stimulates economic growth in regions where it operates. However, hotels, and other companies within the sector, are highly dependent on tourism demand, while factors affecting tourism demand can either cause harm or increase the flow of potential guests. The tourism industry is highly sensitive to changes and tourists are risk-averse . They take their decisions regarding where to travel carefully, once perceived threats to their safety, health, or security are considered

seriously in their decisions. Tourism demand depends on the perceptions about a tourist destination and how tourists assess that destination. Tourists tend to travel to less dangerous destinations as the Arab Spring showed, through a reduction in the flow of tourists in the regions affected and in adjacent countries. However, the reduced demand in those places led to a tourist diversion to outer countries , negatively influencing the demand for tourism-related services such as those provided by the hospitality industry within the former countries, and increasing the flow of guests to hotels in the latter, which are perceived as more secure countries.

The positive effect of the tourism industry and tourism demand on societies is observed through its economic contributions. The tourism industry is a significant economic driver in many regions, by generating revenue and jobs, stimulating local economies, and fostering societal well-being. In 2023, the industry contributed to 3% of global Gross Domestic Product (GDP) and an estimated total export revenues of USD 1.6 trillion . Nevertheless, the tourist sector is not confined to a specific economic sector. It encompasses a wide set of economic and social activities generally seen on a global scale, including goods and services from various economic sectors . The hospitality industry is among the relevant industries within the tourism industry, on which the sector depends. The hospitality industry contributes heavily to the role of the tourism industry within the economy, providing a set of services that tourists are counting on, such as accommodation, food, transportation, or entertainment. The overall set of services provided serves as a catalyst for regional development, through factors such as job creation. However, the tourism industry, in a broader sense, as well as the hospitality industry, are also dependent on the idiosyncrasies of places visited. Hence, factors affecting tourism demand are critical to implementing tourism and hospitality policies and achieving efficiency in the tourism and hospitality industries globally .

Factors affecting tourism demand as well as hospitality industry demand vary. Yet, tourism demand impacts the hospitality industry's demand, as fewer visitors imply fewer potential clients for hospitality companies. For instance, a place or region's perceived risk affects the flow of tourists to that place or region , and factors such as GDP per capita, real exchange rate, population, and trade openness also affect the flow of tourists . Additionally, the hospitality industry affects a place or region's tourism demand, particularly through prices , and the lack of services provided by hospitality firms would also affect tourism demand.

Following the argument regarding the interrelations between tourism demand and hospitality industry demand, this study seeks to examine the intricacies of tourism demand and hotel industry demand, and how they interact with each other. Both tourism and hospitality industries have a major impact in supporting a region's, country's, and place's economy, where a decline in the number of visitors can cause severe economic stress, as recent events such as the COVID-19 pandemic or war and terrorism have shown. Furthermore, demand analysis in tourism and hospitality supports optimal resource allocation due to demand fluctuation, helps marketing and business managers plan offers and operations, and enables firms and policy makers to address the environmental and cultural impacts of tourism, managing crises, and investment decisions. Consequently, to accomplish the proposed objective, this study uses a content analysis technique that, based on an automated co-word analysis, seeks to explore and understand the conceptual structure of tourism and hospitality demand and how the research stream evolved over the recent periods of global turmoil such as the pandemic, geopolitical tensions, economic crisis, and war. This paper further aims to provide an interpretation of how the conceptual structure interrelates and delivers guidance for practitioners, companies, and academics. The previous literature employing thematic analysis in tourism and hospitality demand addressed specific arguments such as crisis management , human resources , or online comments . However, a search targeting specifically tourism demand and hospitality demand yielded no results.

This study includes five sections. **Section 2** presents a literature review concerning tourism demand and hospitality industry demand, and **Section 3** presents the study methodology. Results are shown in **Section 4** and **Section 5** presents the results discussion and main conclusions.

Demand variability in the tourism industry depends on trust from tourists, as recent events have shown, including pandemics, war, and terrorism. These events cause severe harm to tourist flow and risk the overall industries within the tourism sector. The sustainability of places, including the environmental effects of the tourism industry , terrorism as a source of insecurity , and crime affect tourism demand, since tourists perceiving potential risk may consider not visiting a place again and advise friends and relatives about that place .

Relevant theories within the tourism and hospitality demand literature involve aspects such as economic and socio-psychological factors, including income and prices, motivations and destination choices , consumer choice theory, addressing consumer decision-making processes, budget constraints, characteristics of destination , like stereotypes and prejudices , non-economic determinants like cultural background , and structural changes in market dynamics, driven by economic trends such as change in consumer spending, preferences, and rising incomes. Furthermore, previous research showed that demand variability in the tourism industry witnessed relevant

fluctuations due to external events in the industry. The 11 September 2001 terrorist attack in the US drove tourism demand to a downturn . The 2003 Severe Acute Respiratory Syndrome (SARS) had similar effects; the Middle East Respiratory Syndrome (MERS) outbreak , first detected in 2012, and the 2004 Indian Ocean tsunami, which affected countries such as India, Indonesia, Malaysia, the Maldives, Myanmar, Sri Lanka, Seychelles, Thailand, and Somalia , showed how tourism industry demand is sensitive to external factors, and the COVID-19 pandemic emphasised it . However, tourism and hospitality demand depend on further issues than tourist security risks.

Financial risk is also among the relevant factors considered by tourists when deciding to travel. The 2007–2008 global financial crisis, for instance, caused a severe downturn in the tourism industry. Due to the economic consequences of the crisis, including unemployment and a decrease in purchasing power, the industry observed a decrease in tourist flow . These types of crises reduced customers' purchasing power for tourist services, causing major challenges to the sector , with a critical impact on the employment, opportunities, and income of tourism employees, and disparate effects on tourism industries .

Tourism demand is critical to the hospitality industry as the number of guests, among other aspects, depends on the number of tourists visiting a place. The specificities of the hospitality industry put it among the ones more at risk of being hit by the factors influencing the tourism industry, despite some previous evidence showing the resilience of hotels and restaurants facing crises, particularly financial crises. When comparing these companies with the remaining tourism sector , the industry is not immune to the consequences of consumer assessment of risk, neither to pressure or guidance from regulatory entities, nor mandatory measures targeting customers and general public safety, as the COVID-19 pandemic demonstrated.

The hospitality sector presents traits that make it prone to demand fluctuation. Among them, customer trust, economic crises, and safety are critical. However, they are mostly out of the control of hospitality firms. Conversely, other aspects such as cleanliness , prices , perceived quality , convenient location, service quality, reputation, and friendliness of staff can be addressed by companies, individually, and are capable of making a relevant influence on firms' demand.

The previous literature addressing hospitality firms' demand was focused on topics ranging from internal and external aspects of the industry. From an external standpoint, environmental and policy factors, including climate change and uncertainty , the influence of environmental policies, and sustainability concerns of the industry , as well as the impact of natural disasters , were highlighted in the literature. In recent years, the literature addressing hospitality demand, presented an emphasis on economic instability, fostered by the most recent pandemic and war, the impact of global events, geopolitical tensions, and crises such as the Russia–Ukraine war . However, the adaptation of companies during the COVID-19 pandemic , particularly through service innovation, emphasised the importance of demand to the hospitality industry and how these events affect demand forecasting .

The effect of consumer behaviour on hospitality companies' demand has been addressed from various perspectives, including an economic approach , clients' consumption patterns , and authentic experiences . Economic instability factors impact and its effect on revenue management due to demand , including trends and individual financial circumstances, namely household income and prices , have also been emphasised. However, hospitality demand and consumer behaviour also depend on firms' innovation and on services provided to customers, marketing, and branding strategies towards competitiveness to meet customer needs, in which creative marketing and innovative branding play a relevant role. Yet, customers decide where to stay based on further aspects and are not immune to pricing strategy, which plays a great deal in demand modelling to forecast and optimise hotel performance . In fact, the relevance of forecasting hotel demand has gained relevance in the literature through machine learning, technology, and big data.

Among the main characteristics of service provided by hospitality firms is the quality of personnel. Employees play a critical role in providing superior value to customers. Nevertheless, the industry usually suffers from a shortage of skilled personnel and needs adequate educational programs aligned with the industry requirements, which can be solved through the development of hospitality, as well as tourism management programs . Qualified managers are likewise critical to addressing the idiosyncrasies affecting hospitality companies' demand.

Therefore, the effects of tourism demand and hospitality demand on societies are critical and usually observed from several scopes including economic contributions and employment, social development and cultural exchange, impact on local communities, effects on sustainable development, and social responsibility.

In the Philippine context, the tourism industry's shift toward digitalization is increasingly evident through the adoption of online platforms. Hotel companies that prioritize technology are better positioned to disrupt the traditionally slow-moving and inefficient operations of some luxury hotel brands. Considering this, this study aimed to examine the system quality of digitalized hotel booking in two four-star hotels located in Pasig City, and to assess its impact on guests' acceptance of these digital booking systems. Anchored on the Technology Acceptance Model (TAM) and the SERVQUAL model, this study employed a descriptive-correlational research design to determine the extent of acceptance of digitalized hotel booking systems, as well as the perceived quality of such systems. A total of 382 guests participated in the study. The relationship between the extent of acceptance and the perceived quality of digitalized hotel booking was analyzed using Pearson's r. Furthermore, the relationship of perceived usefulness and ease of use on the quality of the booking application was examined through linear regression analysis. The digitalized hotel booking systems are perceived to be of very high quality across key service dimensions, including tangibles, reliability, responsiveness, assurance, and empathy. Guests demonstrate a very high level of acceptance toward these systems, particularly in terms of their perceived usefulness and ease of use. Furthermore, a strong positive correlation was found between the perceived system quality and guest acceptance of digital booking platforms. The quality of the system significantly influences both the perceived usefulness and ease of use, underscoring the critical role of system quality in enhancing user experience and acceptance.

There has also been a rise in technological innovations in the hospitality industry that aim at improving the performance of the industry and customer satisfaction. The use of online reservation systems to oversee bookings in hotels using online platforms, which are not dependent on human involvement, is one of the most revolutionary trends noticed in the past few years. Such systems would facilitate booking of rooms quickly and real-time room booking, as well as instant confirmation of rooms available to guests. Barua and Kaiser (2024) state that online reservation is now an indispensable tool of hotel management that is characterised by such benefits as enhanced accessibility, increased rates of booking, and centralised control of inventory. The adoption of these kinds of technologies is especially essential in cities such as Accra, where tourism and business travel are consistently rising, thereby subjecting hotel services to increased pressure.

Under pressure to keep up with emerging trends, star-rated hotels, which are expected to provide higher levels of service, are increasingly turning to digital solutions that help satisfy customers and stay competitive. According to the study by Oduro (2024), it is noteworthy that in Ghana, hotels equipped with the latest digital systems, including online reservation platforms, are more likely to attract international guests who prioritize convenience and efficiency. The hospitality industry in Accra is growing rapidly, and by adopting online booking technology, workflow can be improved, and guest experiences enhanced as booking errors and queues are reduced. However, the adoption rate and success of these systems vary among hotels, depending on managerial attitudes, financial

resources, and staff's digital literacy (Azevedo and Osorio, 2020).

As much as there are numerous positive aspects, a number of obstacles exist that prevent the massive use of online reservation technologies by hotels in Accra. They are low ICT systems, high installation expenses, security threats on the internet, and opposition to shift change by employees and leaders (Kwarteng, 2022). We must comprehend how these challenges combine with features of hotels, including star rating, ownership structure, and market orientation, to come up with focused interventions that can be used to encourage technology adoption. Hotels with star ratings are supposed to provide high-quality services and are subjected to pressures of changing technologies so as to be at the same level as international service standards. Of such innovations, the online reservation system has become an important instrument in the management of customer booking solutions, in addition to service delivery and profitability (Fredrick and Authority, 2019).

Some of the benefits that come with adoption of ORS include; increase in revenue management, guest satisfaction and low operational costs. Research conducted in the past has underscored that hotels who employ ORS have an improved occupancy rate, faster check payments, and greater accessibility to foreign tourists (Faliha et al., 2021). Not only that, ORS also gives businesses the opportunity to stay open 24/7, which is a competitive feature in a market where customer convenience should be of paramount importance (Işkın et al., 2024). Nonetheless, regardless of these advantages, the rate of ORS usage and its perceived utility might differ among hotel categories, managers, and technology systems and therefore, it becomes critical to examine the particular ORS advantages achieved by star-rated in Accra.

Even though the world is transforming digitally in the hospitality industry, there has been no clear appreciation of the extent to which these systems have been adopted by star-rated hotels in Accra and what factors dictate their adoption and successful performance (Alshammare et al., 2022; Sepula, 2019). Thus, the research on digital integration of the hotel industry in Accra at the level of star-rated hotels will be a great contribution concerning the question of the contemporary position of the digitalization of the hospitality industry in Ghana and will serve as guidance in making future regulatory and operational policies.

Online Reservation Systems (ORSs) are digital systems and platforms that customers can use to make bookings of services, hotels, tables at restaurants, or packages online (through the internet) without necessarily having face-to-face communication with another party (Zhao et al., 2022). These are solutions that are usually incorporated in the websites of businesses or through third-party websites, such as Booking.com, Expedia, or Airbnb, allowing immediate availability/price/confirmation information to be accessed (Saura et al., 2024). ORSs play a prominent role in the hospitality sector to facilitate efficiency in the overall booking process; minimising labour input by occurrences of manual mistakes; heightening the convenience of the clients, and the access to bookings all over the globe (Spanaki and Durmishllari, 2019).

The components that constitute online reservation systems are usually regarded as the following key functionalities: customer relationship management, inventory management, provision of a secure payment gateway, automated confirmation messages, and data analytics tools (Sadreddini, 2020). In terms of hotels, this implies that they have an opportunity to check the use of rooms, control the pricing policy, create and launch promotional offers, and gather

feedback, all performed through a unified online portal. Such a level of integration helps to increase operational efficiency, improve customer service, and enhance revenue optimization methods (Natalchuk and Brylinska, 2020).

The online reservation technologies are an indispensable part of the competitive hospitality market nowadays due to the development of digitalization and the rising consumer interest in self-service technologies (Park and Kim, 2024). The systems not only meet the needs of the market in terms of faster and convenient booking methods, but they also ensure that the systems are preventive in terms of marketing and branding the product, as well as interaction with customers. Consequently, their combination has now become the criterion of quality and professionalism, in particular, in those star-rated hotels interested in improving the customer experience and operational perfection (Park and Kim, 2024).

The integration of online reservation systems (ORSs) in the hospitality industry can be understood through various theoretical lenses. Among the theories are the Technology Acceptance Model (TAM), Diffusion of Innovations (DOI), and Resource-Based View (RBV).

Technology Acceptance Model

Davis (1989) devised the Technology Acceptance Model (TAM). TAM assumes that perceived usefulness and perceived ease of use are central in influencing the intention of an individual to embrace an emerging technology. Regarding the application of ORS in hotel services, management and employees will embrace its implementation when they realize that it is useful in the delivery of the services and is simple to use. The application of this model has been popular in the research on the realization of digital technology in the service sector, such as that of the hotel industry (Granić and Marangunić, 2019).

TAM will aid the study because it highlights the psychological and functional requirements of adopting ORS. The relevance of the constructs and an assessment of the usefulness and ease of using TAM can be emphasized by the perception of the responses of hotel managers and employees regarding the efficiency and quality of service when using the system (Gupta et al., 2022). Regarding star-rated hotels, where the intention to provide high-quality customer service is usually considered a priority, the belief that by using ORS, booking processes can be simplified, mistakes can be resolved, and customer satisfaction can be increased, becomes a socially compelling reason to implement it. The user friendliness also defines the rate at which the staff would be able to settle into the system, and thus the higher the chances of the system being put to use and becoming permanent in the hotel.

Diffusion of Innovations (DOI)

The Diffusion of Innovations (DOI) introduced by Rogers (2003) explains the scope, issues, and pace at which new technologies are propagated in the organisation or society. DOI focuses on some aspects, which include relative advantage, compatibility, complexity, trialability, and observability (García-Avilés, 2020). DOI helps in the study by expounding on how the whole process of ORS adoption operates in hotel networks. DOI postulates that the hotel star-rating systems will embrace ORS in case they feel that they have a relative advantage concerning traditional booking channels, for example, enhanced competitiveness and enhanced occupancy rates (Li et al., 2025). It also depends on the compatibility with the existing practice and even the chance to test the technology applied or see it in operation. Hotels that observe the peer institution undertaking the implementation of ORS successfully are more likely to take the same step, particularly when the shift happens to be manageable and worth it (Chan et al., 2023).

Resource-Based View (RBV)

A strategic management paradigm known as the Resource-Based View (RBV) emphasizes how crucial an organisation's internal resources and competencies are to obtaining a competitive advantage. This method has become popular for examining decisions about technology adoption, particularly in the hospitality industry. To better understand how organisational resources affect hotels' ability to accept and successfully implement cutting-edge technological solutions for accommodation management, researchers have used the RBV. According to RBV, hotels with unique and abundant resources are better positioned to implement technology and gain a competitive edge (Lubis, 2022).

Finding and using valuable, rare, unique, and non-substitutable (VRIN) resources is a crucial component of RBV. VRIN resources include a variety of assets, including financial capital, technological infrastructure, and experienced human resources, when it comes to the application of technology in accommodation management. Hotels with significant financial resources can invest in state-of-the-art technological solutions that can improve operational efficiency and guest experiences, including property management systems (PMS), customer relationship management (CRM) software, and data analytics tools. Additionally, hotels can easily integrate and implement cutting-edge technology when they have a strong technical infrastructure that includes hardware, cloud-based services, and dependable internet access (Ofori and Appiah-Nimo, 2022).

RBV also highlights how crucial human capital is in influencing decisions about technology adoption. Hotels with informed and competent staff are better able to comprehend, use, and reap the benefits of sophisticated technological solutions, especially in fields like digital marketing, data analytics, and information technology (IT). These human resources support the creation of creative plans, efficient use of technological instruments, and ongoing enhancement of accommodation management procedures (Kruesi and Bazelmans, 2023).

Empirical Review

The empirical investigations of the adoption of online reservation technologies have demonstrated that the level of integration differs significantly in geographic and economic contexts. Goh and Law (2019) highlight that hotels with integrated ORSs tend to report improved occupancy rates, optimized revenue streams, and heightened customer satisfaction. This underscores a direct correlation between the effective integration of digital tools and enhanced performance outcomes in the hospitality industry. Likewise, Amoako and Asare (2021) observe similar benefits within the Ghanaian context, noting that ORTs have facilitated market expansion and operational efficiency. However, they also identify persistent challenges such as financial limitations and technical difficulties that hinder full utilization, suggesting that adoption alone does not equate to deep integration.

Further extending the discussion, Bediako and Odoom (2022) focus on the disparity in ORS usage between internationally affiliated and locally owned hotels in Ghana. Their findings show that larger, globally linked hotels are more likely to integrate ORSs due to better infrastructure, stronger ICT support, and higher staff competence in digital technologies. They point out that major barriers include unreliable internet access, inadequate ICT infrastructure, and limited digital literacy among hotel employees. Interestingly, they also identify key enablers like committed leadership and prior digital experience as crucial for successful ORS integration. This adds an organizational and human dimension to the discussion, suggesting that managerial attitude and staff preparedness are

instrumental in determining integration success.

Nyamekye and Mensah (2023) add a temporal and situational perspective by examining the effects of the COVID-19 pandemic on ORS adoption. Their study reveals that health-related concerns and the demand for contactless services accelerated the adoption of ORSs, particularly in urban star-rated hotels. This finding emphasizes the importance of external pressures, such as public health concerns, in driving technological transformation in service delivery. Collectively, these studies illustrate that ORS integration is shaped by a dynamic interplay of internal capabilities and external influences, such as infrastructure, customer expectations, and crisis events.

Despite the rich insights provided by previous research, a notable gap persists in understanding the depth and sustainability of ORS integration, especially within the context of star-rated hotels in Accra. Most prior studies either generalize findings across the entire hospitality sector or focus predominantly on adoption rather than long-term integration and optimization. The proposed study aims to address this gap by examining not just whether ORTs are adopted, but how deeply they are embedded into hotel operations, what barriers persist, and what contextual opportunities can be leveraged to enhance their long-term utility. This targeted approach will provide a more nuanced and actionable understanding of ORS integration within a specific and significant segment of Ghana's hospitality industry.

[1]Bernard Lane & Elisabeth Kastenholz,Rural Tourism: New Concepts, New Research, New Practice; Guest Editors,Volume 23, 2015 and thank Bill Bramwell, co-editor of the Journal of Sustainable Tourism / *Societies* **2024**, *14*(10), 207; **https://doi.org/10.3390/soc14100207 /** https://doi.org/10.53378/jhtcr.353367 **.**

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2.2 Local Studies

The LakByahe Application offers a ground-breaking solution to revolutionize tourism in Nasipit, Agusan del Norte, Philippines. It is a unified application aiming to address existing challenges by incorporating user-centric features. Through advanced location-based services and streamlined booking processes, the application intends to enhance efficiency, accessibility, and sustainability in the tourism sector. By providing real-time tracking, efficient transit solutions, and a comprehensive directory of establishments, the LakByaheapp fills information gaps and fosters economic growth by connecting tourists with local businesses. The study adopts a mixed-method research approach, combining quantitative techniques with qualitative methods to comprehensively investigate and enhance the booking and reservation system. The software development life cycle for the application involves critical stages such as requirement analysis, design, implementation, testing, deployment, and maintenance. Suggestions and recommendations based on survey responses include incorporating augmented reality experiences, offering curated tour packages, creating community forums, and ensuring offline access to maps. The ongoing collaboration with local businesses and tourists, regular updates, user training sessions, and a well-crafted marketing strategy are recommended to amplify the app's impact. Overall, the LakByahe App presents a comprehensive solution to improve the province's tourism industry, fostering economic growth and sustainable development in the region.

This study carried out a literature review focusing on scientific publications from the Web of Science database between 2018 and 2023, which sought to identify patterns and trends that characterise the evolution of this research stream and to reveal the current state of the art in this field of knowledge.

Offering a theoretical vision associated with demand in the hotel and tourism sector, the thematic analysis reveals relevant insights into market trends over the analysed periods, leading us to a reflection exercise on the fluctuation of demand. The digital transition, with the corresponding integration of technology, the green transition, and the underlying focus on environmental sustainability, or even social balance and political and economic stability, require decision-makers to create efficient and effective strategies in the context of a highly dynamic market.

It must be emphasised that both tourism and hospitality make an important contribution to the economic prosperity of economies, whether at a macro (country), meso (regional), or micro (local) level. From this perspective, a drastic reduction in consumption or the number of visitors will not only cause serious economic tensions, but also losses in terms of wealth creation, employability, and quality of life in the territories .

In this line of thinking, over the years, global crises represented points of materialised threats to the tourism industry demand and the hospitality sector. Crises such as the 2007–2008 global financial crises created the economic setup that led to a reduction in tourist demand. Particularly in the Western hemisphere, people who used to travel drastically reduced their consumption of tourist services, with consequences on the demand in tourist industries . On the other hand, diseases such as the 2003 Severe Acute Respiratory Syndrome (SARS), the Middle East Respiratory Syndrome (MERS) outbreak , and the 11 September 2001 terrorist attack in the US had similar effects, driving the tourism demand to a downturn.

Despite the previous crises affecting tourism and hospitality demand, recent events, starting with the COVID-19 pandemic, which led the World Health Organization (WHO) to classify the outbreak as a Public Health Emergency of International Concern and, on 11 March 2020, to determine the classification of the pandemic, put the industry in an unprecedented situation . Over more than two years, the tourism industry was halted due to the consequences of

the pandemic. Hence, the literature on tourism and hospitality demand was mostly focused on the pandemic and its effects on tourist demand.

Following the end of the pandemic crisis, global supply chains and war drove the world economy to a path of high inflation and high interest rates. More recently, terrorism, which, despite being of a limited and regional spectrum, also highlighted the fragile situation that tourist industries face . However, despite the number of events potentially putting the industry demand at risk, statistical data show a recovery in world tourism , closing the gap to the period before the pandemic.

This study used a co-word analysis to analyse the thematic field affecting tourist and hospitality demand, aiming to identify trends and emerging areas. A thematic analysis was carried out based on a retrospective analysis, starting with the most recent period (2022–2023), followed by the analysis of the 2020–2021 period, ending in the 2018–2019 period. Based on the 2022–2023 period, the five most relevant themes identified were "COVID-19" highly emphasised in the literature, "demand forecast", "sustainability", and "tourism demand", with "COVID-19" standing out as the "driving theme" (a theme with greater centrality and scientific density). These themes highlight the effects of external factors on the industry, on tourist demand , and its forecasting . Additionally, the use of "machine learning", "big data", and technology, emerging as key themes in tourism and hospitality demand data from the literature , emphasise the critical importance of demand forecasting to the industry.

The role of "sustainability", "climate change", "sustainable tourism", and related themes in recent years shows the path of the research field. Transforming events such as the ones the industry passed in the last few years have the potential to reshape the way people, communities, and companies behave. Sustainable tourism was addressed early in the literature in the 2000s, mostly focused on strategic planning , environmental management and ecological sustainability , and fire disasters in tourist regions . However, the emphasis on sustainable tourism gained traction with the pressure of overtourism, particularly due to environmental challenges , such as the degradation of marine ecosystems, environmental resources, such as groundwater ecosystems, and pollution . The obtained results show that the post-pandemic period is characterized by a reassessment of "sustainability" as a driver of tourist and hospitality demand, which highlights the potential threats the industry faces.

These themes are highly aligned with tourist regions and hospitality firms' image and with customer behaviour. Tourist customers expect authentic experiences . However, firms are expected to be environmentally sustainable and provide innovative services at the same time. Therefore, successful tourist regions and firms would be the ones capable of complying with both aspects, providing innovative services with a reduced environmental footprint at the same time. Additionally, customer behaviour also depends on economic stability , financial circumstances, income, and tourist service prices , which are critical to forecasting demand and optimising hotel performance. From a customer's perspective, price is what a person gives to obtain a product or service, and people tend to evaluate purchases on the basis of how they assess prices . Therefore, despite responding differently according to the stimulus, travellers choose them based on a complex network of factors and the compounding effect of risk, physical or financial, sustainable practices from companies and regions, as well as companies' innovative services, and marketing and branding strategies influence in a decisive way potential customers' decision where to go and to stay. Companies, regions, and policy makers addressing these issues will be able to tackle the negative effects of tourist demand in downturn periods.

While there are similarities with previous research, particularly in the recognition of key themes such as the impact of economic crises , health pandemics , sustainability , and the role of technology , this study also presents primary differences in the scope, focus, and methodological approaches. This study provides a comprehensive and heuristic approach to understanding the interaction between external factors and their implications for future research in tourism and hospitality demand, emphasising the importance of adaptability and continuous innovation in a dynamic and uncertain global environment.

The tourism and hospitality industries are crucial for economic development. However, they are also highly sensitive to external shocks, which can drastically change demand patterns. The increasing emphasis on sustainability stresses the importance of eco-efficient business practices and resource conservation and underscores a research line targeting how sustainable tourism practices are implemented and their long-term impacts on both the environment

and the hospitality industry. On the other hand, technological advancements, particularly in machine learning and big data, are transforming how demand is forecasted and managed. This research area seems to be established to a continuous development, as more sophisticated models that can better predict and respond to dynamic market conditions evolve. Moreover, due to the role of economic stability and social factors, such as income levels, consumer behaviour, and safety perceptions on tourism, future research could investigate how these factors influence each other in different contexts.

Consequently, regarding the theoretical consequences of this study, results show several relevant themes for future research, with critical effects on the management of tourist and hospitality firms. On the one hand, the recent reassessment and emphasis on sustainable tourism demonstrate that this topic could be among the critical drivers of tourist demand, particularly the negative effects of overtourism, the degradation of environmental ecosystems, and tourism as a source of environmental ecosystem recovery. On the other hand, the use of "machine learning", "deep learning", and "big data", joined with recent developments in Artificial Intelligence, seem to be vital in forecasting tourist and hospitality demand. Hence, the use of these technologies by regions and companies in forecasting demand, targeting customer behaviour, implementing marketing and branding strategies, and targeting purchasing intentions, through electronic word-of-mouth (eWOM) and online reviews, are set to drive future research in tourist and hospitality management.

Despite the enlightenment this study brought, it presents some limitations that must be acknowledged, particularly the use of a heuristic analysis to predict future research paths. The dynamic nature of global events means that the industry is continuously facing uncertainty and unforeseen occurrences, such as new economic crises, health pandemics, or geopolitical changes, potentially derailing current forecasts. Consequently, researchers and industry stakeholders must remain adaptable and continuously update their models and strategies to account for these potential disruptions. Departing from this perspective, future research should also consider more robust frameworks that can better address the volatility and uncertainty in tourism and hospitality sections.

The co-word analysis provides interesting and relevant information about the unit in the analysis. However, an automated technique suffers the shortcoming of potentially omitting relevant information. On the other hand, relevant data can also be missed due to the use of a single database as a source of information. Despite these potential limitations, given the nature of the analysis, the authors believe that the methods used fully address the proposed objective. Moreover, the WoS database, due to its coverage and reliable metrics, provides strong evidence of the robustness of the methods to assure the coherence of the obtained results.

The Immersive Hotel Booking System with a 360° Virtual Tour is a web-based platform developed for The First Islatel in Alaminos City, Pangasinan, to address challenges associated with traditional hotel reservation processes. Existing booking methods, such as call-in, walk-in, and basic online reservations, provide limited opportunities for potential guests to explore hotel facilities before making a booking decision. This lack of visual interaction often results in booking uncertainty, unmet guest expectations, manual data-entry errors, and inefficiencies in reservation management. To address these concerns, the study aimed to design and develop an interactive and user-centered hotel booking system that enhances customer engagement and streamlines hotel operations. The system was developed using the Rapid Application Development (RAD) model to enable iterative design, user feedback integration, and accelerated system deployment. It is built on a Three-Tier Architecture consisting of presentation, application, and database layers to ensure scalability, maintainability, and security. Key features of the system include an immersive 360° virtual tour viewer, a self-service digital Customer Information Sheet (CIS), a secure online booking and payment module, and an administrative control panel for reservation and customer management. The system was evaluated by 34 respondents composed of hotel management personnel, staff, guests, and information technology experts. Evaluation results produced an overall weighted mean of 4.13, interpreted as Very Good. Among the quality attributes assessed, Usability obtained the highest rating of 4.71, interpreted as Excellent, followed by Reliability with 4.11 and Functionality with 4.06. Findings indicate that the system is functional, reliable, user-friendly, and ready for deployment. Furthermore, it offers a practical and replicable digital solution for hospitality establishments seeking to modernize reservation processes, improve guest experiences, and support digital transformation initiatives

[2]Reginald Ryan U. Gosela,International Journal of Advanced Research in Science Communication and Technology,Volume 4, Issue 2, June 2024 / https://doi.org/10.54499/UIDB/05583/2020. Furthermore, we would like to thank the Research Centre in Digital Services (CISeD) and the Instituto Politécnico de Viseu for their support.

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Peng, X., Zhu, J., Lee, S., Zhou, D., Song, W., & Ying, T. (2024). Digital transformation in the hospitality industry: A bibliometric review from 2000 to 2023. International Journal of Hospitality Management, 120, 103761. https://www.sciencedirect.com/science/article/abs/pii/S0278431924000732 [Google Scholar] [Crossref]

2.3. Related Literature

Tourism has become an important contributor to many sustainability issues. In addition, online booking platforms have grown to become an increasingly used means of booking accommodation, which can be crucial for encouraging sustainability. Furthermore, the promotion of sustainable tourism (ST) and accommodation services at an international level has become essential. Although the number of sustainability-certified accommodation solutions has been increasing, limited attempts have been made to promote ST using online booking platforms. This study aims to identify whether the concept of sustainability has been integrated into online booking platforms. For the objective to be met, a representative sample of platforms were identified and then analysed both in terms of their content and by testing them. The results highlight that this concept has not been adequately integrated into such platforms and that the COVID-19 pandemic has shifted the priority from sustainability to health and safety issues. However, the opportunity to take advantage of the pandemic to promote ST is still possible.

In global terms, with respect to rapid growth and social, economic, and environmental impact, the tourism industry is one of the most important industries in the world . In recent years, the hospitality sector has faced the challenges that come with being part of the trends of globalization, localization, personalization, and concern for the environment . Many studies have focused on improving the performance of tourism and hotels . In addition, there is a positive relationship between quality of service and customer satisfaction in the hotel industry , as well as between quality of service and the consumer's perception of quality . Consequently, to survive and achieve successful results, quality improvement is key in a sector as competitive as the hospitality industry . Therefore, hospitality companies' good management of these quality of service practices will allow their development and success .

Additionally, sustainability is considered an important element in terms of reputation. In the hospitality sector, certifications of sustainable practices form a competitive advantage , and they give businesses a better reputation among customers . Previous studies show a positive association between environmental practices and business success in the hospitality sector . Sustainable development has been a growing topic in the years since the origination of this concept in 1987 from the World Commission on Environment and Development (WCED) .

The importance of this sector is reflected in the global economy, as it accounts for 10.4% of all global economic activity; moreover, it represents one in ten jobs worldwide . The importance of sustainable tourism is reflected by its inclusion in the Sustainable Development Goals (SDGs) set by the United Nations (UN) for the year 2030 . Specifically, SDG 8 deals with sustainable economic growth, and full occupation is one of its indicators, including policies that help promote sustainable tourism and create local employment .

The inclusion of sustainability in quality management should have a positive impact on the results of the company, but in no case should it lead to a reduction in the quality of other services.

Given the aspects described above, the present systematic review of the literature aims to identify the main factors and sustainable practices of quality of service to determine how they impact the development and success of companies in the hospitality sector (hotels and restaurants). In addition, the findings in this study will allow us to detect future lines of research to explore and provide a research agenda for future researchers.

This paper is divided into five sections. The introduction describes the theoretical framework of the topic. The second section explains the methodology used: qualitative research was done through a systematic literature review. The content analysis of the selected literature review is described in the third section which is divided in three areas. Then, a discussion is provided through the topic and several research questions are identified for creating a future research agenda. Lastly, conclusions are presented with implication for academics and practitioners.

In this study, we propose a systematic review that addresses the management of service quality and the effect of sustainable practices in the hospitality sector. This review offers a general description of the different scientific contributions made to date that adopt reproducible methods . Defining this systematic methodology in ten steps, beginning with the identification of keywords and even validating the documents with the citation method . Petticrew and Roberts suggest a conceptualization of the systematic review focused on striving to identify, evaluate, and synthesize all relevant studies on the defined topic; they propose a structured review in twelve steps . Easterby-Smith et al. define two main processes for the systematic review . The first attempts to define the review protocol and the relevance of the research studies in its specific field of research. The second process identifies the main findings to define gaps in research in this field of knowledge. Aiming to see the complete panorama, excluding these methodologies, the diagram of systemic revisions and metanalysis PRISMA were taken in consideration in order to select the articles . The PRISMA method allows us to identify and select the paper with a higher quality and interest, through 4 phases: identification, screening, eligibility and inclusion .

After the previous contributions and in accordance with Centobelli et al. and Cerchione and Esposito , and with the contributions of the previous authors, the literature review was organized in two stages. In the first stage, by PRISMA, a flow chart was built in order to identify and select the articles included in the analysis. In the second stage, the analysis of the included papers was carried out.

This article conducted a systematic review of the literature on service quality management and sustainability in the field of hospitality. This review has allowed us to present a unified contextual framework in which certain gaps in the literature can be identified and with which we can define new lines of research to expand the literature.

Regarding the research question, this review's main objective is to identify the sustainable factors that positively influence the quality of hospitality service, sustainable practices, and service quality and the impact of service quality and sustainability in service on the success of hospitality.

A descriptive analysis was carried out that provides an overview of the articles selected in the literature review. This analysis offers a summary of the documents that address the subject of the study. In reference to the methodology used, most articles are based on quantitative methodologies, and a few have a more qualitative or mixed point of view. This content analysis of the articles included has provided a description of the main problems covered by the research on quality of service and sustainability in the hospitality sector. The research agenda proposed based on our analysis allows us to provide guidance for future lines of research and to draw conclusions for academics and professionals.

A content analysis has also been performed, that allows us to detect and delimit literature in three areas. The first one refers to critical factors of sustainable success that affect the quality of service in the hospitality sector; five have been identified: (1) environmental factors, (2) business factors, (3) human factors, (4) motivational and customer factors, and (5) relational factors. The second area refers to ten main practices that companies do to improve their quality of sustainable services in the hospitality sectors. Finally, in the third area critical factors of environmental practices and quality of service are analyzed. We can determine that the only factors that can be found in both categories are (1) decision-making in the purchase and (2) customer satisfaction.

For academics, the highlighted and identified research gaps and the consequent questions proposed represent possible lines of research to improve and contribute knowledge to the content of these research areas. Studies in these research areas should not only investigate the impact of sustainable practices and quality of service in a company in a generic way but should also consider differentiation according to the companies' differing characteristics.

Therefore, the proposed research agenda with 6 research questions should offer future researchers the opportunity to develop a comprehensive framework of sustainable practices and quality of service, in addition to the ability to study the impact and influence such practices have on companies' development.

Regarding professionals, the review of the literature has shown that there are joint environmental practices and quality of service that positively influence (1) purchase decision making and (2) customer satisfaction. However, there remains much to demonstrate from other practices and key factors of the company that can positively influence development factors. This study has shown how such practices influence company development factors, but not in conjunction with sustainable practices and service quality. This avenue can mark the future lines of action that companies in the hospitality sector must pursue to obtain greater success from these practices, which are increasingly important in the sector. However, the professional will also be interested in the results of studies of practices and impacts segmented by size of the company, quality, and other characteristics, to implement those that best suit the company and sector.

All these conclusions invite us to pursue these new lines of research to obtain results and thus expand knowledge in the area and sector of the hotel trade, specifically in restaurants and hotels.

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2.4 Synthesis of the Reviewed Literature

This study aims to review tourism and hospitality management research on online satisfaction, offline satisfaction and booking intentions published in numerous recognised tourism and hospitality journals from 2000 to 2023. This article studied and reviewed 62 published articles that emerged in the past 22 years in selected top 10 tourism, leisure and hospitality journals. Various variables influencing customer satisfaction, that is, online satisfaction and offline satisfaction leading to the intention to book a hotel online, were identified based on the previous studies.

A conceptual model is proposed, which can be empirically tested while conducting future studies in the tourism and hospitality sector, specifically with respect to hotel booking intention. This study offers a conceptual framework for the antecedents used to measure online and offline satisfaction while booking a hotel online.

The study ascertained information quality, system quality and service quality as independent variables to measure online satisfaction and perceived values like perceived functional value, emotional value, social value, and monetary value as independent variables to measure offline satisfaction. This study provides a detailed literature review on hotel booking intention and variables that influence overall satisfaction, that is, online and offline satisfaction published in specific hospitality and tourism journals over the past 22 years.Information Communication Technologies, especially the Internet, is leading to great developments in the tourism industry (Buhalis and Law, 2008). The Internet has come as a new way of communication and selling for travel companies (Law and Wong, 2003, Llach et al., 2013). In recent years, Egyptian hotels have faced massive challenges due to the changing character of the travel industry. For decades, the hotel sector industry has been dependent on intermediaries to sell their products to consumers. The Internet as a new distribution channel will help travel providers, particularly the hotel industry, to reach consumers directly and help travel providers to save money (Zhou, 2004). Furthermore, the emergence of the internet brought lower prices and time savings for consumers (Heung, 2003).

The Internet is now a paramount distribution channel for travel companies (Lee and Morrison, 2010). Travel business on the internet accounts for 15 percent of overall travel sales (US Census Bureau, 2003). A forecast from the Market Intelligence Centre (MIC) (2009) reported that the online travel product category is the Internet's largest commercial area (48.9%), generating a worldwide revenue of over 446 billion United States dollars (USD) in 2014. Sales of online travel worldwide grew 10% between 2011 and 2014 and predictions until 2016 have shown that Sales of online travel worldwide will grow at 8% yearly (Statista, 2015).

A survey research reveals that the success of online shopping is determined mostly by consumer intentions to purchase (Park, 2010). Unlike Internet consumers in Egypt and other emerging economies, however, Egyptian consumers are well known for fickle consumption patterns and lack of e-commerce loyalty, both of which pose major challenges to online shopping businesses (El Ansary and Roushdy, 2013). Egypt is currently one of the leading nations especially in the Middle East area with a well-established e-commerce environment and advanced IT infrastructure, but rapid growth of e-commerce will soon occur in other nations with similar consumption patterns.

Thus, examining and understanding how to maximize Egyptian consumers' intention to book hotels online is critical to the success of Egyptian online hotels and can help firms to develop a general reference model for online hotel shopping business success.

Numerous studies have attempted to identify factors leading to consumers' intentions to purchase travel online, emphasizing customer value creation (Francis and White, 2004), consumer trust and commitment (Chen, 2006, Kim and Chihyung, 2009, Kim et al., 2011a, Kim et al., 2011b, Ponte et al., 2015, Mukherjee and Nath, 2007), perceived usefulness (Amaro and Duarte, 2015), and attitude (Ayeh, 2015, Amaro and Duarte, 2015). However, little attention has paid to the integration of these factors into a comprehensive model (Wang, 2008, Kim et al., 2012). Such an integrated approach is also lacking in the studies of Egyptian online hotels. Furthermore, Law et al. (2009) noted that relationship marketing research in the tourism and hospitality industry focuses mainly on the supplier marketing activities of firms and less attention has been paid to the consumer side of the exchange process. Research addressing online travel shopping presents contradictory results and is typically fragmented (Amaro and Duarte, 2013). There is a scant amount of research on trust in online context for tourism products (Kim et al., 2011a, Kim et al., 2011b). Therefore, the findings of the preceding studies support the significance of the current study.

Previous studies pointed out that the effects of the antecedents of intention to purchase may be contingent on online shopping habits (Khalifa and Liu, 2007). The moderating influence of habit on the relationship between trust and intention to purchase has been examined by many studies (Chiu et al., 2012, Hsu et al., 2015). Therefore, in this study, habit is included in our model to test its moderating effects on the relationships between intention to book a hotel online and its determinants (i.e., commitment, trust, and attitude).

This research adopts a distinctive way to analyse the factors influence consumers intentions to book hotel online, by proposing and empirically testing an integrated model, with contributions from well-grounded theories, namely Commitment–Trust theory (KMV) (Morgan and Hunt, 1994) and the Technology Acceptance Model (TAM) (Davis, 1986) contributing to the current literature since, to the best of knowledge, this has not been done in any other study. Therefore, The current study aims to contribute the following to the literature of tourism and hospitality: (1) identify the determinants that affect consumer intention to book hotel online; (2) by integrating two well-recognized technology adaption theories: the commitment–trust theory and the Technology Acceptance Model (TAM), we help to understand the intention of consumers to book hotel online; (3) We also examine the moderating role of habit on the association between consumer intention to book hotel online and its determinants. The findings will help online hotels' managers to evolve strategies that enhance the intention of consumers to book hotels online.

Our study is organized as follows; the next section represents literature pertaining to the study variables and theories as well as the hypotheses development. Then we demonstrate our data collection and measure operationalization. Finally, we explain the study results, discussion, and managerial implications as well as demonstrating the limitations and future research.

Luxury hospitality is a dynamic and fast-growing niche in the overall hospitality market (Back et al., 2021, Correia et al., 2022), with recent reports suggesting that the market size of the luxury hospitality segment will increase from USD 93.4 billion in 2022 to USD 238.5 billion by 2028 (Fortune Business Insights, 2023). These numbers demonstrate strong consumer interest in luxury experiences (Bain & Company, 2021, Forbes, 2022). Examples of successful players include Park Hyatt, a luxurious hotel in the USA offering superior experiences attuned to consumers' lifestyles, values, and aspirations (The Luxury Travel Expert, 2021). Academic research also suggests that luxury hospitality has become an integral part of the overall hospitality industry, and its contributions to delivering premier and authentic consumer experiences should not be underestimated (Holmqvist et al., 2022; Shahid and Paul, 2022). Luxury hospitality provides exceptional experiences through emotional engagement and tangible and intangible service encounter attributes such as opulent ambient conditions, guest-to-staff interactions, employees' luxurious appearances, and personal touch during a service interaction (Heo and Hyun, 2015; Peng et al., 2019a).

The concept of luxury hospitality, however, remains subjective and elusive. At the same time, luxury hospitality can be considered a sub-domain of luxury services (Wirtz et al., 2020), concerned with providing accommodation, food, drink, and beverages. It encompasses a prestigious image, superior service quality, premium pricing, exclusivity,

elegance, unique and personalized consumer experiences, authentic value, and deep and intimate connections during guest-staff interactions (Ko et al., 2019, Correia et al., 2022; Shahid and Paul, 2022). However, we identify two tensions in the literature that inform the need for the current study.

First, while luxury hospitality conceptually differs from other hospitality services, this distinction is not always clear and explicit. As for the high-level luxury branding field, there is no generally accepted definition of luxury hospitality. While much of the luxury hospitality literature has been context-focused on luxury hotels and restaurants, it has not theoretically delineated its research from related fields. For example, what are the differences between luxury, upscale, and heritage hospitality? As such, the question arises what exactly is luxury hospitality? Although upscale hospitality also offers high-quality services and may provide some luxury elements, it is not as extravagant and exclusive as luxury hospitality (Chevalier and Mazzalovo, 2008). For example, Ritz Carlton is a luxury hotel, and Courtyard by Marriott is an upscale hotel operating under Marriott's international brand. There is also a perception that hotels awarded five stars or their equivalent by agencies like Forbes Travel Guides are luxury hotels (Lau et al., 2017). Such hotels may, however, not fit into the conceptual umbrella of luxury hospitality. The latter term, therefore, can easily be confused with other hospitality services.

Moreover, heritage hospitality generally encompasses cultural and historical elements of significance. Such elements can be part of luxury hospitality but do not have to be (Chhabra, 2015). For example, the ultra-luxe Raffles London at The OWO in the UK is a heritage hotel that carefully restored the Edwardian Old War Office as luxury. In contrast, the Covent Garden in London, with its food and beverage outlets, offers heritage hospitality that is not luxurious. Thus, there is a differentiation between the various hospitality segments, and this study focuses on luxury hospitality. Second, most researchers use the term luxury hospitality, luxury travel, luxury hotels, and luxury tourism interchangeably (e.g., Iloranta, 2019a 2019b, 2022; Amatulli et al., 2021), posing conceptual difficulties for researchers. Therefore, studies that explore the concept of luxury hospitality in its own right and the light of a clear conceptual definition is needed (Back et al., 2021).

In this research, we draw on the conceptualizations of the terms' luxury brand', 'luxury services,' and 'hospitality' from the literature to define luxury hospitality in Section 2. In light of this definition, we conduct a comprehensive systematic review of existing studies on luxury hospitality to answer the following research questions: (1) What is the intellectual and conceptual structure of the luxury hospitality literature? (2) Which theoretical perspectives have been used to study luxury hospitality? (3) What are promising future avenues for research in luxury hospitality?

We provide a comprehensive understanding of research on luxury hospitality published from 2000 to 2023. Using the techniques of bibliometric analysis (Donthu et al., 2021, Kraus et al., 2022), we present and discuss the most prominent themes and theories used in luxury hospitality research. Finally, we develop future research directions based on present themes through bibliographic coupling. By providing a retrospective overview of research in luxury hospitality (Lim et al., 2022a), we consolidate knowledge on this topic and, therefore, help scholars identify progress to-date, emerging, and unexplored areas in luxury hospitality and position their work in ways that they promote theoretical and practical advancements (Donthu et al., 2021, Hunt, 2010).

The remainder of this article is organized into six sections. Section 2 enunciates the conceptual underpinnings of luxury hospitality. Section 3 presents the methodology adopted for the systematic literature review. Section 4 outlines the findings of the descriptive analysis, including publication trends, the top ten globally cited articles, and author and keyword occurrences. Section 5 discusses the emerging themes resulting from the bibliographic coupling analysis. Section 6 identifies and describes the most prominent theories used. Section 7 summarizes our findings and discusses further research opportunities and limitations. Section 8 concludes with managerial implications.

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CHAPTER 3

METHODOLOGY

3.1 Research Design

The study utilized a **Descriptive-Developmental Research Design** in the development and implementation of the **Online Booking System for Avianna's Inland Resort** located in Sta. Barbara, Iloilo. This research design was considered appropriate because the study involved two major components: first, the examination and documentation of the existing reservation and operational processes of the resort; and second, the development of a computerized system designed to address the problems and limitations identified in the existing procedures. The combination of descriptive and developmental approaches allowed the researchers to systematically investigate the current condition of the resort while developing a practical technological solution based on the actual needs of the organization and its users.

The **descriptive component** of the research focused on understanding, documenting, and analyzing the existing operational procedures of Avianna's Inland Resort, particularly those related to reservation management, customer inquiries, record keeping, accommodation monitoring, and booking transactions. The researchers gathered information regarding how reservations had been processed, how customer information had been recorded, how accommodation availability had been monitored, and how booking-related inquiries had been handled by resort personnel. Through observation, interviews, and document analysis, the researchers were able to obtain relevant information regarding the actual processes followed by the resort before the implementation of the proposed system.

The descriptive approach was important because it provided the researchers with a clear understanding of the existing situation and served as the basis for identifying the problems that needed to be addressed. By examining the manual reservation process, the researchers identified several challenges, including the possibility of double bookings, delayed reservation confirmations, inaccurate or incomplete customer records, difficulty in monitoring accommodation availability, excessive paperwork, and time-consuming record retrieval. These observations helped the researchers determine which processes needed to be improved through the proposed Online Booking System.

The descriptive component also enabled the researchers to gather information from the resort owner, administrators, reception personnel, and intended users of the system. Their experiences, observations, and requirements provided valuable information regarding the functionality and usability expected from the proposed application. The gathered information was analyzed and used as a basis for identifying the necessary system features, user requirements, data requirements, and operational processes that needed to be incorporated into the system. This ensured that the proposed system was developed according to the actual needs of the resort rather than based solely on theoretical assumptions.

The **developmental component**, on the other hand, focused on the design, development, implementation, testing, and evaluation of the proposed Online Booking System. Developmental research was appropriate because the study did not merely describe the existing problems but also produced a functional technological solution intended to address those problems. The researchers applied software development principles and information technology concepts to create a web-based platform that supported reservation management, customer information management, accommodation availability monitoring, online inquiries, booking status monitoring, and report generation.

The developmental process began with the identification and analysis of the system requirements based on the information gathered during the descriptive phase. The researchers examined the existing reservation procedures and translated the identified operational needs into specific system requirements. These requirements served as the foundation for the planning and design of the system. The researchers then developed the overall system structure, user interface, database design, navigation flow, and functional components necessary to support the identified customer and administrative activities.

During the development stage, the researchers designed and implemented the system's customer and administrator functions. The customer-related functions included registration and login, viewing resort information, viewing accommodation details and rates, checking availability, submitting reservations, monitoring booking status, viewing

reservation history, and sending inquiries. The administrator-related functions included customer record management, reservation monitoring and approval, accommodation management, availability monitoring, resort information management, inquiry management, and report generation. These functionalities were developed to address the operational requirements identified during the investigation of the resort's existing processes.

Database development was also an important component of the developmental phase. The researchers designed a centralized database capable of storing and organizing customer information, user accounts, accommodation records, reservation details, booking status, and other relevant information. The database was developed to support efficient data retrieval, updating, and management while reducing the problems associated with scattered and manually maintained records. The centralized storage of information also helped ensure that reservation data could be accessed by authorized personnel when needed.

After the development of the major system components, the researchers conducted testing and evaluation to determine whether the system functioned according to the identified requirements. Testing was performed to identify errors, inconsistencies, usability concerns, and functional limitations that could affect system performance. The researchers reviewed the results of the testing process and made necessary modifications to improve the system's functionality, reliability, usability, and overall performance. This iterative process helped ensure that the developed application was capable of performing the required reservation and administrative functions.

The developmental approach also supported the evaluation of the system based on relevant software quality characteristics. The system was assessed in terms of functionality, usability, reliability, efficiency, and security. Functionality focused on whether the system performed its intended functions correctly. Usability examined whether customers and resort personnel could use the system easily and effectively. Reliability considered the consistency and dependability of system operations, while efficiency focused on the system's ability to process transactions and information within an acceptable period. Security was considered in relation to user authentication, account protection, and the safeguarding of customer and reservation information.

The use of the Descriptive-Developmental Research Design was particularly appropriate for the study because it combined systematic investigation with practical application. The descriptive component allowed the researchers to understand the actual operational conditions and problems of Avianna's Inland Resort, while the developmental component allowed them to transform the identified needs into a functional software solution. Unlike a purely descriptive study that primarily focuses on describing existing conditions, the developmental approach provided an opportunity to create, test, and evaluate a technological intervention designed to improve the identified processes.

In the context of the study, the developed Online Booking System served as the primary output of the developmental phase. The system was designed to improve the accuracy and efficiency of reservation transactions, reduce administrative workload, organize customer and booking records, provide updated accommodation availability, and improve customer accessibility to resort services. Through the automation of reservation-related activities, the system addressed several limitations associated with the resort's previous manual procedures and provided a more organized approach to booking management.

The research design also supported continuous improvement throughout the development process. Information obtained during interviews, observations, and document analysis was used to identify areas that required improvement in the system. Feedback and results obtained during testing and evaluation were also considered in refining the system's functions and interface. This allowed the researchers to make necessary adjustments to ensure that the final system was aligned with the requirements of both customers and resort personnel. Such an approach promoted user-centered development and increased the suitability of the system for actual implementation.

Furthermore, the Descriptive-Developmental Research Design provided a systematic framework for connecting the identified research problems with the developed technological solution. The information obtained during the descriptive phase served as the foundation for the developmental phase, ensuring that system features were directly related to the operational concerns of the resort. The approach therefore helped maintain consistency between the research objectives, identified problems, system requirements, development activities, and final system output.

The researchers considered this research design appropriate because it allowed the study to address both the investigative and practical aspects of information system development. Through the descriptive process, the researchers gained an understanding of the existing reservation procedures and the challenges experienced by the resort. Through the developmental process, they were able to design, develop, implement, test, and evaluate a system that addressed those challenges. The combination of these approaches contributed to the development of a

system that was based on actual organizational requirements and intended to provide measurable improvements in reservation management and service delivery.

Descriptive research was one of the approaches utilized in this study to systematically gather, analyze, interpret, and present information regarding the existing conditions, processes, practices, and challenges experienced by Avianna's Inland Resort in Sta. Barbara, Iloilo. Descriptive research focused on presenting an accurate representation of the current situation without manipulating the existing variables or conditions. It allowed the researchers to examine the actual reservation procedures, customer service practices, record-keeping methods, accommodation monitoring, and other operational activities of the resort. Through this approach, the researchers were able to obtain a clearer understanding of the existing situation and identify areas that required improvement.

In this study, the descriptive approach was primarily used to examine the current reservation management process of Avianna's Inland Resort. Before developing the proposed Online Booking System, the researchers first examined how the resort handled customer inquiries, reservation transactions, accommodation scheduling, customer information, and booking records. This assessment was necessary because understanding the existing workflow allowed the researchers to identify the specific problems and limitations that needed to be addressed through the proposed system. The descriptive research process therefore served as an important foundation for determining the appropriate system requirements and functionalities.

The existing reservation process of the resort primarily relied on traditional methods such as walk-in inquiries, telephone calls, text messages, social media communications, and manual record keeping. These methods had allowed the resort to manage its reservation activities during its initial operations; however, they had also created challenges as the number of customers and reservation requests increased. The researchers examined how these manual procedures affected the efficiency of reservation processing, accuracy of customer information, monitoring of accommodation availability, communication with customers, and overall management of booking activities.

Through interviews, the researchers gathered firsthand information from individuals who were directly involved in the operations of the resort. The interviews provided the researchers with information regarding the actual experiences and difficulties encountered during reservation processing, customer inquiries, accommodation monitoring, and record management. The resort owner, administrators, reception personnel, and other relevant individuals provided information regarding the existing procedures and identified problems that they had experienced in their daily activities. Their responses also provided insights into the features and functions they considered necessary for the proposed Online Booking System.

Observation was another important data-gathering activity used during the descriptive phase of the study. The researchers observed how reservation requests were received, recorded, verified, processed, and monitored by resort personnel. They also examined how employees checked accommodation availability and updated reservation information. Through direct observation, the researchers were able to identify actual workflow practices, repetitive tasks, delays, and other operational difficulties that might not have been fully identified through interviews alone. Observation provided a more realistic understanding of how the manual reservation process operated in the actual resort environment.

The researchers also examined existing documents and records related to reservation management. These included reservation logs, customer records, booking schedules, accommodation information, and other documents used by the resort in managing its reservation activities. Document analysis allowed the researchers to examine how information had been recorded, organized, updated, and retrieved. It also helped identify potential problems related to incomplete information, duplicate records, inaccurate entries, and difficulty in locating previous reservation details.

The descriptive research process revealed several operational concerns associated with the existing manual reservation system. One of the major concerns was the possibility of double bookings and scheduling conflicts. Since reservation information had been manually recorded and availability had been checked by resort personnel, there had been a possibility that two or more customers could request the same accommodation for the same date without the availability information being updated immediately. Such situations could create conflicts between reservations and negatively affect the customer experience.

Another concern identified through the descriptive investigation was the possibility of inaccurate, incomplete, or misplaced customer records. Manual recording procedures had been susceptible to human errors, including incorrect encoding of customer information, missing details, duplicate entries, and difficulty in locating previous records.

These issues could affect the accuracy of reservation information and create additional work for resort personnel who needed to verify or correct customer records.

The researchers also identified delays in reservation confirmation as another challenge associated with the manual process. Since reservation requests required resort personnel to manually check availability, record booking information, communicate with customers, and confirm the reservation, the process could take considerable time. During periods when reservation requests increased, such as weekends, holidays, and peak seasons, the workload of resort personnel could become heavier, resulting in slower responses to customer inquiries and booking requests.

The descriptive research also examined the challenges associated with monitoring the availability of cottages, rooms, pavilions, and other resort facilities. Manual monitoring required resort personnel to review reservation records and determine which facilities had been available, reserved, or occupied. As the number of reservations increased, this process became more difficult and increased the possibility of inaccurate availability information. The lack of a centralized system for monitoring facility schedules also made it more difficult for management to obtain an updated overview of the resort's accommodation status.

In addition, the researchers examined the workload experienced by resort personnel in handling customer inquiries and reservation requests. Employees had to respond to questions regarding accommodation rates, facility availability, reservation schedules, and other resort information while simultaneously recording and updating booking information. These repetitive manual activities required considerable time and effort and could reduce the productivity of resort personnel. The descriptive findings therefore indicated the need for a system that could automate routine reservation-related activities and provide customers with direct access to essential resort information.

The descriptive approach also provided information regarding the needs and expectations of the intended users of the proposed system. Customers required a more convenient way to access resort information, check accommodation availability, and submit reservations without personally visiting the resort or repeatedly contacting personnel. Resort administrators and receptionists, on the other hand, required a more organized method of managing customer records, monitoring reservations, updating accommodation availability, and generating reports. These different user requirements were considered in determining the functionalities that were incorporated into the proposed system.

The information obtained through descriptive research served as the foundation for the developmental phase of the study. The problems, operational procedures, user requirements, and limitations identified during the investigation were translated into system requirements and functional specifications. For example, the identified problem of double bookings supported the need for an accommodation availability monitoring feature, while the difficulty of maintaining manual records supported the development of a centralized database. The delays in reservation processing supported the inclusion of online reservation and booking status functions, while the difficulty in preparing reports supported the development of automated report generation.

The descriptive research approach also helped ensure that the proposed Online Booking System was based on actual organizational needs rather than assumptions. By examining the existing practices and gathering information from individuals directly involved in the resort's operations, the researchers were able to develop a more relevant and practical technological solution. The findings provided a basis for determining which features were necessary, which processes needed to be automated, and how the system could be designed to support both customers and resort personnel.

Furthermore, the descriptive component contributed to the validity and relevance of the study because it established a clear connection between the identified problems and the proposed technological intervention. The researchers were able to demonstrate that the system was developed in response to actual challenges encountered by Avianna's Inland Resort. This approach reduced the possibility of developing unnecessary features and ensured that the system focused on reservation management, customer accessibility, accommodation monitoring, record organization, communication, and administrative reporting.

Descriptive research played an important role in the study because it provided a comprehensive understanding of the existing reservation processes and operational conditions of Avianna's Inland Resort. Through interviews, observations, and document analysis, the researchers identified the strengths and weaknesses of the existing manual procedures and determined the major concerns that affected reservation management and customer service. The information gathered during this phase served as a strong foundation for the succeeding stages of system

development and helped ensure that the Online Booking System was designed according to the actual requirements of the resort and its intended users. Through the descriptive approach, the study was able to establish a clear basis for developing a more organized, efficient, reliable, and user-friendly online reservation solution for Avianna's Inland Resort.

Developmental Research

Developmental research focused on the systematic design, development, implementation, testing, evaluation, and improvement of products, systems, processes, and technological solutions intended to address specific problems or organizational needs. Unlike purely descriptive research, which primarily focused on observing, documenting, and describing existing conditions, developmental research emphasized the creation of practical solutions that could improve efficiency, effectiveness, and overall performance within a particular environment. In this study, the developmental approach provided the framework for transforming the operational problems identified at Avianna's Inland Resort into a functional web-based technological solution.

The developmental research approach was appropriate for this study because its primary objective was to develop and implement an **Online Booking System for Avianna's Inland Resort** in Sta. Barbara, Iloilo. The proposed system was developed to address the limitations of the resort's existing manual reservation procedures and to provide a computerized solution that improved reservation management, customer convenience, accommodation monitoring, information organization, and administrative operations. Through the developmental approach, the researchers were able to convert the identified operational problems and user requirements into specific system features and functionalities.

The developmental process allowed the researchers to move from the analysis of the existing reservation procedures toward the creation of an operational software application. The information obtained through interviews, observations, and document analysis during the descriptive phase served as the basis for identifying the requirements of the proposed system. These requirements guided the researchers in planning, designing, coding, testing, deploying, and evaluating the Online Booking System. Each stage was conducted systematically to ensure that the developed system was aligned with the actual operational needs of the resort and the expectations of its intended users.

The development of the Online Booking System involved several major activities, including system analysis, system planning, database development, user interface design, system coding and development, system testing, and system deployment.

System Analysis

The first stage of the developmental process involved a comprehensive analysis of the existing reservation and management procedures of Avianna's Inland Resort. During this stage, the researchers examined how reservations had been received, recorded, processed, and monitored. They also analyzed how customer information had been maintained, how accommodation availability had been checked, and how resort personnel had handled customer inquiries and booking requests.

The system analysis focused on identifying the strengths and weaknesses of the existing manual reservation process. The researchers examined the difficulties encountered by both customers and resort personnel, including delayed reservation confirmation, difficulty in monitoring accommodation availability, possible double bookings, inaccurate or incomplete customer records, and the time required to retrieve and update reservation information. The analysis also considered the administrative activities involved in managing customer records, reservations, accommodation information, inquiries, and reports.

The information obtained during the analysis served as the foundation for identifying the functional and non-functional requirements of the Online Booking System. Customer requirements included user registration and login, access to resort information, viewing accommodation details and rates, checking availability, submitting reservations, monitoring booking status, viewing reservation history, and sending inquiries. Administrative requirements included customer account management, reservation monitoring and approval, accommodation management, availability monitoring, inquiry management, and report generation.

Through system analysis, the researchers were able to establish a clear connection between the problems experienced by the resort and the features required in the proposed system. This ensured that the system was

designed to address actual operational needs rather than unnecessary or unrelated functions.

System Planning

After identifying and analyzing the system requirements, the researchers proceeded with the planning stage. System planning involved defining the objectives, scope, limitations, users, resources, major functions, and development activities of the proposed Online Booking System. The researchers also established the general workflow of the system to ensure that customer reservations and administrative transactions could be processed in an organized manner.

During this stage, the researchers determined the major modules that needed to be included in the system. Separate customer and administrator functions were planned according to the responsibilities and requirements of each user group. The customer module was planned to provide convenient access to resort information and reservation services, while the administrator module was designed to support the management of bookings, customer records, accommodations, inquiries, and reports.

System planning also helped the researchers organize the development activities and determine the sequence in which the different system components would be developed and integrated. Potential technical and operational issues were considered during the planning process, allowing the researchers to establish appropriate approaches for addressing challenges that could arise during development.

Proper planning helped maintain the direction of the project and ensured that the development activities remained consistent with the objectives, scope, and requirements of the study.

Database Development

Database development was an essential component of the Online Booking System because the system required an organized method of storing, retrieving, updating, and managing information. The researchers developed a centralized database that served as the primary repository for customer information, user accounts, accommodation details, reservation records, booking dates, reservation status, and other information necessary for the operation of the system.

The database structure was designed to organize related information systematically and support efficient data management. Appropriate relationships were established among the relevant records to reduce duplication and maintain data consistency. The researchers also considered data integrity and security to help ensure that stored information remained accurate and protected from unauthorized access.

The database supported the reservation process by allowing booking information to be stored and retrieved efficiently. It also supported accommodation availability monitoring by providing updated information regarding existing reservations and the status of reservable facilities. This helped administrators determine which cottages, rooms, and pavilions had been available or reserved for specific dates.

The centralized database also reduced the resort's dependence on scattered manual records. Instead of relying primarily on handwritten documents and separate reservation records, authorized personnel were able to access organized information through the system. This improved the efficiency of record retrieval and provided a more systematic approach to managing customer and reservation information.

User Interface Design

The developmental process also included the design of the user interface of the Online Booking System. The interface was designed to provide a clear, organized, accessible, and user-friendly environment for customers and administrators. The researchers considered ease of navigation, readability, accessibility, and functionality to ensure that users could perform the required activities with minimal difficulty.

For customers, the interface provided access to resort information, facilities, accommodation details, rates, availability, reservation forms, booking status, reservation history, and inquiry functions. The information was organized into appropriate sections to allow customers to navigate the website and obtain the information they needed before making a reservation.

For administrators, the interface provided access to customer records, reservation management, accommodation

information, availability monitoring, customer inquiries, and report generation. Administrative functions were organized to allow authorized personnel to monitor and manage reservation activities efficiently.

Since the proposed system was developed as a web-based platform, the user interface was also designed to be accessible through internet-enabled devices such as desktop computers, laptops, tablets, and smartphones. This provided users with greater flexibility in accessing the system without requiring the installation of a separate mobile application.

An effective user interface was important because it directly influenced the usability and accessibility of the system. The researchers therefore ensured that navigation elements, forms, buttons, menus, and other interactive components were arranged logically and consistently.

System Coding and Development

The coding and development stage involved translating the system requirements, database design, and user interface specifications into an operational software application. During this stage, the researchers utilized appropriate programming languages, frameworks, and development tools to implement the various functionalities of the Online Booking System.

Several key features were developed to support the operational and administrative requirements of Avianna's Inland Resort. User registration and authentication functions were implemented to allow customers and authorized personnel to create accounts and securely access the appropriate system features. User management functions were also developed to allow administrators to manage registered accounts and user-related information.

Accommodation management was implemented to allow administrators to manage information regarding the cottages, rooms, and pavilions available for reservation. Relevant accommodation details, rates, and availability information could be maintained through the administrative functions of the system.

The reservation processing function was developed to allow customers to submit booking requests through the online platform. Customers could provide the necessary reservation information and select the appropriate accommodation based on the available schedules. On the administrative side, reservation approval and monitoring functions allowed authorized personnel to review, manage, and track booking requests.

An availability-checking feature was also incorporated into the system to provide updated information regarding the availability of cottages, rooms, and pavilions. This feature supported the prevention of scheduling conflicts and double bookings by allowing reservation information and accommodation status to be monitored systematically.

The system also included report generation capabilities that allowed administrators to produce organized reservation-related reports. These reports provided useful information for monitoring bookings, reviewing reservation activities, and supporting administrative decision-making.

An administrative dashboard was also implemented to provide administrators with a centralized view of important reservation and operational information. Through the dashboard, authorized personnel could monitor relevant activities and access major administrative functions without having to manually search through separate records.

The integration of these features transformed the conceptual system design into a functional web-based application. The coding and development stage therefore served as the primary technical phase in which the identified operational requirements were converted into actual system functionalities.

System Testing

System testing was an important component of the developmental process because it was necessary to determine whether the developed system functioned according to its intended requirements. Various testing activities were conducted to identify errors, verify system functionality, assess performance, and evaluate usability before the system was deployed.

Unit testing was conducted to examine individual modules and functions of the system. The researchers tested components such as user registration, login authentication, accommodation management, reservation processing, availability checking, and report generation individually to determine whether each function operated correctly.

Integration testing was then conducted to verify that the different system modules worked properly when combined. This testing focused on the interaction between the user interface, database, reservation module, accommodation management, and administrative functions. It ensured that information entered or updated in one component was properly transferred and reflected in other related components.

System testing was performed to evaluate the Online Booking System as a complete application. The researchers examined whether the integrated system performed according to the requirements identified during system analysis. Important processes such as customer registration, reservation submission, booking approval, availability monitoring, reservation status updates, and report generation were tested as part of the complete system.

User Acceptance Testing (UAT) was also conducted with the intended users of the system. This testing allowed users to interact with the developed application and provide feedback regarding its functionality, usability, accessibility, and suitability for actual resort operations. Their observations and recommendations were considered in identifying areas that required further improvement.

Performance testing was conducted to assess the responsiveness and stability of the system during expected user activities. The researchers examined whether the system could process common transactions within an acceptable period and whether the major functions remained operational during normal usage.

Security-related testing was also considered to help protect user accounts and reservation information. Authentication and access controls were examined to ensure that system functions were appropriately restricted according to user roles. These measures helped protect customer and administrative information from unauthorized access.

Compatibility testing was conducted to determine whether the web-based system functioned properly across different devices and commonly used web browsers. The researchers examined the system using desktop computers, laptops, tablets, and smartphones to ensure that users could access the platform through different internet-enabled devices.

The results of the testing activities were used to identify errors, inconsistencies, and areas requiring improvement. Necessary corrections and modifications were made to improve the functionality, reliability, usability, security, and overall quality of the Online Booking System before deployment.

System Deployment

After the system had undergone development, testing, and necessary improvements, it was prepared for deployment within the intended operational environment of Avianna's Inland Resort. System deployment involved configuring the developed application and database, preparing user accounts, organizing the necessary system information, and introducing the system to its intended users.

The Online Booking System was configured in the intended operating environment to ensure that its required components functioned properly. The database was also configured to support the storage, organization, retrieval, and management of customer, accommodation, and reservation information.

User accounts were prepared according to the roles and responsibilities of the intended users. Customers were provided with account functions that allowed them to access reservation-related services, while authorized administrators and resort personnel were provided with appropriate administrative access for managing bookings, accommodations, customer records, inquiries, and reports.

Relevant existing information was organized and entered into the system as necessary to support the initial operation of the application. The researchers also familiarized the resort administrators and authorized personnel with the major system functions and procedures. The orientation covered activities such as managing accommodation information, processing reservation requests, monitoring booking status, checking availability, managing user accounts, and generating reports.

An operational readiness assessment was also considered to determine whether the system, data, users, and supporting procedures were adequately prepared for actual utilization. Any remaining issues identified during the preparation and initial use of the system were reviewed and addressed where necessary.

The deployment stage marked the transition of the Online Booking System from development and testing toward actual utilization. Through proper configuration, user preparation, and system orientation, the researchers ensured that the developed application was ready to support the reservation and administrative operations of Avianna's Inland Resort.

Overall Developmental Process

Overall, the developmental research process provided a systematic framework for transforming the operational problems of Avianna's Inland Resort into a functional web-based solution. The researchers began by analyzing the existing manual reservation procedures and identifying the requirements of customers and resort personnel. These findings guided the planning, database development, user interface design, coding, testing, and deployment of the Online Booking System.

Each developmental stage contributed to the overall quality and effectiveness of the system. System analysis established the actual needs of the resort, system planning organized the development activities, database development provided structured information management, user interface design supported usability, coding transformed the requirements into functional features, testing verified the system's performance and reliability, and deployment prepared the system for actual use.

Through this developmental process, the researchers were able to produce an Online Booking System that addressed the major limitations of the resort's manual reservation procedures. The system provided customers with more convenient access to resort information and reservation services while giving administrators a centralized platform for managing bookings, customer records, accommodation availability, inquiries, and reports. The developmental approach therefore contributed to the modernization of the resort's reservation process and supported the improvement of its operational efficiency, customer service, and digital presence.

System Maintenance and Improvement

Developmental research does not end after deployment. Continuous maintenance and improvement are necessary to ensure the long-term effectiveness of the Online Booking System.

Various activities were performed to ensure the continued functionality, reliability, security, and efficiency of the Online Booking System. Bug fixing was conducted to identify and resolve errors or issues encountered during actual system use. Security updates were also applied to address potential vulnerabilities and help protect user accounts, reservation records, and other important system information from unauthorized access.

Database optimization was performed to maintain the proper organization and efficiency of stored information. This helped improve data retrieval and supported the smooth operation of reservation and administrative functions. Performance improvements were also implemented to enhance the system's responsiveness and reduce delays when users accessed or performed different activities within the system.

Feature enhancements were introduced based on identified needs, user feedback, and changes in operational requirements. These improvements helped ensure that the system remained useful and aligned with the resort's reservation and administrative processes. In addition, user support and troubleshooting were provided to assist administrators and other users in addressing difficulties encountered while using the system. Through these maintenance activities, the researchers helped ensure that the Online Booking System remained functional, secure, efficient, and responsive to the changing needs of its users.

Feedback gathered from users may be utilized to identify opportunities for system enhancement and future development.

Developmental research played a vital role in this study because it focused not only on identifying and analyzing existing problems but also on designing, developing, implementing, testing, and evaluating a practical solution that directly addressed the identified needs of Avianna's Inland Resort. Unlike research approaches that primarily described or explained existing situations, developmental research emphasized the creation of a functional system that could improve existing processes and provide practical benefits to its intended users. In this study, the developmental research approach served as the foundation for the design and development of a web-based Online Booking System for Avianna's Inland Resort in Santa Barbara, Iloilo.

The primary purpose of the study was not merely to investigate the weaknesses of the resort's existing manual reservation process but also to transform the findings into a functional computerized system that could improve reservation management and overall business operations. Through developmental research, the researchers were able to examine the resort's existing procedures, identify operational difficulties, determine user requirements, and translate these requirements into system features and functionalities that addressed the needs of both resort management and customers.

The developmental research approach enabled the researchers to follow a systematic process that included requirement analysis, system planning, interface design, database development, coding, testing, implementation, and evaluation. Each stage contributed to the development and improvement of the proposed system. The researchers used the information gathered during the descriptive phase as a basis for determining the necessary functions and components of the system. This structured approach helped ensure that the developed application was aligned with the objectives of the study and the actual operational requirements of the resort.

One of the major contributions of developmental research was its ability to provide a practical solution to real-world organizational problems. In the case of Avianna's Inland Resort, the developed Online Booking System addressed limitations associated with the existing manual reservation procedures. The system provided an automated platform for managing booking transactions, organizing customer information, monitoring accommodation availability, and generating reservation reports. Through these functions, the system helped minimize problems such as double bookings, inaccurate or incomplete reservation records, delayed booking confirmations, difficulty in monitoring accommodation schedules, and time-consuming administrative activities.

Developmental research also supported the integration of information technology into the daily operations of the resort. By applying web development technologies, database management principles, and software engineering practices, the researchers were able to develop a digital solution that improved the organization and accessibility of reservation information. The web-based nature of the system allowed customers to access resort information and reservation services using internet-enabled devices, thereby providing a more convenient alternative to traditional walk-in inquiries, telephone calls, and manual reservation procedures.

Another important contribution of developmental research was its emphasis on continuous evaluation and improvement. Throughout the development process, the researchers conducted testing and gathered feedback from intended users to determine whether the system functions met their requirements. The results of the testing and user feedback were used to identify errors, usability concerns, and other areas requiring improvement. Necessary modifications were then made to improve the functionality, usability, reliability, efficiency, and security of the developed system before its implementation.

The developmental research process also provided valuable learning experiences for the researchers. It allowed them to apply theoretical concepts and skills acquired from the Bachelor of Science in Information Technology program to an actual organizational setting. Through the development of the Online Booking System, the researchers gained practical experience in systems analysis and design, web application development, database management, user interface design, software testing, system implementation, and project management. The development process also strengthened their problem-solving, analytical thinking, communication, teamwork, and decision-making skills.

Furthermore, the study demonstrated the practical application of information technology in addressing operational challenges within the tourism and hospitality industry. The developed system showed how digital technologies could be utilized to improve reservation management, customer accessibility, information organization, and administrative efficiency. By providing an online platform for resort services, the system also supported the resort's efforts to establish a stronger digital presence and respond to the increasing importance of technology in tourism and hospitality operations.

The developmental research approach also provided a basis for evaluating the effectiveness and quality of the developed system. The researchers considered important software quality characteristics such as functionality, usability, reliability, efficiency, and security in assessing the system. These characteristics helped determine whether the system performed its intended functions, was easy to use, operated reliably, responded efficiently, and provided appropriate protection for user and reservation information.

The findings and outputs of the developmental research may also serve as a useful reference for future researchers and developers who intend to create similar information systems for resorts, tourism establishments, and other service-oriented organizations. The development process, system features, database structure, testing procedures,

and implementation experience presented in the study may provide useful guidance for future projects involving online booking and reservation management systems. Future developers may also use the developed system as a foundation for additional features and technological enhancements.

Specifically, the developmental research enabled the researchers to design and develop a web-based Online Booking System that was tailored to the operational requirements of Avianna's Inland Resort. It allowed them to apply information technology concepts, software engineering principles, and database management techniques to address actual business problems. The research also enabled the automation of manual reservation procedures, thereby reducing paperwork, repetitive administrative tasks, and the possibility of human errors.

The developmental approach further enabled the researchers to organize customer information, reservation records, and accommodation availability through a centralized database. It provided customers with an online platform through which they could access resort information, view accommodation details, check availability, and submit reservations using internet-enabled devices. It also improved communication between customers and resort management through reservation-related inquiries, booking confirmations, and updates.

The developmental research supported the implementation of authentication and access-control features to help protect user accounts and system information. It also enabled the development of automated reservation reports that provided administrators with organized booking information for monitoring and operational decision-making. These functions contributed to the overall efficiency and reliability of the resort's reservation management process.

The study also demonstrated how digital technology could support the digital transformation of small and medium-sized tourism establishments. Through the development of the Online Booking System, Avianna's Inland Resort was provided with a technological solution that improved the accessibility of its services and supported more organized management of reservation activities. The system therefore represented an important step toward modernizing the resort's existing manual processes.

Developmental research served as an important foundation for producing a functional, efficient, reliable, and user-centered Online Booking System for Avianna's Inland Resort. It enabled the researchers to move beyond identifying existing problems and to develop a practical technological solution that addressed the identified operational challenges. Through systematic analysis, planning, development, testing, and evaluation, the study contributed to the improvement of reservation management, customer convenience, administrative efficiency, and the resort's digital presence. In the long term, the developed system was expected to support the modernization, competitiveness, operational improvement, and continued growth of Avianna's Inland Resort within the tourism and hospitality industry.

Although the primary research design of the study was **Descriptive-Developmental Research Design**, an experimental component was incorporated during the testing and evaluation of the developed Online Booking System. The experimental component focused on determining whether the developed system provided measurable improvements compared with the existing manual reservation process of Avianna's Inland Resort. Rather than treating experimental research as the primary research design, it was used as a supplementary approach to evaluate the effectiveness and efficiency of the developed system.

In this study, the existing manual reservation process served as the basis for comparison, while the developed Online Booking System served as the computerized intervention. The researchers compared selected reservation-related activities performed through the traditional manual process and the developed system. This comparison allowed the researchers to determine whether the system improved the speed and efficiency of reservation transactions and reduced the amount of manual work required from resort personnel.

The experimental evaluation focused primarily on **reservation processing time** because the time required to complete a reservation directly affected customer convenience, employee productivity, and the efficiency of resort operations. The researchers observed selected reservation activities under both methods and recorded the amount of time required to complete each activity. The results were then compared to determine the differences between the manual and computerized processes.

Comparison of Reservation Processing Time

Reservation processing time was used as one of the indicators for evaluating the effectiveness of the Online Booking System. Under the existing manual reservation process, customers had made inquiries through walk-in transactions,

phone calls, text messages, and social media platforms. Resort personnel had manually checked the availability of cottages, rooms, and pavilions by reviewing reservation records, logbooks, or other manually maintained schedules. Customer information had then been recorded, reservation schedules had been updated, and booking confirmations had been communicated to customers.

Because these activities depended heavily on manual intervention, the reservation process had required additional time and effort from resort personnel. The process had also been more difficult to manage when several customers submitted inquiries or reservation requests at the same time. During weekends, holidays, and other periods of increased customer demand, the amount of time required to process reservations could increase because resort personnel had to handle multiple transactions manually.

In contrast, the developed Online Booking System provided a centralized web-based platform that automated several reservation-related activities. Customers were able to access the system, view resort information, browse available accommodations, select preferred dates, provide the required reservation information, and submit booking requests online. The system stored reservation information in the database and allowed administrators to monitor and manage booking requests through the administrative interface.

The comparison of the two processes allowed the researchers to determine whether the developed system reduced the time required to complete reservation-related activities. The same or similar reservation tasks were performed using both methods whenever possible, and the corresponding processing times were recorded for comparison.

Time Required to Check Accommodation Availability

The first activity evaluated was the time required to check accommodation availability. This involved determining whether a cottage, room, or pavilion was available for a customer's preferred date and reservation period.

Under the manual process, resort personnel had to review reservation logbooks, schedules, or other manually maintained records to determine whether the requested accommodation was available. This process required personnel to search through existing records and verify whether another customer had already reserved the selected facility.

Through the developed Online Booking System, accommodation availability was stored and monitored through the centralized database. Customers were able to view available accommodations and reservation schedules through the online platform, while administrators could monitor the status of accommodations through the administrative functions of the system. This reduced the need for repeated manual checking and made availability information more accessible.

The researchers compared the amount of time required to determine accommodation availability under both processes. The results provided an indication of whether the developed system improved the speed and efficiency of availability checking.

Time Required to Record Customer Information

The second activity evaluated was the time required to record customer information. Under the existing manual process, resort personnel had manually written or encoded customer details in reservation records. This required additional effort and increased the possibility of incomplete, duplicated, or incorrectly recorded information.

Through the Online Booking System, customers were able to provide their information through registration and reservation forms. Required information was submitted electronically and stored in the centralized database. This reduced the need for resort personnel to repeatedly encode customer information manually.

The researchers compared the amount of time required to collect and record customer information using the manual process and the developed system. The comparison helped determine whether the computerized process reduced administrative workload and improved the efficiency of customer record management.

Time Required to Process Reservation Requests

The third activity evaluated was the time required to process a reservation request. Under the manual process, several activities had to be performed by resort personnel, including checking accommodation availability, recording customer information, updating reservation schedules, and communicating with the customer regarding the booking.

These activities had been performed separately and depended heavily on the availability of resort personnel. As a result, the complete reservation process could take longer, particularly when several reservation requests were received simultaneously.

With the developed Online Booking System, reservation information was submitted through the online platform and stored in the database. Administrators were able to access booking requests through the administrative interface and manage the reservation status. The system also supported the updating of accommodation availability based on reservation information.

The researchers compared the processing time required to complete reservation requests using both methods. The comparison provided an indication of whether the developed system improved reservation processing efficiency.

Time Required to Confirm Reservations

The fourth activity evaluated was the time required to confirm reservation requests. Under the manual process, customers had to wait for resort personnel to verify availability, review reservation details, and communicate the booking status through phone calls, text messages, social media messages, or other communication methods.

The amount of time required for confirmation depended on the availability of resort personnel and the volume of reservation requests being handled. This could result in delays, particularly during periods of high customer demand.

The developed Online Booking System provided administrators with a centralized interface for reviewing and managing reservation requests. Once a booking request had been reviewed and its status updated, the corresponding reservation information could be reflected in the system. Customers could also monitor the status of their reservation through their account.

The researchers compared the time required to process and confirm reservations under the manual and computerized methods. The results helped determine whether the developed system reduced delays associated with reservation confirmation.

Experimental Comparison and Evaluation of Results

After the reservation activities had been evaluated using both methods, the researchers compared the recorded processing times. The comparison focused on identifying differences in the amount of time required to perform availability checking, customer information recording, reservation processing, and reservation confirmation.

The researchers used descriptive statistical measures such as average processing time and total processing time to summarize the results. The differences between the manual and computerized processes were then examined to determine whether the developed Online Booking System provided improvements in reservation processing efficiency.

The percentage improvement or time reduction could also be determined using the recorded results. This allowed the researchers to present a clearer comparison between the existing manual process and the developed computerized system. The results could be presented through tables and graphical representations to make the differences in processing time easier to interpret.

The experimental evaluation provided evidence regarding the effectiveness of the developed Online Booking System in improving reservation processing. A reduction in processing time indicated that the system was able to perform reservation-related activities more efficiently than the existing manual method. It also indicated that automation could reduce repetitive administrative tasks and minimize the time required to process customer requests.

The comparison also provided an opportunity to observe the practical benefits of the developed system. Faster availability checking allowed customers to obtain information about accommodations more conveniently. Faster recording and processing of reservation information reduced the administrative workload of resort personnel. More organized reservation information also helped administrators monitor bookings and accommodation schedules more efficiently.

The experimental component was important because it provided a practical basis for evaluating the effectiveness of the developed Online Booking System. While the descriptive component of the study identified the existing

problems and operational conditions of Avianna's Inland Resort, and the developmental component focused on designing and developing the technological solution, the experimental component provided an opportunity to compare the performance of the existing manual reservation process with the developed computerized system.

Through this comparison, the researchers were able to determine whether the developed Online Booking System provided improvements in reservation processing time, accuracy, and operational efficiency. The evaluation also supported the determination of whether the system achieved one of the major objectives of the study, which was to reduce the time, effort, errors, and manual workload associated with reservation management.

The experimental component complemented the descriptive-developmental research design by providing measurable indicators of the system's performance. The comparison between the manual and computerized reservation processes demonstrated the potential of the developed Online Booking System to improve reservation processing, provide more accessible accommodation information, reduce administrative workload, minimize reservation errors, and organize booking records more efficiently. These improvements supported the overall objective of modernizing the reservation procedures of Avianna's Inland Resort through the application of web-based information technology.

Comparison of Accuracy

Another important aspect of the experimental evaluation was the comparison of the accuracy of the manual reservation process and the developed Online Booking System. Accuracy was considered an important factor in reservation management because it affected the reliability of customer information, booking schedules, accommodation availability, and other reservation records. Inaccurate reservation information could result in customer dissatisfaction, scheduling conflicts, operational difficulties, and additional administrative work. Therefore, evaluating the accuracy of the developed system was necessary to determine whether it provided improvements over the existing manual reservation process.

Under the existing manual reservation process of Avianna's Inland Resort, reservation information had been recorded through handwritten logs, notebooks, and other manual record-keeping procedures. Although these methods had been used to manage reservations, they depended heavily on the accuracy, attentiveness, and consistency of resort personnel. Errors could occur during the recording and updating of reservation information, particularly when personnel handled several booking requests simultaneously.

Possible errors included incorrect customer information, incorrect reservation dates, incomplete booking details, duplicate entries, incorrect accommodation assignments, and delayed updates of reservation records. These problems could become more difficult to manage as the number of reservation requests increased, particularly during weekends, holidays, and periods of high customer demand.

The developed Online Booking System addressed these concerns by providing a centralized database for storing and managing reservation and customer information. The system utilized standardized online forms, required fields, database storage, and reservation management functions to reduce unnecessary manual encoding. The availability-monitoring function also supported the checking and updating of accommodation schedules based on recorded reservations.

Through the computerized approach, customer and reservation information was stored in an organized manner and could be retrieved and updated by authorized users. This reduced the dependence on handwritten records and provided administrators with a more systematic method of maintaining reservation information.

To evaluate the effectiveness of the developed system, the researchers compared reservation records processed through the existing manual method and those processed through the Online Booking System. Selected reservation transactions were examined under comparable conditions, and the researchers recorded and analyzed the errors encountered in each process. The comparison provided a basis for determining which method produced more accurate, complete, and reliable reservation information.

The accuracy evaluation focused on several indicators.

Frequency of Booking Errors

The researchers examined the frequency of errors encountered during reservation processing under both the manual and computerized methods. Booking errors included incorrect reservation dates, inaccurate customer information,

incomplete reservation details, incorrect accommodation assignments, duplicate entries, and missing information.

The manual process was more dependent on human recording and verification, which increased the possibility of errors during reservation processing. In contrast, the Online Booking System used standardized forms and database-based record management to minimize unnecessary manual encoding and improve the consistency of stored information.

The number of errors recorded under each method was compared to determine whether the developed system reduced the occurrence of booking-related inaccuracies.

Incidence of Double Bookings

The occurrence of double bookings was another important indicator of reservation accuracy. A double booking occurred when the same accommodation was assigned to more than one customer for overlapping reservation dates or schedules.

Under the manual reservation process, accommodation availability had to be checked and updated by resort personnel. Delayed updates, overlooked entries, or simultaneous reservation requests could increase the possibility of scheduling conflicts.

The developed Online Booking System provided a centralized reservation database and an availability-monitoring function. Reservation information was recorded within the system, allowing administrators to monitor the status of cottages, rooms, and pavilions. The availability information supported the identification of already reserved accommodations before processing additional booking requests.

The researchers compared the incidence of double bookings under the manual and computerized processes to determine whether the developed system contributed to more accurate accommodation scheduling.

Accuracy of Customer Records

The accuracy and completeness of customer records were also evaluated. Customer information included details required for registration and reservation processing, such as the customer's name, contact information, account information, and other necessary booking details.

Under the manual process, customer information had been recorded through handwritten or manually encoded records. This method could result in incomplete information, duplicate entries, recording mistakes, or difficulty in retrieving previous records.

The Online Booking System provided standardized registration and reservation forms through which customers entered their information directly into the system. Required fields and structured data entry helped ensure that the necessary information was provided before the transaction was submitted. The information was then stored in the centralized database, allowing authorized administrators to retrieve and update customer records more systematically.

The researchers compared the completeness and accuracy of customer records generated through both methods to determine whether the developed system improved the quality and organization of customer information.

Accuracy of Accommodation Schedules

Accurate accommodation scheduling was another important factor in evaluating the system. The researchers examined the accuracy of information regarding the availability and reservation status of cottages, rooms, and pavilions.

Under the manual process, resort personnel had to update reservation schedules manually whenever a booking was received or confirmed. Failure to update records promptly could result in outdated availability information and possible scheduling conflicts.

The developed Online Booking System centralized accommodation and reservation information, allowing administrators to monitor the status of available and reserved facilities through the system. The reservation records were used to support the updating of accommodation availability and improve the consistency of scheduling

information.

The researchers compared the accommodation schedules maintained through the manual process with those managed through the developed system. The comparison helped determine whether the computerized system provided more accurate and organized accommodation information.

Completeness of Reservation Records

The completeness of reservation records was also considered in the accuracy evaluation. A complete reservation record contained the necessary customer, accommodation, reservation date, and booking-status information required for proper transaction management.

Manual record keeping could result in missing information because of incomplete forms, overlooked entries, or recording errors. In the developed system, standardized reservation forms were used to guide customers in providing the required information. The database then stored the submitted information in an organized manner.

The researchers examined selected reservation records from both methods and determined whether the required information had been properly recorded. This provided another basis for comparing the reliability of the manual and computerized reservation processes.

Evaluation and Analysis of Results

After the necessary data had been collected, the researchers compared the number and types of errors encountered under the manual reservation process and the developed Online Booking System. The results were organized according to the identified accuracy indicators, including booking errors, double bookings, customer record accuracy, accommodation schedule accuracy, and completeness of reservation records.

Descriptive statistical measures such as frequency, percentage, and average values were used where appropriate to summarize the results. The findings were presented through tables and graphical representations to provide a clear comparison between the two reservation methods.

A lower number of errors in the computerized process indicated an improvement in reservation accuracy. The researchers also examined whether the developed system reduced the occurrence of double bookings, improved the completeness of customer records, and provided more consistent accommodation schedules.

The comparison of accuracy complemented the earlier evaluation of reservation processing time. While processing time measured how efficiently reservation activities were completed, accuracy measured how reliably the information was recorded and managed. Together, these indicators provided a broader assessment of the effectiveness of the developed Online Booking System.

Customer Satisfaction Evaluation

The responses collected from the customers were summarized and analyzed using descriptive statistical techniques such as frequency distribution, percentage, weighted mean, and overall satisfaction ratings. These statistical measures enabled the researchers to determine the level of satisfaction experienced by the respondents after using the Online Booking System. The researchers also analyzed the qualitative comments, observations, and suggestions provided by the respondents to identify the strengths of the system and determine areas that could be improved in future versions.

The findings provided information regarding the customers' perceptions of the system in terms of ease of use, accessibility, convenience, booking speed, information availability, and overall satisfaction. Higher satisfaction ratings indicated that the developed system had successfully addressed several limitations associated with the previous manual reservation process. In particular, the system provided customers with easier access to resort information, faster reservation processing, convenient availability checking, and a more organized method of submitting booking requests.

The customer feedback also helped the researchers determine whether the developed system had met the expectations of its intended users. Positive comments regarding the system's interface, accessibility, booking procedures, and availability of information indicated that the system had provided a more convenient reservation experience. Meanwhile, suggestions and identified limitations served as a basis for determining possible

improvements to the system.

Overall, the customer satisfaction evaluation served as an important indicator of the effectiveness and acceptability of the Online Booking System. The results demonstrated the extent to which the developed application had improved the customer reservation experience while supporting the modernization of Avianna's Inland Resort through digital technology. The evaluation also provided evidence that the system had the potential to contribute to improved customer service, stronger customer engagement, and better accessibility of resort reservation services.

System Performance Evaluation

In addition to evaluating reservation processing time, data accuracy, and customer satisfaction, the researchers conducted a comprehensive system performance evaluation to determine whether the developed Online Booking System met the required technical and operational standards. The system performance evaluation focused on measuring the efficiency, reliability, responsiveness, and security of the application under normal operating conditions. Since the proposed system was a web-based application accessed by customers and administrators through internet-enabled devices, maintaining satisfactory system performance was essential to ensure a positive user experience and support the smooth operation of the resort.

The performance evaluation examined the capability of the system to process user requests accurately and efficiently while maintaining data integrity and system stability. Several performance indicators were considered to determine whether the developed application satisfied the requirements of Avianna's Inland Resort and performed according to its intended functions. The results of the evaluation provided the researchers with information regarding the effectiveness of the system and helped identify areas that required improvement or optimization.

The researchers evaluated several important aspects of system performance, including **page loading speed, system response time, database performance, system reliability, and security effectiveness**.

Page Loading Speed

Page loading speed referred to the amount of time required for the system to display a webpage after a user accessed a particular function through a web browser. The evaluation included commonly accessed pages and functions such as the homepage, login page, registration page, accommodation listings, booking forms, administrator dashboard, and report pages.

Page loading speed was evaluated because it directly affected the overall user experience. Faster-loading pages reduced waiting time and allowed customers and administrators to complete their activities more efficiently. Slow-loading pages, on the other hand, could cause inconvenience and discourage customers from completing the reservation process. The researchers observed and evaluated the loading performance of the system under normal usage conditions to determine whether the application provided acceptable responsiveness.

The results of the evaluation indicated whether the developed system was capable of displaying important pages and system components within an acceptable period. Efficient page loading contributed to a smoother interaction between users and the Online Booking System.

System Response Time

System response time referred to the amount of time required for the application to process a user's request and provide the appropriate output. The researchers evaluated common transactions such as logging into the system, searching for available accommodations, submitting reservation requests, updating booking records, retrieving customer information, and generating reports.

The evaluation of response time was important because customers and administrators expected the system to respond promptly when performing reservation-related activities. Efficient response times reduced delays and allowed users to complete their tasks with less interruption.

The researchers observed the response of the system during different transactions and determined whether the application was capable of processing requests efficiently. The evaluation also helped identify functions that required optimization to improve the overall responsiveness of the system.

Database Performance

Database performance referred to the efficiency of the database in storing, retrieving, updating, and managing information required by the Online Booking System. Since the system relied on a centralized database for customer information, reservation records, accommodation details, availability schedules, and administrative records, database performance was an important factor in determining the overall efficiency of the application.

The researchers evaluated the database operations involved in record insertion, data retrieval, record modification, searching, and report generation. The evaluation also considered whether reservation information remained consistent and accurate during different system transactions.

An efficient database allowed the system to retrieve and update information quickly while maintaining organized and reliable records. Proper database performance also supported real-time monitoring of accommodation availability and helped ensure that updated reservation information was immediately reflected in the system.

System Reliability

System reliability referred to the ability of the Online Booking System to perform its intended functions consistently and correctly without unexpected failures. A reliable system was expected to operate without frequent crashes, data corruption, loss of reservation information, or interruptions during normal usage.

The researchers evaluated the stability of the application while performing various reservation and administrative transactions. The evaluation included observing successful transaction completion, system errors, unexpected interruptions, and the consistency of stored information.

System reliability was particularly important because customers depended on the system to access accommodation information and submit reservation requests, while administrators relied on it to manage bookings and customer records. The evaluation results helped determine whether the developed system was sufficiently stable and dependable for the intended operational environment of Avianna's Inland Resort.

Security Effectiveness

Security effectiveness referred to the ability of the developed system to protect user information, reservation records, and administrative data from unauthorized access. Since the Online Booking System stored customer information and account credentials, security was considered an essential component of system performance.

The researchers evaluated the implemented security mechanisms, including user authentication, password protection, role-based access, session management, and access restrictions for administrative functions. These features were assessed to determine whether users were only able to access the functions and information appropriate to their assigned roles.

The system also incorporated security measures provided through the Laravel framework and application-level controls. These measures helped protect the system from unauthorized access and common web-based security risks. The researchers considered the effectiveness of these security mechanisms in protecting customer and administrative information and maintaining the confidentiality and integrity of stored data.

Overall System Performance Assessment

After evaluating the individual performance indicators, the researchers summarized the findings using descriptive statistics, performance tables, and graphical presentations when appropriate. The results from page loading speed, system response time, database performance, system reliability, and security effectiveness were examined to determine whether the developed application satisfied the operational requirements of Avianna's Inland Resort.

The overall assessment provided an indication of the system's ability to support daily reservation and administrative activities. The results demonstrated whether the Online Booking System performed efficiently, responded appropriately to user requests, maintained organized and reliable records, and provided secure access to authorized users.

The system performance evaluation therefore served as an important component of the overall assessment of the developed application. Together with the evaluation of reservation processing time, reservation accuracy, and

customer satisfaction, it provided a more comprehensive basis for determining the effectiveness and suitability of the Online Booking System. The results supported the objective of developing a functional, reliable, efficient, secure, and user-friendly system capable of improving the reservation and management operations of Avianna's Inland Resort.

The responses collected from the customers were summarized and analyzed using descriptive statistical techniques such as frequency distribution, percentage, weighted mean, and overall satisfaction ratings. These statistical measures were used to determine the level of customer satisfaction with the developed Online Booking System. The researchers also analyzed the qualitative comments, observations, and suggestions provided by the respondents to identify the strengths of the system and determine areas that could be improved in future development.

The findings provided information regarding the customers' perceptions of the system in terms of ease of use, accessibility, convenience, booking speed, information availability, and overall satisfaction. The satisfaction ratings indicated the extent to which the developed Online Booking System had addressed the limitations of the previous manual reservation process. The results also provided an assessment of whether the system had provided customers with a faster, more convenient, accessible, and reliable reservation experience.

The qualitative feedback obtained from the respondents further helped the researchers identify specific strengths and limitations of the system. Positive comments provided insights into the features that customers found useful and convenient, while suggestions and observations helped identify areas that required further improvement. These responses were considered in assessing the overall acceptability and usability of the developed system.

Ultimately, the Customer Satisfaction Evaluation served as one of the important indicators of the effectiveness of the Online Booking System. The results provided evidence regarding the system's ability to improve the customer reservation experience while supporting the modernization of Avianna's Inland Resort through digital technology. The evaluation also helped determine whether the system contributed to improved service accessibility, convenience, and customer engagement.

System Performance Evaluation

In addition to evaluating reservation processing time, data accuracy, and customer satisfaction, the researchers conducted a comprehensive system performance evaluation to determine whether the developed Online Booking System met the required technical and operational standards. The system performance evaluation focused on measuring how efficiently, reliably, and securely the application performed under normal operating conditions. Since the developed system was a web-based application accessed by customers and administrators through internet-enabled devices, satisfactory system performance was essential in providing a positive user experience and supporting the smooth operation of the resort.

The performance evaluation examined the capability of the system to process user requests accurately and efficiently while maintaining data integrity and system stability. Several performance indicators were considered to determine whether the developed application satisfied the requirements of Avianna's Inland Resort and performed according to its intended functions. The findings from the evaluation provided objective information regarding the effectiveness of the developed system and helped identify areas that required further improvement or optimization.

The researchers evaluated several important aspects of system performance, including **page loading speed, system response time, database performance, system reliability, and security effectiveness**.

Page Loading Speed

Page loading speed referred to the amount of time required for the system to display web pages after users accessed them through a web browser. The evaluation included commonly accessed pages and system components such as the homepage, login page, registration page, accommodation listings, booking forms, administrator dashboard, and report pages.

Fast page loading was important because it reduced waiting time and contributed to a smoother user experience. Slow-loading pages could cause inconvenience and potentially discourage customers from completing their reservations. During the evaluation, the researchers observed and assessed the loading performance of the system under normal usage conditions to determine whether the website provided acceptable responsiveness for both customers and administrators.

The assessment of page loading speed provided information regarding the system's ability to display requested pages efficiently. Efficient page loading contributed to a more responsive interaction between users and the Online Booking System.

System Response Time

System response time referred to the amount of time required for the application to process a user's request and return the appropriate output. The researchers evaluated common system transactions, including logging into the system, searching for available accommodations, submitting reservation requests, updating booking records, generating reports, and retrieving customer information.

The evaluation of system response time was important because customers and administrators required timely responses when performing reservation-related activities. Faster response times reduced delays and allowed users to complete their tasks more efficiently.

The researchers measured and observed the response of the system during different transactions to determine whether the application processed requests within an acceptable period. The evaluation also helped identify system functions that could be further optimized to improve overall responsiveness and efficiency.

Database Performance

Database performance referred to the efficiency of the database in storing, retrieving, updating, and managing information required by the Online Booking System. Since the application depended on a centralized database for managing customer records, reservation details, accommodation availability, and administrative information, database performance played an important role in the overall operation of the system.

The researchers evaluated database operations involving record insertion, data retrieval, data modification, searching, and report generation. The evaluation also considered the consistency and accuracy of stored information during different reservation and administrative transactions.

An efficient database allowed the system to retrieve and update information in an organized and timely manner. It also supported the proper management of customer records and reservation schedules while helping ensure that accommodation availability information remained updated and reliable.

System Reliability

System reliability referred to the ability of the Online Booking System to perform its intended functions consistently and correctly without unexpected failure. A reliable system was expected to operate without frequent crashes, data corruption, loss of reservation information, or unnecessary interruptions during normal usage.

The researchers evaluated the stability of the application while performing different reservation and administrative transactions. The evaluation included observing successful transaction completion, system errors, unexpected interruptions, and the consistency of stored information.

Reliability was particularly important because customers depended on the system to access resort information, check accommodation availability, and submit reservation requests. Likewise, administrators relied on the system to manage bookings, customer records, accommodation schedules, and reports. The evaluation therefore helped determine whether the developed system was sufficiently stable and dependable for the operational requirements of Avianna's Inland Resort.

Security Effectiveness

Security effectiveness referred to the capability of the developed system to protect sensitive information, maintain user privacy, and prevent unauthorized access to system functions and confidential records. Since the Online Booking System stored customer information, reservation records, user accounts, and administrative data, security was an essential component of the system evaluation.

The researchers evaluated the implemented security mechanisms by examining user authentication, password protection, role-based access control, session management, and database security. These mechanisms were assessed to determine whether users were able to access only the functions and information appropriate to their assigned

roles.

The system also utilized security features provided by the Laravel framework together with application-level security measures. The researchers considered these mechanisms in evaluating the protection of customer and administrative information against unauthorized access and common web-based security risks. The evaluation focused on determining whether the implemented security controls were functioning properly and adequately supported the confidentiality and integrity of system data.

Overall System Performance Assessment

After evaluating the individual performance indicators, the researchers summarized the findings using descriptive statistics, performance tables, charts, and graphical presentations when appropriate. The results from page loading speed, system response time, database performance, system reliability, and security effectiveness were examined to determine whether the developed application satisfied the technical and operational requirements of Avianna's Inland Resort.

The overall assessment provided an indication of the system's ability to support the daily reservation and administrative activities of the resort. The evaluation determined whether the Online Booking System performed efficiently, responded appropriately to user requests, maintained organized and reliable records, and provided secure access to authorized users.

The System Performance Evaluation therefore served as an important component of the overall assessment of the developed application. Together with the evaluation of reservation processing time, reservation accuracy, and customer satisfaction, it provided a comprehensive basis for determining the effectiveness, reliability, efficiency, usability, and security of the Online Booking System. The results supported the study's objective of modernizing the reservation procedures of Avianna's Inland Resort through the use of a functional, reliable, secure, and user-friendly web-based information system.

Significance of Experimental Evaluation

The inclusion of experimental evaluation significantly strengthens the overall quality, credibility, and validity of this research. While descriptive and developmental research provide valuable information regarding existing problems and the process of system development, experimental evaluation provides measurable evidence of the effectiveness of the proposed Online Booking System. Rather than relying solely on observations, opinions, or subjective assessments, experimental evaluation allows the researchers to collect objective quantitative and qualitative data that demonstrate the actual impact of the developed system on the operations of Avianna's Inland Resort.

Experimental evaluation enables the researchers to compare the existing manual reservation process with the proposed computerized reservation system under similar conditions. Through systematic testing and performance measurement, the researchers can determine whether the Online Booking System achieves its intended objectives of improving operational efficiency, reducing processing time, minimizing reservation errors, organizing customer information, and enhancing customer satisfaction. The use of measurable performance indicators makes the findings more reliable and provides stronger evidence to support the conclusions of the study.

Another important contribution of experimental evaluation is its ability to identify the strengths and limitations of the developed application before full implementation. By analyzing the system's performance in terms of processing speed, accuracy, reliability, usability, and security, the researchers can determine which components perform effectively and which areas require further improvement. This evaluation process supports continuous refinement of the system and helps ensure that the final product satisfies both technical requirements and user expectations.

The experimental evaluation also provides valuable information regarding the practical benefits of adopting an online reservation platform within the tourism and hospitality industry. The expected findings may demonstrate significant improvements in reservation processing time, booking accuracy, customer convenience, administrative productivity, database management, and communication between customers and resort personnel. These improvements may contribute to better customer service, increased operational efficiency, improved resource utilization, and stronger business competitiveness.

The experimental findings may assist resort management in making informed decisions regarding the long-term implementation and future enhancement of the Online Booking System. Evidence obtained from system testing and user evaluations can justify the adoption of computerized reservation management and support future investments in information technology. The results may also serve as a reference for other small and medium-sized tourism enterprises that intend to modernize their operations through similar digital solutions.

The significance of the experimental evaluation extends beyond the immediate objectives of the study. It contributes to the body of knowledge in Information Technology by demonstrating the practical application of systems analysis, software engineering, database management, and web development principles in solving real-world business problems. Future researchers and software developers may use the documented evaluation procedures, performance measurements, and research findings as references in developing more advanced online reservation systems for resorts, hotels, and other hospitality establishments.

Ultimately, the experimental evaluation serves as the final verification that the proposed Online Booking System successfully fulfills its intended purpose. Through comprehensive testing and systematic analysis, the researchers can determine whether the developed application provides significant advantages over the existing manual reservation process. The anticipated improvements in efficiency, accuracy, reliability, security, customer satisfaction, and overall operational performance will validate the effectiveness of the proposed solution and demonstrate its potential contribution to the modernization, competitiveness, and sustainable growth of Avianna's Inland Resort.

3.2. System Development Methodology

The researchers selected the **Agile Model** as the System Development Life Cycle (SDLC) methodology for the development of the Online Booking System for Avianna’s Inland Resort in Santa Barbara, Iloilo. The Agile Model was selected because it provided a flexible and iterative approach that allowed the researchers to continuously develop, test, evaluate, and improve the system based on the identified requirements and feedback from the intended users.

The System Development Life Cycle (SDLC) served as a structured framework that guided the researchers throughout the planning, designing, development, testing, implementation, and maintenance of the Online Booking System. The use of the Agile Model allowed the researchers to divide the development process into manageable iterations or sprints. Each iteration focused on specific system requirements and functionalities that were developed, tested, reviewed, and improved before proceeding to the succeeding development activities.

The Agile Model was appropriate for the study because the requirements of the Online Booking System were based on the actual operational needs of Avianna’s Inland Resort. Through continuous communication and feedback, the researchers were able to identify necessary modifications and incorporate improvements during the development process. This approach allowed the researchers to address problems as they were identified and helped ensure that the final system was aligned with the requirements of the resort, its administrators, receptionists, and customers.

Requirements Gathering and Analysis

The Agile development process began with requirements gathering and analysis. During this stage, the researchers gathered information regarding the existing reservation procedures and operational activities of Avianna’s Inland Resort. Interviews, observations, and document analysis were conducted to understand how customer inquiries, reservations, accommodation availability, customer records, and booking schedules were handled.

The researchers analyzed the existing manual reservation process and identified the difficulties experienced by the resort personnel and customers. These included the time required to process reservations, the possibility of inaccurate records, difficulties in monitoring accommodation availability, delayed booking confirmations, and the risk of scheduling conflicts or double bookings.

The researchers also identified the requirements of the intended users. Customers required a convenient platform for viewing resort information, checking accommodation availability, and submitting reservations. Administrators and receptionists required functions for managing customer accounts, accommodations, reservations, booking schedules, inquiries, and reports. The information gathered during this stage served as the basis for determining the features and functionalities that were incorporated into the system.

Sprint Planning

After the requirements had been identified and analyzed, the researchers organized the development activities into a series of manageable sprints. Each sprint contained specific objectives, tasks, and system functionalities that were developed within a particular development cycle.

During sprint planning, the researchers prioritized the system features according to their importance to the resort's reservation operations. Fundamental features such as user registration, authentication, accommodation management, availability monitoring, and reservation processing were prioritized during the initial development stages. Other functions, including reservation management, customer management, report generation, and administrative dashboard features, were subsequently developed and integrated.

The sprint planning process helped the researchers organize the development activities, monitor their progress, and allocate sufficient time and resources to each functionality. It also allowed them to make adjustments whenever changes or additional requirements were identified during the development process.

System Design

After the requirements had been established, the researchers designed the structure and interface of the Online Booking System. The design process included the development of the system architecture, database structure, navigation flow, user interfaces, booking forms, and administrative dashboard.

The researchers designed the customer interface to provide convenient access to resort information, accommodation details, availability schedules, and reservation functions. The administrative interface was designed to provide authorized personnel with access to reservation management, accommodation management, customer records, booking monitoring, and report generation.

The database design was also developed during this stage. The database was structured to store customer information, user accounts, accommodation details, reservation records, booking statuses, and other information required by the system. Relationships between the different database entities were established to maintain data consistency and support efficient retrieval and management of information.

The system designs were reviewed throughout the development process. When usability concerns or changes in requirements were identified, the researchers modified the designs accordingly before proceeding to the succeeding development activities.

System Coding and Development

After the system design had been completed, the researchers proceeded with the coding and development of the Online Booking System. The selected programming languages, frameworks, database technologies, and development tools were used to transform the system designs and requirements into functional software components.

The system was developed incrementally through several development iterations. The researchers initially

developed the user registration and authentication functions. These features allowed customers and authorized personnel to create accounts and access system functions according to their assigned roles.

The accommodation management module was subsequently developed to allow administrators to manage accommodation information and monitor the availability of cottages, rooms, and pavilions. The reservation module was then developed to allow customers to select accommodations, specify booking dates, provide the required information, and submit reservation requests through the online platform.

Administrative reservation functions were also developed to allow authorized personnel to review, approve, reject, update, and monitor reservations. The availability monitoring function was integrated with the reservation module so that accommodation schedules were updated based on reservation records. This helped reduce the possibility of scheduling conflicts and double bookings.

The researchers also developed user management and administrative dashboard functions. These features allowed administrators to manage customer accounts, monitor booking activities, and access important reservation information through a centralized interface. Report generation capabilities were also incorporated to allow administrators to obtain organized information regarding reservations and accommodation occupancy.

Each system component was developed and tested before it was integrated with the other modules. This incremental development process allowed the researchers to identify problems at an early stage and make the necessary corrections without significantly affecting other completed components.

System Testing

Testing was conducted throughout the development process to ensure that each system function operated according to its intended requirements. Unlike a development approach in which testing was performed only after the entire system had been completed, the Agile Model allowed the researchers to conduct testing continuously during each development iteration.

Unit testing was performed to examine individual modules and functions. The researchers tested features such as user registration, login, accommodation management, reservation submission, booking approval, availability monitoring, and report generation to determine whether each component performed correctly.

Integration testing was then conducted to verify whether the different system modules worked properly when combined. The researchers checked whether information entered through the customer interface was correctly stored in the database and whether updates made by administrators were properly reflected in the appropriate system components.

System testing was also performed to evaluate the complete Online Booking System. The researchers examined its functionality, usability, reliability, performance, security, and compatibility with different devices and web browsers.

User Acceptance Testing (UAT) was conducted with the intended users to determine whether the system met their operational requirements. The feedback gathered from the users was reviewed by the researchers and was used to identify areas that required modification or improvement.

Sprint Review and Feedback

At the end of each development iteration, the researchers conducted a review of the completed features. The purpose of the review was to determine whether the developed functionalities satisfied the requirements identified during the earlier stages of the study.

Feedback from the intended users was considered during the review process. Users provided observations and suggestions regarding the system interface, navigation, booking procedures, accommodation information, and administrative functions. The researchers analyzed the feedback and identified features that required modification or further improvement.

The feedback obtained from each review was incorporated into succeeding development iterations. This allowed the researchers to continuously improve the system based on actual user experiences rather than relying solely on the researchers' assumptions.

System Refinement and Improvement

The researchers continuously refined the Online Booking System based on the results of testing and user feedback. Errors and usability issues identified during development were corrected in the succeeding iterations.

System refinement included improving navigation, modifying forms, correcting programming errors, strengthening input validation, improving database operations, and adjusting system functionalities according to the requirements of the resort.

The researchers also evaluated the system's usability, reliability, efficiency, accuracy, and security throughout the development process. Necessary modifications were implemented to improve the quality and functionality of the system before final deployment.

The continuous refinement process helped ensure that the developed Online Booking System remained consistent with the operational requirements of Avianna’s Inland Resort and the expectations of its intended users.

System Integration

After the individual modules had been developed and tested, the researchers integrated the components into a complete Online Booking System. The integration process connected the customer interface, administrative interface, database, reservation functions, accommodation management, user management, and reporting features.

The researchers verified whether customer information and reservation details were correctly transmitted and stored in the database. They also verified whether changes made by administrators, such as approving reservations and updating accommodation availability, were properly reflected in the system.

Integration testing was conducted to identify problems that could occur when the different modules interacted with one another. This process ensured that the system operated as a unified application and that its different components communicated correctly.

Final Evaluation

After the major development iterations had been completed, the researchers conducted the final testing and evaluation of the Online Booking System. The evaluation examined the system's functionality, usability, reliability, efficiency, accuracy, security, and overall performance.

The researchers also compared selected aspects of the existing manual reservation process with the developed computerized system. Reservation processing time and booking accuracy were considered as important indicators in determining whether the system provided improvements over the existing process.

Customer satisfaction was also evaluated to determine the users' perceptions regarding the convenience, accessibility, ease of use, booking speed, information availability, and overall functionality of the system.

The results obtained from testing and evaluation were analyzed and used to determine whether the developed Online Booking System had successfully addressed the identified problems of the existing manual reservation process.

System Deployment and Implementation

After the system had successfully undergone testing and evaluation, the researchers prepared it for deployment and actual implementation. The system was configured in its intended operating environment, and the database was prepared to store the required customer, accommodation, and reservation information.

User accounts were created and configured according to the roles and responsibilities of the intended users. Administrators and authorized resort personnel were oriented on the basic functions and procedures of the system. The orientation included managing accommodations, monitoring reservations, processing booking requests, updating customer information, checking accommodation availability, and generating reports.

The researchers also ensured that the necessary system components were properly configured before the system was introduced for actual use. Existing relevant records were organized and prepared for entry into the computerized system when applicable.

The deployment process marked the transition of the Online Booking System from development and testing into actual operational use. Through proper implementation and user orientation, the researchers helped ensure that the resort personnel were capable of using the system to support their reservation and administrative activities.

Maintenance and Continuous Improvement

After implementation, the system was maintained to ensure its continued functionality and reliability. The researchers monitored the system for possible errors, technical issues, and areas that could be improved based on actual user experience.

Maintenance activities included correcting identified errors, updating resort information, improving system performance, and modifying selected functions when necessary. Feedback from administrators, receptionists, and customers was considered in identifying possible improvements.

Although the system had been completed and implemented, the Agile methodology allowed future enhancements to be introduced through additional development iterations. Possible future improvements included online payment integration, mobile application development, advanced analytics, automated notifications, and other features that were outside the scope of the current study.

Application of the Agile Model in the Study

The Agile Model provided the researchers with a flexible and iterative framework for developing the Online Booking System for Avianna’s Inland Resort. Through repeated cycles of planning, designing, coding, testing, reviewing, and refining, the researchers were able to transform the identified problems of the manual reservation process into functional system features.

The methodology allowed the researchers to accommodate changes and improvements throughout the development process. Instead of completing the entire system before receiving feedback, the researchers presented and evaluated developed features progressively. This approach allowed potential problems to be identified and corrected at an earlier stage.

The Agile Model also supported the different requirements of the system's intended users. Customers required a convenient and accessible platform for making reservations, while administrators and receptionists required functions for managing bookings, accommodations, customer records, and reports. Through continuous evaluation and feedback, the researchers were able to develop functions that addressed these different requirements.

Importance of Agile Methodology in the Study

The Agile methodology played an important role in the development of the Online Booking System for Avianna’s Inland Resort because it provided the researchers with a flexible and systematic approach to planning, developing, testing, and improving the system. Since the system was intended for use by both customers and resort administrators, the researchers considered usability, reliability, efficiency, and adaptability throughout the development process.

The Agile methodology supported continuous interaction between the researchers and the intended stakeholders. Throughout the development process, the researchers gathered feedback from the resort owner, administrators,

receptionists, and potential users regarding the system's features, interface design, navigation, and functionality. The feedback obtained during the development iterations helped the researchers determine whether the developed features were consistent with the actual operational requirements of the resort.

The methodology also helped reduce development risks because problems were identified and addressed during the early stages of development. Instead of waiting until the entire system had been completed before conducting testing, the researchers tested individual features and modules after each development iteration. Errors, inconsistencies, and usability concerns identified during testing were corrected and improved in succeeding iterations.

The use of Agile also improved project transparency because the progress of the system could be monitored throughout the development process. Completed features were reviewed and evaluated, allowing the researchers and stakeholders to determine which components had been successfully implemented and which areas still required improvement. This collaborative process contributed to the development of a system that was more closely aligned with the needs of Avianna’s Inland Resort.

Advantages of the Agile Model

The Agile Model provided several advantages during the development of the Online Booking System for Avianna’s Inland Resort. These advantages contributed to the organized development, continuous improvement, and successful implementation of the system.

Flexible and Adaptable to Changes

One of the primary advantages of the Agile Model was its flexibility in accommodating changes during system development. Software requirements could change as new operational needs, user preferences, or problems were identified. The Agile approach allowed the researchers to modify system requirements and functionalities without significantly disrupting the overall development process.

In this study, the flexibility of Agile allowed the researchers to accommodate modifications requested or identified during the development of the Online Booking System. Changes involving system features, interface design, booking procedures, and administrative functions were considered during succeeding iterations. This allowed the system to remain aligned with the operational requirements of Avianna’s Inland Resort.

Encouraged Continuous Feedback

The Agile Model encouraged continuous communication and feedback between the researchers and stakeholders. At the end of each development iteration, completed features were reviewed and evaluated to determine whether they satisfied the identified requirements.

Feedback obtained from the resort owner, administrators, receptionists, and potential users helped the researchers identify areas that required improvement. The suggestions and observations gathered during the review process were considered in succeeding iterations, allowing the researchers to improve the system based on actual user needs and experiences.

Allowed Incremental System Development

Another advantage of the Agile Model was its support for incremental development. Instead of developing the entire Online Booking System simultaneously, the researchers developed the system through smaller and manageable components.

The development began with essential functions such as user registration and login authentication before proceeding to accommodation management, availability monitoring, reservation processing, booking management, customer management, and report generation. Each module was developed and tested before being integrated with the other components.

This approach allowed the researchers to focus on individual system functions while continuously monitoring their quality and performance.

Improved Collaboration Among Developers and Stakeholders

The Agile Model promoted collaboration among the researchers, resort administrators, management personnel, receptionists, and intended users. Communication and consultation were conducted throughout the development process to discuss requirements, system progress, identified problems, and possible improvements.

This collaboration allowed the researchers to gain a better understanding of the actual reservation procedures and operational needs of the resort. It also helped ensure that the system features were developed based on practical requirements rather than assumptions.

Detected Errors Early Through Regular Testing

Regular testing was another important advantage of the Agile Model. Testing activities were conducted throughout the development process rather than being performed only after the entire system had been completed.

The researchers tested individual modules and integrated system components during the development iterations. Errors involving user authentication, reservation processing, accommodation availability, database operations, and administrative functions were identified and corrected as development progressed.

Early detection of errors helped prevent minor problems from developing into larger system issues. Continuous testing also contributed to the reliability, stability, and overall quality of the developed system.

Produced Functional Modules Incrementally

The Agile Model allowed the researchers to produce functional system modules incrementally. Completed features became available for testing and evaluation before the entire system was finalized.

For the Online Booking System, functions such as customer registration, login authentication, accommodation viewing, availability monitoring, and reservation processing were developed progressively. This allowed the researchers to evaluate the functionality of each component and make improvements before proceeding to additional features.

The incremental development process also provided visible evidence of project progress throughout the development period.

Enhanced Customer Satisfaction

Customer satisfaction was considered throughout the development of the Online Booking System. The researchers incorporated user feedback and evaluation results to improve the system's ease of use, accessibility, convenience, booking speed, and information availability.

The Agile approach helped the researchers create a system that responded to the needs of customers and resort personnel. Customers were provided with an online platform for accessing resort information, checking accommodation availability, and submitting reservations, while resort personnel were provided with tools for managing reservations and customer records.

These improvements contributed to a more convenient and organized reservation experience.

Reduced Project Risks

The iterative nature of Agile helped reduce development risks by allowing the researchers to identify and address problems during the early stages of development. Regular testing, evaluation, and feedback provided opportunities to detect technical issues, usability concerns, and requirement gaps before the final implementation.

Potential problems were addressed through modifications and improvements in succeeding development iterations. This reduced the possibility of major system problems occurring during final deployment and helped maintain the progress of the project.

Improved System Quality

Continuous development, testing, evaluation, and refinement contributed to the overall quality of the Online Booking System. Each completed module underwent testing and review before being integrated into the complete system.

The researchers corrected identified errors, improved system navigation, refined interfaces, strengthened data validation, and addressed other technical and functional concerns throughout the development process.

As a result, the completed system demonstrated improved functionality, reliability, usability, efficiency, and security in accordance with the requirements identified during the study.

Supported Continuous Improvement

The Agile Model supported continuous improvement because each development iteration provided an opportunity to evaluate the existing system and identify possible enhancements. Testing results and stakeholder feedback were used as bases for modifying and refining system features.

This continuous improvement process allowed the researchers to enhance the Online Booking System throughout its development rather than treating system completion as the end of improvement activities. The approach also provided a foundation for possible future modifications and additional functionalities based on the changing needs of Avianna’s Inland Resort.

Relevance of Agile to the Online Booking System

The Agile Model was highly relevant to the development of the Online Booking System because it supported the changing requirements and operational conditions encountered during the development of the system. The reservation procedures of Avianna’s Inland Resort involved different users, including customers, administrators, management personnel, and receptionists, each of whom had specific requirements and responsibilities.

Through iterative development, continuous testing, stakeholder collaboration, and incremental implementation, the Agile methodology allowed the researchers to address these different requirements systematically. The researchers were able to modify system features, correct errors, improve interfaces, and refine system processes based on testing results and user feedback.

The methodology was also relevant because the Online Booking System consisted of several interconnected functions, including user authentication, accommodation management, availability monitoring, reservation processing, booking management, customer record management, and report generation. Developing these components incrementally allowed the researchers to test individual functions before integrating them into the complete system.

Agile supported the researchers in addressing the problems associated with the resort's existing manual reservation process. The developed system provided a centralized platform for managing reservations and customer information, reduced dependence on manual record keeping, improved monitoring of accommodation availability, and provided customers with a more convenient method of making reservations.

The Agile Model provided several advantages throughout the development of the Online Booking System for Avianna’s Inland Resort. It provided the researchers with a flexible and adaptable approach that allowed them to respond effectively to changes in system requirements, user expectations, and operational needs encountered during the development process. This flexibility enabled the researchers to modify and improve system features whenever necessary without significantly disrupting the overall development process.

Advantages of Agile Model

The Agile Model also encouraged continuous feedback from stakeholders throughout the development process. The researchers regularly gathered feedback from the resort administrators and other intended users regarding the system's features, functionality, and interface. The feedback received helped the researchers identify areas that required improvement and ensured that the developed system remained aligned with the actual needs and expectations of its users.

Agile Model supported incremental system development by allowing the researchers to develop the system through smaller and more manageable modules. Instead of completing the entire system at once, individual features were developed, tested, evaluated, and improved before proceeding to the succeeding modules. This approach made it easier for the researchers to monitor development progress, identify problems, and implement necessary adjustments at an early stage.

The methodology also improved collaboration among the researchers, developers, resort administrators, and other stakeholders. Regular communication allowed the project participants to discuss development progress, clarify requirements, evaluate completed features, and provide recommendations for improvement. This collaborative process contributed to a better understanding of the resort's operational requirements and helped ensure that the system was developed according to its intended purpose.

Another advantage of the Agile Model was its support for continuous testing and evaluation. The researchers tested the developed modules during different stages of the development process rather than waiting until the entire system had been completed. Through frequent testing, errors, inconsistencies, and functional issues were identified and addressed at an early stage. This reduced the possibility of major problems occurring during the final stages of development and contributed to the overall reliability and quality of the system.

The Agile Model also allowed functional modules to be developed and evaluated more quickly. As each module was completed, the researchers were able to assess its functionality and make improvements based on testing results and stakeholder feedback. This incremental approach provided a clearer view of the system's progress and allowed the researchers to ensure that each component performed according to its intended function.

The Agile Model contributed to a more organized, collaborative, and efficient development process. Its flexibility, incremental approach, continuous feedback, and regular testing enabled the researchers to continuously improve the Online Booking System based on actual requirements and identified issues. Through the use of Agile, the researchers were able to develop a system that was more responsive to user needs, reliable in its operation, and suitable for the reservation and administrative requirements of Avianna's Inland Resort.

Developmental research focused on the systematic design, development, implementation, evaluation, and improvement of products, systems, processes, and technological solutions intended to address specific problems or needs. Unlike purely descriptive research, which primarily aimed to observe and describe existing conditions, developmental research sought to create practical solutions that could improve efficiency, effectiveness, and overall performance within a particular environment or organization.

In the context of this study, developmental research was appropriate because the primary objective was to design and develop an Online Booking System for Avianna's Inland Resort. The developed system was intended to address the limitations of the resort's existing manual reservation process and provide a technological solution that improved operational efficiency, customer convenience, and information management.

The developmental approach allowed the researchers to transform the identified problems into functional system requirements and eventually into a working software application. Through this process, the researchers analyzed the existing reservation procedures and developed a computerized solution capable of improving the overall management of the resort's reservation and administrative operations.

The development of the Online Booking System involved several systematic activities, including system analysis, system planning, database development, user interface design, system coding and development, system testing, and

system deployment.

System Analysis

The first stage of the developmental research involved conducting a comprehensive analysis of the existing reservation and management procedures of Avianna's Inland Resort. During this phase, the researchers identified the strengths and weaknesses of the existing manual system, determined user requirements, and gathered information necessary for system development.

System analysis focused on understanding how reservations were processed, how customer records were maintained, how accommodation availability was monitored, and what challenges were encountered by both customers and administrators. The information gathered during this stage served as the foundation for designing an effective and user-centered Online Booking System.

The researchers examined the existing reservation procedures and identified areas where manual processes caused delays, errors, and difficulties in managing reservation information. The results of the analysis helped the researchers determine the functions and features that were necessary to improve the resort's reservation and administrative operations.

System Planning

After identifying the system requirements, the researchers proceeded with project planning. This stage involved defining the objectives of the system, determining its scope and limitations, allocating available resources, establishing development schedules, and selecting the appropriate technologies and tools for implementation.

Proper planning helped the researchers organize the development activities and maintain alignment with the objectives of the study. It also allowed them to anticipate possible technical and operational challenges and prepare appropriate strategies for addressing these concerns during development.

The researchers also established priorities for the development of the different system modules. The major functions required for customer reservation and administrative management were identified and organized according to their importance and relationship to the overall system.

Database Development

An essential component of the developmental process was the creation of a database that served as the central repository of information for the Online Booking System. The database was designed to store and manage customer information, reservation records, accommodation details, user accounts, and other administrative information required by the system.

The researchers developed a structured database to maintain data accuracy, consistency, organization, and security. The database design allowed information to be efficiently stored, retrieved, updated, and managed by authorized users.

The database also supported the reservation process by maintaining booking records and accommodation availability information. This centralized approach helped reduce the problems associated with manual record keeping and provided administrators with more organized and accessible information for reservation management and decision-making.

User Interface Design

The developmental research process also included the design of a user-friendly interface that facilitated interaction between customers, administrators, and the system. The researchers designed the interface with consideration for ease of navigation, accessibility, functionality, and clarity of information.

The customer interface was designed to allow users to access resort information, register accounts, view

accommodations, check availability, and submit reservation requests with minimal difficulty. The administrative interface was designed to allow authorized personnel to manage bookings, accommodations, user accounts, and reports efficiently.

The researchers continuously reviewed and improved the interface during development to ensure that the system was understandable and convenient for its intended users. The user interface design contributed to the overall usability and accessibility of the developed Online Booking System.

System Coding and Development

The coding phase involved translating the system design into an actual software application. During this stage, the researchers utilized programming languages, frameworks, databases, and development tools to implement the different functionalities of the Online Booking System.

Several key features were developed to support the reservation and administrative operations of the resort. The user registration and authentication feature was implemented to allow users to create accounts and securely access the system based on their assigned roles. The accommodation management feature was also developed to enable administrators to add, update, and manage information regarding available accommodations, including their status and relevant details.

The reservation processing feature allowed customers to submit booking requests through the system, while the booking approval and monitoring feature enabled administrators to review, approve, and monitor reservations efficiently. An availability-checking function was incorporated to provide updated information about available and occupied accommodations, helping prevent scheduling conflicts and double bookings.

The system also included report generation capabilities that allowed administrators to produce organized reservation and occupancy reports for monitoring and decision-making purposes. User management functions were developed to allow administrators to manage user accounts and relevant user information.

An administrative dashboard was also implemented to provide a centralized view of important reservation, accommodation, and system information. This allowed administrators to monitor system activities more conveniently and manage the resort's reservation operations from a centralized interface.

These features worked together to provide an organized, efficient, and reliable Online Booking System that supported both customer reservation activities and the administrative requirements of Avianna's Inland Resort. Through the coding and development process, the researchers transformed the conceptual system design into a functional application capable of addressing the identified operational needs of the resort.

System Testing

Testing was a critical component of the developmental research because it ensured that the developed system functioned according to its intended purpose. Various testing procedures were conducted to identify errors, verify system performance, evaluate usability, and determine whether the system satisfied the specified requirements.

During the testing phase, several testing activities were conducted to evaluate the functionality, reliability, security, performance, and compatibility of the Online Booking System. Unit testing was first performed to examine individual system components and ensure that each function operated correctly according to its intended purpose. This allowed the researchers to identify errors within individual modules before integrating them with other components.

Integration testing was then conducted to verify that the different modules worked properly when combined. The researchers examined whether information was accurately transferred between connected components and whether the interaction between modules produced the expected results. This testing helped identify issues that were not observable during individual module testing.

System testing was subsequently performed to evaluate the Online Booking System as a complete application. The researchers tested the major system functions and determined whether the features operated according to the specified requirements. The testing covered important processes such as user registration, authentication, accommodation management, availability checking, reservation processing, booking approval, user management, and report generation.

User Acceptance Testing (UAT) was also conducted to determine whether the developed system met the needs and expectations of its intended users. The intended users were given opportunities to interact with the system and perform relevant reservation and administrative tasks. Their feedback and observations were gathered and considered by the researchers in identifying areas that required modification or improvement.

Performance testing was conducted to evaluate the system's responsiveness, stability, and ability to handle expected user activities. The researchers observed the system's response during common operations such as logging in, viewing accommodations, checking availability, submitting reservations, updating booking records, and generating reports. The results helped determine whether the system provided acceptable performance during normal use.

Security testing was also performed to identify potential vulnerabilities and ensure that user accounts, customer information, reservation records, and other important data were appropriately protected. The researchers evaluated authentication, access control, password protection, and other security-related functions implemented in the system.

Compatibility testing was conducted to determine whether the system functioned properly across different devices, web browsers, and screen sizes. This was important because customers could access the Online Booking System using different internet-enabled devices, while administrators could use various computers or other devices to manage resort operations.

Through these testing activities, the researchers were able to identify and correct errors, improve system performance, strengthen security, and enhance the overall functionality and usability of the Online Booking System. The results of the testing activities served as a basis for making necessary modifications before the system was considered ready for deployment.

System Deployment

After the completion of development, testing, and evaluation, the Online Booking System was prepared for deployment. The deployment stage involved configuring the developed application within its intended operating environment and ensuring that all necessary system components were properly installed and functional.

The researchers configured the system and database to support the storage, organization, retrieval, and management of customer, accommodation, and reservation information. User accounts were also configured according to the roles and responsibilities of the intended users.

Relevant existing information was organized and prepared for entry into the system to ensure that necessary records were available for actual operation. The researchers also provided guidance and training to the intended administrators to familiarize them with the system's features and procedures. The training included activities such as managing accommodations, processing reservations, checking availability, managing user accounts, monitoring bookings, and generating reports.

An operational readiness assessment was also conducted to determine whether the system, data, users, and supporting procedures were adequately prepared for actual implementation. Any remaining issues identified during this assessment were addressed before the system was fully utilized.

Through these deployment activities, the researchers ensured that the Online Booking System was properly configured, understood by its intended users, and prepared to support the reservation and administrative operations of Avianna's Inland Resort.

System Deployment

After the successful completion of testing and evaluation, the Online Booking System was prepared for deployment. The deployment phase involved installing the developed application within its intended operational environment and making it available for actual use by the resort administrators and customers.

During the deployment process, the researchers installed and configured the system components required for proper operation. The database was configured to support the storage, retrieval, and management of customer information, accommodation records, and reservation data. User accounts were also established and configured according to the roles and responsibilities of the intended users.

Initial data migration was conducted to transfer relevant existing information into the developed system. This ensured that important records were properly organized and made available within the new computerized reservation platform. The researchers also conducted administrator training to familiarize the resort personnel with the system's features, functions, and procedures. The training included the management of accommodations, processing and monitoring of reservations, checking availability, managing user accounts, and generating reports.

An operational readiness assessment was also conducted to determine whether the system, database, user accounts, data, and supporting procedures were properly prepared for actual use. Any issues identified during the assessment were addressed before the system was fully implemented.

Successful deployment marked the transition of the Online Booking System from the development and testing stage to actual system utilization. Through proper installation, database configuration, user account setup, data migration, administrator training, and operational assessment, the researchers ensured that the system was adequately prepared to support the reservation and administrative operations of Avianna’s Inland Resort.

System Maintenance and Improvement

Developmental research did not end after the deployment of the Online Booking System. Continuous maintenance and improvement were necessary to ensure the long-term functionality, security, reliability, and efficiency of the developed system. After deployment, the researchers monitored the system and addressed issues that were identified during actual use and evaluation.

During the maintenance phase, various activities were carried out to ensure the continued functionality and effectiveness of the Online Booking System. Bug fixing was performed to identify and resolve errors, system issues, and unexpected behaviors encountered during system operation. These activities helped maintain the stability of the system and prevented identified problems from affecting reservation and administrative processes.

Security updates were also implemented to address potential vulnerabilities and protect user accounts, customer information, reservation records, and other important system data from unauthorized access. The researchers reviewed the implemented security mechanisms and made necessary improvements to maintain the protection and integrity of information stored within the system.

Database optimization was conducted to maintain the organization and efficiency of stored information. This activity helped improve data retrieval and ensured that customer, reservation, and accommodation records remained properly organized and accessible. The researchers also monitored the database to identify potential issues that could affect the accuracy and efficiency of information management.

Performance improvements were implemented to enhance the responsiveness and efficiency of the system. The researchers addressed identified performance issues and optimized system processes to minimize unnecessary delays when users performed different activities, such as viewing accommodations, checking availability, processing reservations, managing records, and generating reports.

Feature enhancements were also introduced based on user feedback, identified requirements, and the operational needs of Avianna’s Inland Resort. These improvements allowed the system to remain relevant to the changing requirements of the resort and its users. The researchers considered suggestions and observations from users when determining areas that could be improved or enhanced.

User support and troubleshooting were likewise provided to assist administrators and other users who encountered difficulties while using the Online Booking System. The researchers addressed reported concerns, provided guidance on system functions, and assisted users in resolving operational issues. This support helped users become more familiar with the system and encouraged proper utilization of its features.

Through these maintenance and improvement activities, the researchers ensured that the Online Booking System remained functional, secure, efficient, and reliable after deployment. The continuous monitoring and improvement of the system also helped ensure that it remained capable of supporting the reservation and administrative requirements of Avianna's Inland Resort.

Feedback gathered from users during actual system utilization was also considered in identifying opportunities for further system enhancement and future development. User suggestions, identified limitations, and emerging operational requirements provided useful information that could serve as a basis for future improvements and additional system functionalities.

Importance of Developmental Research in the Study

The developmental research approach is important because it enables the researchers to create a practical solution to an existing organizational problem. Rather than merely describing the challenges encountered by Avianna's Inland Resort, the study actively develops a system that can improve operational performance and customer service.

Through developmental research, the researchers were able to design and develop a solution that was tailored to the specific needs and operational requirements of the resort. The research process allowed them to apply information technology concepts and principles in a real-world environment, particularly in addressing challenges related to reservation and accommodation management. Through the development of the Online Booking System, the researchers were able to improve existing reservation processes by reducing manual procedures, organizing booking information, and providing a more systematic approach to managing reservations.

The developed system also enhanced customer convenience and accessibility by allowing customers to access reservation services and accommodation information more easily. It provided a digital platform that reduced the need for physical visits and lengthy communication processes, making the booking process more accessible and efficient. Furthermore, the project contributed to digital transformation within the hospitality industry by demonstrating how information technology could be applied to improve business operations and customer services.

The developmental research approach also allowed the researchers to produce a system that could be evaluated, modified, and improved continuously based on user feedback, identified issues, and changing operational requirements. Through continuous evaluation and improvement, the system was able to remain responsive to the needs of both the resort administrators and customers. Overall, the research provided the researchers with practical experience in developing technology-based solutions while contributing to a more efficient and accessible reservation management process.

Ultimately, developmental research serves as the foundation for producing a functional, efficient, and reliable Online Booking System that contributes to the modernization and growth of Avianna's Inland Resort.

Experimental Research (If Applicable)

Although the primary research design of the study is descriptive-developmental, elements of experimental research may also be incorporated during the testing and evaluation stages of system development. Experimental research focuses on determining whether a particular intervention or treatment produces measurable improvements compared to an existing method or condition.

In this study, the proposed Online Booking System may serve as the intervention, while the existing manual reservation process serves as the basis for comparison. By comparing the performance of both approaches, the researchers can evaluate the effectiveness of the developed system and determine whether it provides significant operational improvements.

The experimental component of the study may involve conducting controlled evaluations in which selected users perform reservation-related tasks using both the manual process and the computerized system. The results can then be analyzed and compared based on predefined performance indicators.

Comparison of Reservation Processing Time

One possible area of experimental evaluation is transaction speed. The researchers may compare the amount of time required to complete a reservation using the traditional manual process versus the Online Booking System.

Several factors were measured to determine the efficiency of the Online Booking System and to evaluate the improvements it provided compared with the existing reservation procedures. The time required to check accommodation availability was measured to determine how quickly administrators could identify available and occupied accommodations. The time required to record customer information was also evaluated to assess the efficiency of storing and retrieving customer records within the system.

The time required to process reservations was measured to determine how efficiently booking requests could be handled from submission to processing. The time required to confirm bookings was also assessed to determine whether the system was able to provide faster confirmation to customers. These factors provided measurable indicators of the system's effectiveness in reducing processing time, improving workflow efficiency, and making the reservation process more convenient for both administrators and customers.

The results may demonstrate whether the computerized system significantly reduces processing time and improves operational efficiency.

Comparison of Accuracy

Experimental testing may also focus on the accuracy of reservation management.

The researchers evaluated several factors to determine the accuracy and reliability of the Online Booking System in managing reservation and accommodation information. The frequency of booking errors was assessed to identify how often incorrect or incomplete reservation information occurred during the booking process. The incidence of double bookings was also evaluated to determine the system's effectiveness in preventing multiple reservations from being assigned to the same accommodation for the same date or period.

The accuracy of customer records was assessed to determine whether customer information was properly recorded, stored, and retrieved by the system. The accuracy of accommodation schedules was also evaluated to ensure that the availability, reservation, and occupancy status of accommodations were correctly reflected in the system. These factors helped the researchers determine whether the Online Booking System reduced errors, improved data accuracy, and provided more reliable management of reservations and accommodation schedules.

By comparing the results before and after system implementation, the researchers can determine whether automation contributes to improved data accuracy.

Customer Satisfaction Evaluation

Another important aspect of the experimental research was the assessment of customer satisfaction. The customers were asked to evaluate their experiences while using the Online Booking System and compare the system-based reservation process with the traditional reservation methods previously used. This evaluation helped the researchers determine whether the developed system provided a more convenient, accessible, and efficient booking experience for customers.

The evaluation focused on several areas, including ease of use, accessibility, convenience, booking speed, information availability, and overall satisfaction. Ease of use was assessed to determine whether customers were able to navigate the system and complete their reservations without significant difficulty. Accessibility was evaluated based on how easily customers could access the system using available devices and internet connections.

Convenience was assessed by determining whether the system reduced the need for physical visits and lengthy communication processes. Booking speed was also evaluated to determine whether customers were able to complete reservations and receive confirmations more quickly. In addition, information availability was assessed based on the accessibility and clarity of accommodation details, rates, availability, and other relevant booking information.

The results may indicate whether the developed system enhances the customer experience.

System Performance Evaluation

The researchers also evaluated the accuracy and reliability of the Online Booking System by examining several factors related to reservation and accommodation management. The frequency of booking errors was assessed to determine how often incorrect, incomplete, or inconsistent reservation information occurred during the booking process. The incidence of double bookings was also evaluated to determine the effectiveness of the system in preventing multiple customers from reserving the same accommodation for the same date or period.

The accuracy of customer records was evaluated to determine whether customer information was correctly entered, stored, updated, and retrieved by the system. The accuracy of accommodation schedules was also assessed to ensure that the availability, reservation, and occupancy status of each accommodation were properly reflected in the system. These evaluations helped the researchers determine whether the Online Booking System reduced booking errors, prevented scheduling conflicts, improved the accuracy of customer information, and provided reliable accommodation scheduling. Overall, the results served as indicators of the system's effectiveness in improving the accuracy and reliability of the resort's reservation management processes.

These evaluations help determine whether the system meets acceptable performance standards.

Significance of Experimental Evaluation

The inclusion of experimental elements strengthens the study by providing measurable evidence regarding the effectiveness of the proposed system. Rather than relying solely on observations and perceptions, the researchers can collect quantitative and qualitative data that demonstrate the actual impact of the Online Booking System on resort operations.

The findings may reveal improvements in efficiency, accuracy, customer satisfaction, and overall operational performance, thereby validating the usefulness of the proposed solution.

Through experimental evaluation, the researchers can determine whether the Online Booking System offers significant advantages over the existing manual reservation process and whether it successfully achieves the objectives of the study.

3.3. System Architecture

Overview of How the System Works

The proposed Online Booking System for Avianna’s Inland Resort is designed to automate and streamline the resort’s reservation and management processes through a web-based platform. The system serves as a centralized solution that allows customers and resort administrators to perform various booking-related activities efficiently and conveniently. By replacing traditional manual reservation methods with a computerized system, the resort can improve operational efficiency, enhance customer satisfaction, and minimize errors associated with manual record management.

The Online Booking System provides customers with an accessible and user-friendly platform where they can obtain information about the resort's facilities, services, accommodation rates, and availability. Through an internet connection and a web browser, customers can access the system anytime and anywhere without the need to physically visit the resort or contact resort personnel directly. This feature significantly improves customer convenience and accessibility while expanding the resort's ability to serve a larger market.

The system allows customers to browse available accommodations such as cottages, rooms, and pavilions. Detailed information regarding each accommodation, including descriptions, capacities, amenities, pricing, and availability status, is displayed within the platform. Customers can compare available options and select accommodations that best meet their needs and preferences.

Once customers decide to make a reservation, they can complete the booking process by selecting their preferred dates, accommodation type, and the number of guests. The reservation details are then submitted electronically and stored within the system database. This automated process eliminates the need for paper-based reservations and reduces the possibility of human errors that commonly occur in manual booking procedures.

The system is designed to maintain a centralized database that stores and organizes all reservation-related information. Customer records, accommodation details, booking schedules, reservation histories, and transaction records are securely stored within the database. This centralized storage enables efficient retrieval, updating, and management of information whenever needed by authorized users.

On the administrative side, the system provides a comprehensive dashboard that allows administrators to manage resort operations effectively. Through the dashboard, administrators can monitor incoming reservation requests, review booking information, update accommodation availability, manage customer records, and generate operational reports. The dashboard serves as the primary control center for managing all system activities and ensuring smooth resort operations.

The system incorporates authentication and security mechanisms to protect sensitive information. Only authorized users are permitted to access administrative functions, while customers can securely access their personal accounts and reservation details. These security measures help maintain data confidentiality, integrity, and protection against unauthorized access.

The proposed Online Booking System not only simplifies reservation management but also contributes to better communication between customers and resort management. Reservation confirmations, updates, and important notifications can be communicated more efficiently through the system, reducing delays and improving customer service quality.

Client-Server Setup

The proposed Online Booking System follows a **Client-Server Architecture**, one of the most widely used architectural models in web-based applications. This architecture separates the system into two major components: the client side and the server side. The client side is responsible for user interaction, while the server side handles data processing, business logic, and database management.

The client-server architecture allows multiple users to access the system simultaneously while ensuring efficient communication between users and the central server. This architecture promotes scalability, reliability, security, and centralized management of information.

Client Side

The client side refers to the devices and software applications used by customers and administrators to access and interact with the Online Booking System. These devices serve as the interface through which users communicate with the system and perform various tasks.

The Online Booking System was designed to be accessible through various internet-enabled devices, allowing users

to access its features and services according to their needs and preferences. The system could be accessed using desktop computers and laptop computers, which provided a larger screen and convenient interface for administrators and users performing detailed reservation-related tasks. It was also made accessible through tablets, allowing users to manage and view booking information using a more portable device.

The system could be accessed through smartphones and other mobile devices with an internet connection. This provided customers with greater flexibility in making reservations, checking accommodation availability, and viewing relevant booking information without being limited to a specific location or device. The compatibility of the system with different internet-enabled devices improved accessibility and convenience for both customers and administrators, supporting a more flexible and efficient online reservation process.

Users access the system through standard web browsers such as:

Users accessed the Online Booking System through standard web browsers that were commonly available on internet-enabled devices. The system was designed to support widely used browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, Safari, and Opera. This browser compatibility allowed customers, administrators, and staff members to access the system without requiring specialized software or applications.

The use of different web browsers provided users with greater flexibility and accessibility when performing reservation and administrative tasks. Customers were able to access booking services and accommodation information using their preferred browser, while administrators and staff members could manage reservations, customer records, accommodation availability, and other system functions through compatible browsers. This cross-browser accessibility contributed to the convenience, usability, and accessibility of the Online Booking System.

The client side is responsible for displaying web pages, collecting user inputs, and sending requests to the server. Through the graphical user interface (GUI), users can navigate the system, view accommodation information, submit reservation requests, update account information, and perform other system functions.

For customers, the client side provides access to features such as:

For customers, the client side of the Online Booking System provided access to various features designed to make the reservation process more convenient and accessible. The user registration feature allowed customers to create personal accounts by providing the required information, while the user login feature enabled registered customers to securely access their accounts and available system functions.

Customers were also able to view resort information and browse available accommodations through the system. These features allowed them to review relevant details about the resort, accommodations, services, and other information before making a reservation. The availability-checking feature provided customers with updated information regarding available accommodations, helping them select suitable options based on their preferred dates.

The reservation submission feature allowed customers to send their booking requests directly through the system without requiring a physical visit or lengthy communication process. After submitting a reservation, customers were able to view the status of their bookings and monitor whether their requests had been received, approved, or otherwise updated. In addition, the account management feature allowed customers to review and update their personal information and manage relevant account details. Overall, these client-side features provided customers with a more convenient, accessible, and organized way of managing their resort reservations.

For administrators, the client side provides access to management features such as:

For administrators, the client side of the Online Booking System provided access to various management features that supported the efficient operation of the resort. The dashboard access feature provided administrators with a centralized view of important system information, including reservation activities, accommodation availability, and other relevant operational data. This allowed administrators to monitor the overall status of the system more

conveniently.

The reservation monitoring feature enabled administrators to review, track, and manage customer reservations. They were able to view booking details, monitor reservation statuses, and process booking requests more systematically. The accommodation management feature allowed administrators to add, update, and manage accommodation information, including availability and status, ensuring that customers received accurate information when making reservations.

The user management feature enabled administrators to manage registered customer accounts and other authorized users of the system. Report generation was also provided to allow administrators to generate reservation, occupancy, and other relevant reports that supported monitoring and decision-making. In addition, system administration functions allowed authorized administrators to manage and maintain important system settings and operational information.

These management features provided administrators with a centralized and organized platform for handling reservation activities, accommodation records, user information, and other administrative responsibilities.

The client interface is designed using responsive web design principles to ensure compatibility across various devices and screen sizes. This allows users to enjoy a consistent and user-friendly experience regardless of the device being used.

Server Side

The server side serves as the core processing component of the Online Booking System. It is responsible for handling requests received from client devices, executing business logic, managing authentication processes, and communicating with the database.

Whenever a user performs an action, such as submitting a reservation request or logging into the system, the request is transmitted to the server. The server processes the request, performs the necessary operations, retrieves or updates data, and sends the appropriate response back to the client.

The server performs several important functions, including:

User Authentication

The server verifies user credentials during the login process and determines whether access should be granted or denied. Authentication ensures that only authorized users can access protected system resources.

Reservation Processing

The server processes reservation requests submitted by customers, validates reservation details, checks accommodation availability, and records booking information within the database.

Data Management

The server retrieves, updates, and organizes information stored within the database. It ensures that all data transactions are performed accurately and consistently.

Business Logic Execution

The server implements the operational rules and procedures that govern system functionality. These rules include availability checking, booking validation, reservation approval, and report generation.

Security Management

The server protects sensitive information through security measures such as password encryption, access control,

session management, and data validation.

Report Generation

The server automatically generates reports based on stored data, allowing administrators to monitor reservations, occupancy rates, customer records, and system activities.

Centralized Database

A centralized database serves as the primary repository of all information managed by the Online Booking System. The database ensures that data is stored securely, organized efficiently, and accessible whenever needed.

The database stores the following information:

Customer Information

- Full Name
- Contact Number
- Email Address
- Username
- Password
- Account Details

Reservation Records

- Reservation ID
- Booking Date
- Accommodation Type
- Number of Guests
- Reservation Status

Accommodation Details

- Cottage Information
- Room Information
- Pavilion Information
- Accommodation Capacity
- Pricing Information

Booking Schedules

- Reservation Dates
- Check-In Dates

- Check-Out Dates
- Occupancy Schedules

Reports and Transaction Records

- Reservation Reports
- Customer Reports
- Accommodation Reports
- Activity Logs
- Administrative Reports

The centralized database improves information management by reducing redundancy, increasing data accuracy, and enabling real-time access to reservation information.

User Interaction Flow

The user interaction flow describes the sequence of activities performed by customers and administrators when interacting with the Online Booking System. This flow illustrates how information moves through the system from initial access to reservation completion.

Step 1: Website Access

The customer accesses the resort website using a web browser and internet connection. The homepage provides information about the resort, available accommodations, amenities, services, and promotional offers.

Step 2: User Registration

New users are required to create an account by providing personal information such as their name, contact details, email address, username, and password.

The registration process allows the system to maintain customer records and facilitate future reservations.

Step 3: User Login

After registration, customers log into the system using their registered credentials. The authentication process verifies their identity and grants access to reservation features.

Step 4: Accommodation Browsing

Customers browse available cottages, rooms, and pavilions. Detailed accommodation information including pricing, descriptions, capacities, and amenities is displayed.

Step 5: Availability Checking

The system automatically checks accommodation availability based on the selected reservation dates and displays available options to the customer.

Step 6: Reservation Submission

The customer selects the desired accommodation and completes the reservation form by providing booking details such as:

- Reservation Date
- Number of Guests
- Preferred Accommodation
- Contact Information

The reservation request is then submitted electronically.

Step 7: Database Storage

The system stores all reservation information securely within the centralized database. The reservation status is recorded and made available for administrator review.

Step 8: Administrator Review

The administrator accesses the dashboard and reviews incoming reservation requests. The administrator verifies reservation details and checks accommodation schedules before approving or rejecting the booking.

Step 9: Reservation Confirmation

Once approved, the customer receives confirmation regarding the reservation status. The booking information is updated within the system and reflected in accommodation availability records.

Step 10: Reservation Monitoring

Administrators continuously monitor reservation schedules, accommodation occupancy, and customer records through the administrative dashboard.

Step 11: Report Generation

The system automatically generates reports that assist management in evaluating business performance, monitoring reservations, and making operational decisions.

Step 12: System Maintenance and Updates

Administrators maintain system information by updating accommodation details, modifying rates, managing user accounts, and ensuring the accuracy of stored records.

Through this interaction flow, the Online Booking System provides a seamless and efficient reservation experience while improving operational efficiency and customer satisfaction at Avianna’s Inland Resort.

3.4 Tools and Technologies Used

Programming Languages

The system is developed using a combination of programming languages that together provide a complete front-end and back-end development stack. HTML5 (Hypertext Markup Language) serves as the structural foundation of all web pages in the system, defining the content and organization of each page's elements. CSS3 (Cascading Style Sheets) provides the visual styling and layout rules that determine how HTML elements are displayed to users, including colors, typography, spacing, and responsive layout behavior. JavaScript adds dynamic interactivity to the web interface, enabling features such as form validation, availability calendar updates, and modal dialogs without requiring full page reloads. PHP is the server-side scripting language that powers the application layer, processing user requests, executing business logic, and generating dynamic HTML content.

Frameworks and Libraries

Bootstrap is used as the primary CSS framework, providing a responsive grid system, pre-styled UI components, and cross-browser compatibility that significantly accelerates the development of a professional, consistent visual interface. Bootstrap's mobile-first design philosophy ensures that the resort's website is fully usable on smartphones and tablets — an increasingly important consideration given the prevalence of mobile browsing among the resort's target audience.

Laravel is the PHP framework that structures the back-end development work. Laravel provides a comprehensive set of tools for modern web application development, including an elegant routing system, powerful middleware for request processing, robust authentication scaffolding, a database migration system for version-controlled schema management, Eloquent ORM for streamlined database interaction, and Blade templating for clean, maintainable view code. Laravel's security features, including built-in protection against cross-site request forgery (CSRF), SQL injection, and cross-site scripting (XSS) attacks, are particularly valuable for a system that will handle sensitive booking and personal data.

Database Management System

MySQL is used as the relational database management system for the project. MySQL is an open-source, widely adopted RDBMS known for its reliability, performance, and compatibility with the PHP and Laravel development ecosystem. The choice of MySQL reflects both its technical strengths and its practical advantages — it is well-supported by the XAMPP development environment used during development, is compatible with the web hosting environments where PHP/Laravel applications are typically deployed, and is familiar to the development team from prior coursework and projects.

Development Tools

Visual Studio Code is used as the primary integrated development environment (IDE) for the project. VS Code offers a rich feature set for web development, including syntax highlighting for all languages used in the project, intelligent code completion (IntelliSense), built-in Git integration for version control, a rich extension ecosystem, and an integrated terminal for running development commands. Its lightweight performance and extensive customization options make it well-suited to the collaborative development workflow employed by the team.

XAMPP (Cross-Platform, Apache, MariaDB, PHP, Perl) serves as the local development server environment, providing a complete web server stack that runs on the development team's local machines. XAMPP enables the team to develop and test the system in a local environment that closely mirrors the conditions of a production web server, without requiring internet access or the cost of cloud hosting during the development phase. phpMyAdmin, included as part of the XAMPP package, provides a web-based graphical interface for managing the MySQL database, facilitating the design, testing, and inspection of the database schema during development.

Category	Technology	Purpose
Front-End Language	HTML5	Web page structure
Front-End Language	CSS3	Visual styling and layout
Front-End Language	JavaScript	Dynamic interactivity
Back-End Language	PHP	Server-side processing
CSS Framework	Bootstrap	Responsive UI components
PHP Framework	Laravel	MVC architecture and back-end tools
Database	MySQL	Data storage and management
IDE	Visual Studio Code	Code writing and management
Dev Environment	XAMPP	Local web server
DB Management UI	phpMyAdmin	Database management interface

3.5 Data Gathering Techniques

Interviews

Personal interviews were conducted with Mr. JJ S. Layson, the owner and manager of Avainna's Beach Resort, as the primary data gathering method for understanding the resort's operational context, existing challenges, and system requirements. Both structured interviews with pre-defined question sets and semi-structured interviews allowing for more open-ended exploration were conducted at various stages of the project. Initial interviews focused on gathering a comprehensive picture of the resort's operations, revenue streams, staffing arrangements, and the specific pain points in the current manual processes. Follow-up interviews were conducted at key milestones in the development process to review design decisions, demonstrate prototypes, and gather feedback for system refinement.

The interview process generated rich qualitative data about the resort's operational environment that would not have been accessible through surveys or document analysis alone. Sir JJ's descriptions of his day-to-day management challenges — managing booking inquiries across multiple channels while also attending to his other businesses, the difficulty of tracking cottage availability during busy periods, the frustration of missed booking opportunities when customers could not reach him directly — provided the kind of concrete, experiential detail that informed the most important design decisions in the project.

Surveys

Surveys were administered to a sample of potential resort customers to gather quantitative data about their online booking preferences, their expectations for hospitality websites, and their likelihood of making reservations through an online platform. The survey instrument included both Likert-scale items, which allowed for quantitative measurement of attitude strength, and open-ended questions, which provided space for respondents to articulate preferences and concerns that the survey designers had not anticipated.

Survey results were analyzed to identify patterns in customer expectations and to inform decisions about the features and design of the public-facing components of the system. The survey data also provided baseline measurements against which the outcomes of user acceptance testing could be compared, enabling the researchers to assess whether the developed system met the expectations established prior to its construction.

Observation

Direct observation of resort operations was conducted during visits to Avainna's Beach Resort over the course of the study. The observation process focused on documenting the actual workflows involved in managing guest check-in, cottage assignment, inventory management, and reservation handling. Observational data provided an important corrective to purely interview-based data, as it allowed the researchers to verify that the processes described by Sir JJ in interviews reflected actual operational practices and to identify aspects of the operational environment that had not been explicitly described but were nonetheless relevant to system design.

Observation also provided the researchers with a visceral understanding of the resort's physical environment and guest experience — an understanding that informed the tone, imagery, and content decisions made in designing the public-facing website. Experiencing the resort as a visitor, with its natural setting, its mini zoo, its welcoming atmosphere, and its range of recreational options, gave the researchers an authentic perspective on the resort's appeal that enhanced their ability to design a website that genuinely represents and promotes the resort.

Document Analysis

Document analysis involved the systematic review of existing business records and materials maintained by the resort. These included handwritten booking logs, pricing sheets, promotional flyers, event planning records, and inventory lists. The analysis of these documents served several important functions in the research process. First, it provided concrete data about the format and content of the information that the system would need to manage — informing decisions about the database schema, the administrative interface design, and the data migration strategy. Second, it revealed the limitations and inefficiencies of the current manual recording systems — providing a clear picture of the specific operational improvements the digital system needed to deliver. Third, it provided samples of

the resort's existing promotional communication style that informed the tone and content of the website.

3.6 Testing Procedures

Testing is one of the most important phases in the development of the Avianna's Inland Resort Online Booking System. It serves as a quality assurance process that evaluates whether the developed system functions according to its intended purpose and satisfies the requirements identified during the analysis and design phases. The testing process helps ensure that the system is reliable, accurate, secure, efficient, and capable of meeting the needs of both customers and resort administrators.

The primary goal of testing is to identify errors, defects, vulnerabilities, and performance issues before the system is deployed for actual use. Through comprehensive testing procedures, the researchers can verify that each component of the system performs correctly, that all modules work together effectively, and that users can successfully complete their tasks without experiencing technical difficulties.

For the Avianna's Inland Resort Online Booking System, several testing activities were conducted throughout the development process to ensure the quality and effectiveness of the application. These testing activities focused on validating system functionality, verifying system integration, assessing usability, and determining whether the system fulfills user expectations.

The testing procedures employed in this study include Unit Testing, System Testing, and User Acceptance Testing (UAT).

Unit Testing

Unit Testing is the first level of software testing performed during the development phase. This type of testing focuses on evaluating individual units, components, functions, or modules of the system independently. The objective is to ensure that each part of the system performs its intended task accurately before it is integrated with other modules.

In software development, a unit refers to the smallest testable component of an application. Examples include functions, methods, forms, scripts, and database procedures. By testing these units separately, developers can identify errors early in the development process and prevent defects from affecting other components of the system.

For the Online Booking System, the researchers conducted unit testing on the major functionalities and modules of the application.

Testing of the Registration Module

The registration module was tested to verify that customers could successfully create accounts by providing valid personal information. The researchers evaluated whether the system properly validated user inputs, checked required fields, prevented duplicate registrations, and stored customer information accurately within the database.

The testing process included verifying:

The testing process included verifying the functionality and reliability of the user registration feature. Required field validation was tested to ensure that users could not submit the registration form without providing the necessary information. Email format validation was also performed to verify that users entered valid and properly formatted email addresses before their accounts were created.

Password requirements were tested to ensure that user passwords complied with the specified security criteria. The system was also tested for duplicate account prevention to determine whether it could identify and reject registration attempts using information that was already associated with an existing account. Successful account creation was verified to ensure that valid registration information resulted in the proper creation of a user account.

Database storage accuracy was tested to confirm that the information submitted during registration was correctly stored in the database and could be retrieved when needed. These tests helped ensure that the registration process

operated correctly, maintained data accuracy, and provided reliable account creation and information storage.

The researchers ensured that the registration process was secure, efficient, and capable of handling different input scenarios.

Testing of the Login Module

The login module was tested to determine whether users could successfully access the Online Booking System using valid credentials. The researchers evaluated the authentication process to ensure that legitimate users were properly authenticated and that unauthorized users were prevented from accessing protected areas of the application. The testing focused on the accuracy, reliability, and security of the login and authentication process.

Valid username and password verification was conducted to confirm that users with correct credentials were successfully granted access to the system. Invalid login detection was also tested to determine whether the system properly rejected incorrect or unrecognized credentials. Password matching validation was performed to ensure that the entered password corresponded correctly with the stored account credentials. The researchers also tested session creation to verify that an authenticated user was provided with an appropriate active session after successful login.

Logout functionality was tested to ensure that users could securely end their sessions and prevent unauthorized access from the same device. Access restriction enforcement was also evaluated to confirm that users without proper authentication could not directly access protected pages or administrative functions. These tests helped ensure that the login module provided reliable authentication, proper session management, and appropriate access control for the Online Booking System.

The results confirmed that the authentication system functioned properly and protected sensitive information from unauthorized access.

Testing of the Booking Module

The booking module was considered one of the most critical components of the Online Booking System because it handled the processing and recording of customer reservation transactions. The researchers conducted extensive testing to ensure that reservation requests were processed accurately, efficiently, and consistently. The tests focused on verifying that the information entered by customers was properly validated, recorded, and reflected in the system.

Accommodation selection was tested to ensure that customers could select an appropriate accommodation from the available options. Date selection was also tested to verify that customers could choose valid reservation dates and that the system properly considered the selected dates when determining accommodation availability. Guest information submission was evaluated to ensure that customer details were correctly entered and associated with the appropriate reservation.

Reservation confirmation was tested to determine whether valid booking requests were properly confirmed and communicated through the system. Booking status updates were also evaluated to ensure that changes in reservation status, such as pending, approved, or cancelled bookings, were accurately reflected in the system. Finally, reservation storage was tested to verify that completed booking information was correctly stored in the database and could be retrieved when needed.

Through these testing activities, the researchers were able to verify the reliability and accuracy of the booking module and ensure that reservation transactions were properly processed from accommodation selection through the storage and monitoring of booking information.

Testing ensured that reservation details were accurately recorded and properly linked to customer accounts.

Testing of Availability Monitoring

The availability monitoring module was tested to verify that accommodation availability was accurately updated whenever a reservation was made, approved, canceled, or modified. The researchers evaluated the module to ensure that the availability information displayed to customers and administrators remained consistent with the actual status

of each accommodation. This testing was important in preventing inaccurate availability information and reducing the possibility of reservation conflicts.

The system was tested to ensure that available accommodations were displayed correctly and could be selected by customers for their preferred reservation dates. Once an accommodation was reserved and the booking was properly processed, it was automatically marked as unavailable for the corresponding dates. The cancellation process was also tested to verify that canceled reservations restored the accommodation's availability, allowing other customers to reserve it when appropriate.

The researchers tested the system's ability to prevent booking conflicts by ensuring that an accommodation could not be reserved by multiple customers for the same date and period. Availability updates were also evaluated to determine whether changes were reflected in real time after reservation transactions were completed. Through these tests, the researchers verified that the availability monitoring module provided accurate and timely accommodation information and effectively supported the prevention of double bookings and scheduling conflicts.

This testing helped minimize the risk of double bookings and scheduling conflicts.

Testing of Report Generation

The report generation module was tested to determine whether the Online Booking System could produce accurate, complete, and organized reports based on the information stored in the database. The researchers evaluated the module to ensure that the generated reports reflected the actual reservation, customer, accommodation, and transaction records maintained by the system. The testing also verified whether the reports contained the necessary information and could be generated according to the required criteria.

Reservation reports were tested to ensure that booking details, reservation dates, accommodation information, and booking statuses were accurately presented. Customer reports were evaluated to verify that customer information was properly retrieved and displayed. Occupancy reports were also tested to determine whether they accurately reflected the number and status of occupied, reserved, and available accommodations.

Booking summaries were tested to provide an organized overview of reservation activities within a specified period. Transaction reports were evaluated to ensure that relevant booking and transaction information was correctly recorded and presented. Through these testing activities, the researchers verified that the report generation module provided reliable and organized information that could support administrative monitoring, record keeping, and data-driven decision-making.

The researchers verified that generated reports contained complete, accurate, and organized information useful for administrative decision-making.

Testing of Database Functions

Database testing was conducted to evaluate the accuracy, reliability, and consistency of how information was stored, updated, retrieved, and deleted within the Online Booking System. The researchers examined the database operations to ensure that customer records, accommodation information, reservation details, user accounts, and other relevant data were properly managed and maintained.

Data insertion was tested to verify that newly submitted information was correctly stored in the appropriate database tables. Data retrieval was evaluated to ensure that stored information could be accessed accurately whenever it was required by the system. Data modification was also tested to confirm that changes made to existing records were properly updated without affecting unrelated information. In addition, data deletion was tested to ensure that records intended for removal were properly deleted according to the system's functions and requirements.

Database connectivity was tested to verify that the application could consistently communicate with the database and perform required operations without interruption. Data consistency was also evaluated to ensure that information remained accurate and synchronized across different system modules. Through these testing activities, the researchers verified that the database provided reliable storage and management of information and supported the proper operation of the Online Booking System.

The researchers ensured that all database operations were performed correctly without compromising data integrity.

Importance of Unit Testing

Unit testing played a significant role in the software quality assurance process because it allowed the researchers to identify and correct defects at an early stage of system development. Each module of the Online Booking System was tested independently to determine whether its functions operated according to the specified requirements. By examining individual components separately, the researchers were able to isolate problems more easily and resolve them before the modules were integrated into the complete system.

The use of unit testing provided several advantages throughout the development process. It enabled early error detection, allowing defects to be identified before they affected other system components. It also contributed to improved code quality by ensuring that individual functions and modules operated correctly. Debugging became easier because identified problems could be traced to specific components rather than the entire system. As a result, development costs and time were reduced by addressing errors at an earlier stage.

Unit testing contributed to better system reliability by verifying the proper operation of individual components before integration. It also simplified system maintenance because errors or issues could be traced and addressed within specific modules without significantly affecting other parts of the application. Overall, unit testing helped ensure that the Online Booking System was developed with reliable, functional, and maintainable components.

Through comprehensive unit testing, the researchers ensured that all system modules functioned correctly before proceeding to the next testing phase.

System Testing

After completing unit testing and integrating all modules, the researchers conducted System Testing. System testing evaluates the behavior of the entire application as a complete and integrated system.

Unlike unit testing, which focuses on individual modules, system testing examines how all components interact with one another. The objective is to verify that the entire system functions according to the specified requirements and performs effectively in realistic operating conditions.

System testing is essential because even if individual modules work correctly, integration issues may arise when those modules interact within the complete application.

Reservation Process Testing

The researchers tested the entire reservation workflow from beginning to end to determine whether the different processes of the Online Booking System worked together correctly and consistently. The testing focused on verifying that information was properly transferred between the customer and administrator functions and that each stage of the reservation process was completed in the correct sequence.

The process began with customer registration, where users created accounts by providing the required information. After registration, the user login process was tested to ensure that customers could securely access their accounts using valid credentials. The accommodation browsing feature was then evaluated to verify that customers could view available accommodations and their relevant details.

Availability checking was tested to ensure that customers received accurate information regarding accommodations available for their selected dates. After selecting an appropriate accommodation, the reservation submission process was tested to verify that booking requests were properly entered and recorded in the system. The reservation was then forwarded to the administrator for review and approval.

Administrator approval was tested to ensure that authorized administrators could review booking requests and update their status accordingly. Finally, the confirmation notification process was tested to verify that customers received appropriate confirmation after their reservations were approved. Through this end-to-end testing, the researchers evaluated the complete reservation workflow and verified that each stage functioned properly and

communicated accurately with the succeeding process.

The objective was to ensure that all reservation activities flowed smoothly without interruptions or errors.

Administrative Function Testing

The administrative functions of the Online Booking System were evaluated to determine whether management tasks could be completed accurately, efficiently, and successfully. The researchers tested the different administrative features to ensure that authorized administrators were able to manage reservations, user accounts, facility information, reports, and other important system activities.

Reservation approval was tested to verify that administrators could review booking requests and approve valid reservations properly. Reservation cancellation was also evaluated to ensure that administrators could cancel reservations when necessary and that the corresponding booking status and accommodation availability were updated correctly. User management was tested to determine whether administrators could view, update, and manage registered user accounts according to their assigned responsibilities.

Facility management was also evaluated to ensure that administrators could add, update, and manage information regarding the resort's facilities and accommodations. Report generation was tested to verify that administrators could produce accurate reservation, occupancy, customer, and transaction reports for monitoring and decision-making purposes. Finally, dashboard monitoring was evaluated to determine whether administrators could view important system information and reservation activities through a centralized dashboard. These tests helped verify that the administrative functions effectively supported the resort's daily management and operational requirements.

The researchers confirmed that administrators could effectively manage resort operations using the system.

Database Integration Testing

Database integration testing was conducted to evaluate the interaction between the Online Booking System and its database. The researchers verified that the application properly communicated with the database during different operations and that information remained accurate and consistent throughout the process. This testing was important in ensuring that customer records, reservation information, accommodation details, and other system data were properly managed.

The researchers tested whether data entered through system forms were correctly stored in the appropriate database records. Retrieved information was also evaluated to ensure that the data displayed by the application matched the records stored in the database. Database updates were tested to verify that any changes made through the system were immediately and accurately reflected in the corresponding records.

Deleted records were tested to ensure that information removed through authorized system functions was properly deleted from the database. Database transactions were also evaluated to determine whether multiple related operations maintained data consistency and prevented incomplete or conflicting records. Through these tests, the researchers verified that the application and database worked together reliably and that information remained accurate, synchronized, and consistent across the Online Booking System.

This testing ensured the reliability of information throughout the system.

Security Testing

Security testing was conducted to evaluate the protection mechanisms implemented within the Online Booking System and to determine whether user information and system resources were adequately protected. The researchers examined different security features to identify potential weaknesses and ensure that only authorized users could access functions and information appropriate to their assigned roles.

User authentication was tested to verify that the system properly identified and authenticated registered users before granting access. Password security was also evaluated to ensure that user credentials were handled appropriately and that unauthorized individuals could not easily gain access to user accounts. Session management was tested to

determine whether user sessions were properly created, maintained, and terminated after logout or inactivity.

Input validation was also evaluated to ensure that information entered into forms was properly checked before being processed or stored by the system. This helped reduce the risk of invalid or potentially harmful input affecting system operations. Access control restrictions were tested to verify that users could only access functions and pages permitted by their assigned roles. Through these security testing activities, the researchers evaluated the system's ability to protect user accounts, reservation information, and other important data from unauthorized access and improper use.

The researchers attempted to identify potential vulnerabilities that could compromise system security.

Security testing is particularly important because the system handles sensitive customer information and reservation records.

Compatibility Testing

Compatibility testing was conducted to determine whether the Online Booking System functioned properly across different web browsers and internet-enabled devices. The researchers tested the system using commonly used browsers, including Google Chrome, Mozilla Firefox, and Microsoft Edge. The testing focused on verifying that the system's features, interfaces, and functions remained accessible and operational across the selected browsers.

The system was also tested on various devices, including desktop computers, laptops, tablets, and smartphones. The researchers evaluated the system's display, navigation, responsiveness, and functionality on each device to ensure that users could access and utilize the system regardless of the device being used. Particular attention was given to the proper presentation of pages, forms, buttons, accommodation information, and reservation features on different screen sizes.

Through compatibility testing, the researchers were able to identify and address potential browser- and device-related issues. The results helped ensure that the Online Booking System provided a consistent, accessible, and user-friendly experience across the tested browsers and devices.

The researchers verified that all pages, forms, menus, and functions displayed properly regardless of device type or browser.

Performance Testing

Performance testing was conducted to evaluate the efficiency, speed, and responsiveness of the Online Booking System when processing different user requests and system operations. The researchers assessed the system's performance to determine whether its functions could be completed within an acceptable period and whether the system remained responsive during regular use.

Page loading speed was evaluated to determine how quickly system pages and information were displayed to users. Database response time was also measured to assess how efficiently the system retrieved and processed stored information. Reservation processing speed was tested to determine how quickly booking requests could be submitted, processed, and recorded in the database.

Report generation speed was evaluated to determine how efficiently the system produced reservation, customer, occupancy, and transaction reports. Overall responsiveness was also assessed by observing how quickly the system responded to user actions such as navigating between pages, submitting forms, checking availability, and updating information. Through these performance testing activities, the researchers were able to identify potential delays and evaluate whether the Online Booking System operated efficiently and provided a responsive experience for its users.

The researchers monitored system performance to ensure a smooth user experience.

Error Handling Testing

Error handling testing was conducted to evaluate how the Online Booking System responded to incorrect,

incomplete, or unexpected user actions. The researchers assessed whether the system could properly identify errors, display appropriate messages, and prevent invalid information from being processed or stored. This testing helped ensure that users were guided toward correcting errors without negatively affecting the system's normal operations.

Empty form submission was tested to verify that the system prevented users from submitting forms without completing the required fields. Invalid email formats were also tested to determine whether the system properly identified and rejected incorrectly formatted email addresses. Incorrect passwords were tested during the login process to ensure that unauthorized access was prevented and that users received appropriate feedback when invalid credentials were entered.

The researchers also tested duplicate bookings to determine whether the system could detect attempts to reserve an already-booked accommodation for the same date or period. Invalid dates were tested to ensure that the system prevented users from selecting or submitting inappropriate reservation dates. Through these error handling tests, the researchers verified that the system responded appropriately to invalid inputs and unexpected actions while maintaining data accuracy, security, and system stability.

Appropriate error messages were displayed to guide users and prevent incorrect transactions.

Importance of System Testing

System testing was conducted to ensure that the entire Online Booking System functioned reliably as an integrated solution. The researchers evaluated the application as a complete system rather than testing individual modules separately. This process helped determine whether the different components, features, and processes worked together according to the specified requirements and operational objectives.

System testing provided several benefits throughout the evaluation process. It allowed the researchers to verify the complete functionality of the application and determine whether all major features operated properly when used together. It also helped identify integration issues that may not have been detected during individual module testing. By addressing these issues, the researchers improved the overall reliability and consistency of the system.

Furthermore, system testing contributed to a better user experience by ensuring that customers and administrators could perform their respective tasks without significant functional problems. It also increased system stability by identifying errors that could affect the application's overall operation. Through comprehensive system testing, the researchers were able to enhance the quality of the Online Booking System and ensure that it provided a functional, reliable, stable, and user-friendly solution for the resort's reservation and administrative needs.

Successful system testing demonstrated that the Online Booking System met its functional and technical requirements.

User Acceptance Testing (UAT)

User Acceptance Testing (UAT) was conducted as the final stage of testing before the system was officially implemented. During this phase, actual users evaluated the developed system to determine whether it had met their expectations, requirements, and operational needs.

The UAT was conducted to determine whether the system had been acceptable for actual use and whether it had effectively supported the daily operations of the resort. The evaluation also allowed the researchers to identify any issues or areas that needed improvement based on the experiences and feedback of the users.

For this study, the UAT involved individuals who represented the primary stakeholders of the system. The participants included the resort owner, resort administrators, resort staff, and potential customers. Each participant evaluated the system based on their respective roles and expected interactions with its features and functionalities.

The resort owner evaluated the system in terms of its ability to support overall resort management and operational requirements. The resort administrators assessed the administrative functions, record management, and other system controls. The resort staff evaluated the system based on its usability and effectiveness in assisting with their daily tasks and resort operations. Meanwhile, potential customers evaluated the system from the perspective of

prospective users, particularly its ease of use, accessibility, and overall user experience.

The feedback and results gathered from the participants were used to determine the overall acceptability of the system. The findings also served as a basis for identifying areas that required further improvement before the system was fully implemented.

Participants were given access to the system and instructed to perform common tasks.

Customer Evaluation Activities

During the User Acceptance Testing, the potential customers were asked to perform several activities to evaluate the system from a user's perspective. They first registered their accounts and logged into the system using their registered credentials. After logging in, they browsed the available resort facilities and reviewed the information provided by the system.

The participants then checked the availability of accommodations based on their preferred dates and proceeded to submit reservations through the system. After completing the reservation process, they viewed the booking confirmations to verify whether their reservation details had been properly recorded and displayed.

These activities allowed the potential customers to evaluate the system's usability, accessibility, and functionality throughout the reservation process. Their experiences and feedback were considered in determining whether the system had adequately met the needs and expectations of its intended users.

These activities allowed the researchers to observe how customers interacted with the platform.

Administrator Evaluation Activities

During the User Acceptance Testing, the administrators were asked to perform several activities to evaluate the system's administrative functions. They first accessed the dashboard to review the information and features available for managing the resort's operations. They then managed reservations by reviewing, updating, and processing booking records through the system.

The administrators also updated accommodation information to ensure that the details displayed in the system were accurate and up to date. They generated reports to evaluate the system's reporting functionality and to determine whether the necessary information could be retrieved and presented appropriately. In addition, they monitored booking records to verify that reservations were properly recorded and that relevant booking information could be accessed when needed.

These activities allowed the administrators to evaluate the system's functionality, accuracy, and effectiveness in supporting the management and monitoring of resort operations. Their observations and feedback were considered in determining the system's overall acceptability and identifying areas that required further improvement.

These activities evaluated whether management functions supported operational requirements.

Evaluation Criteria

Participants assessed the system based on the following criteria:

Functionality

Measures whether the system performs required tasks accurately and completely.

Usability

Evaluates ease of learning, navigation, and overall user-friendliness.

Reliability

Measures consistency and dependability of system performance.

Efficiency

Assesses speed, responsiveness, and productivity improvements.

Security

Determines whether user information and records are adequately protected.

Collection of Feedback

After completing the assigned tasks during the User Acceptance Testing, the participants completed evaluation questionnaires to provide feedback regarding their experience with the developed system. They evaluated the system based on its ease of use, appearance, functionality, and overall performance.

The participants also provided comments and suggestions regarding areas that they believed could be improved. Their feedback included observations about the system's interface, features, and usability during the performance of their assigned activities. In addition, they assessed their overall satisfaction with the system based on their experience using its functions.

The collected responses, comments, and suggestions were reviewed and used to determine the participants' level of acceptance of the system. The feedback also served as a basis for identifying improvements that could further enhance the system's usability, functionality, and overall user experience.

The researchers analyzed the collected feedback and implemented necessary revisions before finalizing the system.

Importance of User Acceptance Testing

User Acceptance Testing was considered an important part of the study because it provided direct validation from the intended users of the developed system. Through the evaluation process, the participants were able to assess whether the system had met their expectations, requirements, and operational needs based on their actual experiences while using its features and functions.

The UAT results helped verify the participants' satisfaction with the system and confirmed whether the identified business and operational requirements had been addressed. The testing process also helped identify usability issues and other areas that required improvement before the system was fully implemented.

The participation of the resort owner, administrators, staff, and potential customers increased stakeholder confidence in the developed system. Their feedback contributed to greater system acceptance by demonstrating that the system had been evaluated by its intended users. The findings also helped reduce implementation risks by allowing potential problems to be identified and addressed before the system was deployed for actual use.

Through User Acceptance Testing, the researchers confirmed that the Online Booking System was capable of meeting the needs and expectations of Avianna’s Inland Resort and its customers.

Summary of Testing Procedures

The testing procedures conducted in this study ensured that the Avianna’s Inland Resort Online Booking System met established quality standards before deployment. Unit Testing verified individual modules, System Testing evaluated the integrated application, and User Acceptance Testing validated the system from the perspective of actual users.

The results of these testing activities demonstrated that the developed system was functional, reliable, efficient, secure, and user-friendly. Through comprehensive testing and continuous improvement, the researchers were able to develop a high-quality Online Booking System capable of supporting the reservation management needs of Avianna’s Inland Resort and enhancing the overall customer experience.

3.7 Evaluation Criteria

The developed system was evaluated against five quality criteria drawn from international software quality standards, specifically ISO/IEC 9126, which provides a widely recognized framework for assessing software quality across multiple dimensions. These criteria were selected for their direct relevance to the functional and user experience objectives of the system.

Functionality

Functionality refers to the degree to which the system provides the features and functions required by its users and meets the stated specifications. A functionally complete system is one in which all specified features work as intended, all user workflows can be completed without encountering errors or dead ends, and the system correctly processes all expected input types and produces the correct outputs. Functionality is the most fundamental quality dimension — a system that is visually attractive and easy to use but fails to perform its core functions is not fit for purpose.

Functionality was evaluated through the unit and system testing phases, which verified that each function of the system produced the correct outputs for both standard and edge-case inputs. UAT participants also assessed functionality by attempting to complete representative tasks and rating the degree to which the system enabled them to accomplish their objectives.

Usability

Usability refers to the ease with which users can learn to use the system, navigate its interface, and accomplish their tasks without confusion or frustration. A highly usable system is one where users can find the information they need quickly, complete transactions with minimal steps, understand system feedback and error messages clearly, and develop proficiency in system use with minimal formal training. Usability is particularly important for the public-facing components of this system, where customers with no prior familiarity with the resort's digital platform must be able to complete a booking independently.

Usability was evaluated primarily through UAT, where participants' navigation behaviors were observed and their subjective ratings of ease of use were collected. The usability evaluation considered both the learnability of the system for first-time users and the efficiency of use for users who had developed familiarity with the system over multiple interactions.

Reliability

Reliability refers to the consistency and dependability of the system's performance across time and across different usage conditions. A reliable system is one that produces consistent results for equivalent inputs, maintains data integrity across all operations, handles unexpected inputs or conditions gracefully without crashing or producing incorrect results, and recovers cleanly from any errors that do occur. Reliability is essential for a booking system,

where inconsistent behavior or data errors could result in double bookings, missed reservations, or customer data loss — all of which carry serious reputational and operational consequences for the resort.

Reliability was evaluated through system testing, which included scenarios designed to test the system's behavior under boundary conditions, with invalid inputs, and following simulated error conditions. The system's error handling and data validation mechanisms were specifically scrutinized during this phase.

Efficiency

Efficiency refers to the system's ability to perform its functions within acceptable time frames and with reasonable resource consumption. An efficient system is one where pages load quickly even on standard internet connections, forms process and submit without noticeable delays, search and availability checking operations complete within a time frame that does not frustrate users, and the database queries underlying these operations are optimized to minimize server load. Efficiency is important for the booking system in particular, where slow response times can cause user abandonment and frustration.

Efficiency was evaluated through performance testing during the system testing phase, which measured page load times, form processing times, and database query execution times under simulated usage conditions. The results informed optimization efforts in the final stages of development.

Security

Security refers to the system's ability to protect sensitive data and system functions from unauthorized access, data breaches, and malicious attacks. Security is a particularly critical consideration for a system that handles personal customer data, reservation records, and administrative access to all aspects of the resort's operations. The key security requirements for this system include secure user authentication, role-based access control, protection against common web application vulnerabilities (including SQL injection, XSS, and CSRF attacks), and secure handling of all data in transit.

Security was evaluated through targeted security testing during the system testing phase, which included attempts to exploit common vulnerability types against the deployed prototype. The results were used to verify that the security controls implemented in the application — including Laravel's built-in security features and additional application-level controls developed by the team — were functioning as intended.

SUPPORTING DIAGRAMS (Included within Chapter 3)

Data Flow Diagram (DFD – Level 0)

Customer → Online Booking System → Database

Administrator → Online Booking System → Database

Data Flow Diagram (DFD – Level 0)

The Data Flow Diagram (DFD) Level 0, also known as the **context diagram**, provides a high-level overview of the proposed Avianna's Inland Resort Online Booking System. It illustrates how external entities interact with the system and how data flows between users and the central system. This diagram simplifies the entire system into a single process to show its overall function and boundaries.

In this level, the system is represented as one main process called the **Online Booking System**, which interacts with two primary external entities: the **Customer** and the **Administrator**. Both entities exchange data with the system, which is then processed and stored in the **Database**.

1. Customer Interaction

The customers were considered the primary users of the system. They interacted with the system to access resort information and perform online reservation transactions. Through the system, the customers were able to view relevant information about the resort and submit their booking requests conveniently.

The customers viewed resort details, including available amenities, rates, and facilities. They also submitted booking requests for cottages, rooms, and event spaces based on their preferred schedule. During the reservation process, they provided the necessary personal information, such as their name, contact number, and preferred booking date and schedule.

After submitting their booking requests, the customers received booking confirmations and status updates through the system. These interactions allowed the customers to access resort information, complete the reservation process, and monitor the status of their bookings.

Once the customer submits a booking request, the system processes the information and stores it in the database for record keeping and availability checking.

2. Administrator Interaction

Administrator Interaction

The administrator was the authorized user responsible for managing and supervising the overall operations of the Online Booking System. As the primary system manager, the administrator played an important role in ensuring that reservation transactions, customer records, accommodation schedules, and resort information were properly maintained and updated. The administrator served as the central authority who oversaw the day-to-day operation of the system and ensured that the data stored in the database remained accurate, secure, and reliable.

Unlike customers, who primarily accessed booking-related services, the administrator was given higher-level privileges that allowed the management of functions essential to the operation of the resort. Through the administrative dashboard, the administrator accessed different system modules that supported reservation management, accommodation monitoring, customer record maintenance, and report generation.

One of the primary responsibilities of the administrator was managing customer reservations. Whenever a customer submitted a booking request, the reservation information was stored in the system database and became accessible through the administrative dashboard. The administrator reviewed the reservation details to ensure that the information provided by the customer was complete and accurate. After reviewing the request, the administrator approved, rejected, or modified the reservation depending on accommodation availability and the resort's policies.

The administrator also monitored booking schedules to prevent scheduling conflicts and double bookings. Through the booking management module, the administrator viewed reservation schedules, occupancy information, and accommodation availability. This function helped the administrator properly allocate resort facilities and maintain organized booking records.

Another important responsibility of the administrator was updating resort information. Since accommodation rates, promotional offers, facility descriptions, and operating schedules could change periodically, the administrator was authorized to modify the information displayed in the system. This ensured that customers were provided with accurate and updated information when making reservations.

The administrator was able to update various types of resort information through the system. These included cottage rates and descriptions, room rates and amenities, pavilion rates and capacities, resort announcements, promotional offers and discounts, operating schedules, facility availability, contact information, and resort policies and guidelines. By managing and regularly updating these details, the administrator ensured that the information presented in the system remained accurate, complete, and up to date. This allowed customers to access reliable information when viewing resort facilities and making reservations through the online booking system.

By regularly updating these details, the administrator helped maintain the accuracy, reliability, and credibility of the resort's online booking platform. The administrator's functions also contributed to the efficient management of reservations and resort information, which supported the overall operation of the system.

In **addition** to managing reservations, the administrator was responsible for maintaining customer records stored within the system. The customer management module allowed the administrator to access customer profiles, review reservation histories, verify account information, and manage user accounts. Proper management of customer information helped improve customer service and enabled the resort to maintain accurate and organized business records.

The administrator also supervised inventory and accommodation management through the system. The administrator monitored the availability of cottages, rooms, and pavilions to ensure that accurate accommodation information was maintained. Whenever a reservation was approved, the system automatically updated the accommodation schedules and occupancy records. The administrator was also able to manually adjust accommodation availability when facilities underwent maintenance, required repairs, or were temporarily closed.

The inventory management process involved monitoring accommodation occupancy, updating room availability, managing cottage schedules, tracking pavilion reservations, recording maintenance activities, and updating facility statuses. Through these functions, the administrator was able to maintain accurate accommodation records and

ensure that availability information remained updated and readily accessible for customer inquiries and reservation transactions. This helped prevent scheduling conflicts and supported the efficient management of the resort's facilities.

Another significant responsibility of the administrator was generating reports through the Online Booking System. The system automatically compiled reservation data and transaction records into organized reports that assisted the resort management in evaluating its overall performance. These reports provided relevant information that supported planning, decision-making, monitoring, and operational improvements.

The system generated various types of reports, including daily reservation reports, weekly reservation reports, monthly booking reports, customer information reports, accommodation utilization reports, occupancy rate reports, revenue and transaction reports, and activity monitoring reports. These reports enabled the management to monitor booking trends, identify peak and low seasons, evaluate the utilization of resort facilities, and review transaction activities. The information gathered from these reports also helped management make informed decisions regarding resort operations, resource allocation, and future planning.

The administrator was responsible for maintaining the accuracy and consistency of the database. Since the database served as the primary repository for reservation and customer information, it was important that all records remained complete, organized, and free from errors. The administrator regularly reviewed the stored data, corrected inconsistencies, updated records when necessary, and ensured that information was properly maintained.

The data maintenance activities included verifying customer information, updating reservation records, correcting data entry errors, removing duplicate records, monitoring database integrity, and managing user accounts and permissions. These activities helped ensure that the information stored in the system remained accurate, consistent, and reliable. Proper database management contributed significantly to the reliability and operational efficiency of the Online Booking System by ensuring that administrators could access accurate information when performing reservation management, customer management, reporting, and other administrative functions.

The administrator also monitors overall system performance and security. Through the administrative dashboard, administrators can track user activities, monitor login attempts, review system logs, and ensure that security measures are functioning correctly. These monitoring activities help protect sensitive customer information and prevent unauthorized access to the system.

Administrator serves as the central authority responsible for coordinating, monitoring, and managing all activities within the Online Booking System. Through effective reservation management, accommodation monitoring, customer record maintenance, report generation, and database administration, the administrator ensures the smooth operation of the resort's online services. The administrator’s interaction with the system is essential for maintaining operational efficiency, improving customer service, and supporting the successful implementation of the Online Booking System for Avianna’s Inland Resort.

3. Online Booking System (Main Process)

The Online Booking System serves as the central processing unit and core component of the proposed Avianna’s Inland Resort Online Booking System. It functions as the primary platform that receives, processes, stores, and manages information exchanged between customers, administrators, and the database. As the heart of the entire application, the system acts as an intermediary that facilitates communication among all components while ensuring that transactions are performed accurately, efficiently, and securely.

The system receives input data from both customers and administrators. Once data is submitted, the system processes the information according to predefined rules and procedures before storing it in the centralized database. The system also retrieves information from the database whenever requested and provides appropriate responses to users.

As the main process within the Data Flow Diagram, the Online Booking System performs several critical functions necessary for the successful operation of the resort’s booking services.

User Input Validation

One of the primary functions of the Online Booking System was validating the information submitted by users. Before processing any transaction, the system verified whether all required fields had been completed and whether the information provided complied with the necessary requirements. This validation process helped ensure that only accurate and appropriate information was accepted and stored in the system.

The system checked whether all required fields had been completed, verified the proper format of email addresses, validated contact numbers, ensured that usernames were unique, and confirmed that passwords met the required criteria. It also checked the validity and format of booking dates and prevented duplicate reservation entries. These validation procedures were applied during account registration, login, and reservation transactions to minimize errors and ensure the accuracy of user-submitted information.

The input validation process helped maintain the accuracy and consistency of the data stored in the database. It also prevented incomplete, invalid, or duplicate information from being recorded in the system. Through these validation measures, the Online Booking System maintained database integrity and improved the overall reliability and efficiency of the application.

Reservation Processing

The Online Booking System managed the complete reservation process from the submission of a booking request to its confirmation. Whenever a customer submitted a booking request, the system processed the provided information and verified the reservation details before recording the transaction. This process helped ensure that reservations were properly documented and that the information submitted by customers was accurate and complete.

The reservation process involved receiving booking requests, verifying customer information, checking accommodation availability, recording reservation details, updating booking schedules, and generating reservation records. After the reservation information had been processed, the booking request was forwarded to the administrator for review and appropriate action. The administrator then reviewed the reservation based on the customer's information, selected accommodation, preferred schedule, and availability.

Through this process, the system helped organize reservation transactions and maintain updated booking records. It also reduced the possibility of scheduling conflicts and duplicate reservations by checking accommodation availability before recording the booking. The reservation processing function contributed to the efficient management of customer bookings and supported the overall operations of the resort.

By automating reservation processing, the system significantly reduces the time required to handle booking requests and improves overall service efficiency.

Availability Checking

One of the most important functions of the Online Booking System was monitoring the availability of cottages, rooms, and pavilions. Before accepting a reservation, the system automatically compared the requested booking schedule with the existing reservations stored in the database. This process allowed the system to determine whether the selected accommodation was available for the customer's preferred date and schedule.

The availability checking process helped prevent double bookings and avoid scheduling conflicts by comparing new booking requests with existing reservation records. It also maintained accurate occupancy records and provided updated accommodation status to users. Through this function, the system supported more efficient reservation management and reduced the possibility of errors caused by overlapping bookings.

Customers were able to view the availability of accommodations through the system without waiting for manual confirmation from resort personnel. This provided customers with more convenient access to updated accommodation information and allowed them to make reservation decisions based on the available cottages,

rooms, and pavilions.

User Account Management

The Online Booking System managed customer and administrator accounts through authentication and authorization procedures. User account management allowed individuals to access system features and services based on their assigned roles and permissions. This function helped ensure that users could only access the information and functionalities appropriate to their respective accounts.

The system handled user registration, login, password management, profile updates, access control, and account verification. During registration, users provided the required account information, which was validated before an account was created. The login process authenticated users before granting access to the system. Users were also able to manage their passwords and update relevant profile information when necessary. Access control mechanisms were applied to restrict users from accessing functions that were not authorized for their assigned roles, while account verification helped confirm the validity of user accounts.

Through these functions, the system maintained controlled access to its resources and protected customer and administrative information. The user account management process contributed to the security, reliability, and proper operation of the Online Booking System by ensuring that only authorized individuals could access protected system features and data.

Data Storage and Retrieval

The Online Booking System communicated directly with the database to store, manage, and retrieve information required for its various functions. Every reservation, customer profile, accommodation record, and administrative transaction was securely maintained within the database. The centralized storage of information allowed the system to organize and manage important records efficiently.

The database stored various types of information, including customer profiles, reservation records, accommodation information, booking schedules, system logs, and transaction histories. These records were maintained to ensure that the necessary information was available whenever it was required by customers or administrators. The system also updated stored records whenever changes were made to reservations, customer information, accommodation availability, or other system transactions.

Whenever users requested information, the system retrieved the appropriate data from the database and displayed it through the user interface. This process allowed customers and administrators to access relevant information without manually searching through separate records. The centralized approach to data storage and retrieval supported efficient information management, improved data accessibility, and provided quick access to important records needed for reservation and administrative activities.

Administrative Processing

The Online Booking System supported administrative activities by providing tools that assisted resort management in monitoring, managing, and controlling daily operations. These functions allowed the administrator to perform important management tasks through the administrative dashboard and ensured that reservation and resort information was properly maintained.

The administrative actions processed by the system included reservation approval, reservation cancellation, accommodation updates, customer management, content management, report generation, and inventory monitoring. The administrator used the system to review and approve reservation requests, cancel bookings when necessary, and update accommodation information based on availability and resort operations. The system also supported the management of customer records and resort content, allowing the administrator to maintain accurate and updated information.

The system provided report generation and inventory monitoring functions that assisted the administrator in

reviewing operational data and monitoring the availability of resort facilities. These administrative functions helped streamline management tasks, reduce manual work, maintain organized records, and improve the overall productivity and efficiency of resort operations.

Notification and Confirmation Generation

To improve communication between customers and resort management, the Online Booking System automatically generated notifications and confirmations whenever important reservation-related events occurred. These automated messages provided users with timely information regarding their transactions and helped reduce the need for manual communication between customers and resort personnel.

The system generated reservation confirmations, approval notifications, reservation status updates, cancellation notices, administrative alerts, and system reminders. Reservation confirmations were provided after booking transactions had been recorded, while approval notifications informed customers when their reservation requests had been reviewed and approved. Reservation status updates provided information regarding changes to the booking, while cancellation notices informed users when reservations had been cancelled. Administrative alerts and system reminders were also generated to provide relevant information to administrators and users when necessary.

Through these automated notifications and confirmations, the system helped ensure that customers and administrators remained informed about the status of reservation transactions. This functionality improved communication, reduced delays in providing updates, and contributed to a more organized and efficient reservation process.

Report Generation

The Online Booking System included a reporting feature that automatically compiled and organized relevant data into meaningful reports. This feature assisted administrators in monitoring resort operations, reviewing business performance, and making informed decisions based on the information recorded in the system.

The reports generated by the system included daily reservation reports, weekly reservation reports, monthly booking reports, occupancy reports, customer reports, revenue reports, and accommodation utilization reports. These reports provided administrators with organized information regarding reservation activities, customer records, occupancy levels, revenue, and the utilization of resort facilities.

Through the reporting feature, administrators were able to review reservation trends, monitor accommodation occupancy, evaluate facility utilization, and assess revenue-related information. The generated reports supported management in identifying operational patterns and making appropriate decisions regarding resort operations, resource allocation, and future planning. This functionality also reduced the need for manual compilation of records and improved the efficiency of administrative reporting.

The reporting function eliminates the need for manual report preparation and improves management efficiency.

Security Management

Security was an essential responsibility of the Online Booking System because the application stored customer, reservation, and administrative information that required protection from unauthorized access. The system implemented several security mechanisms to protect stored information and ensure that users could only access functions appropriate to their assigned roles.

The security functions included user authentication, password encryption, role-based access control, data validation, session management, and activity monitoring. User authentication verified the identity of individuals before they were granted access to the system, while password encryption helped protect user credentials from unauthorized disclosure. Role-based access control restricted access to specific features based on the user's assigned role. Data validation helped prevent invalid or potentially harmful information from

being stored in the system, while session management controlled user sessions and helped prevent unauthorized access to active accounts. Activity monitoring also allowed administrative activities and system transactions to be tracked when necessary.

These security measures helped maintain the confidentiality, integrity, and availability of information stored within the system. By implementing these security functions, the Online Booking System provided a more secure environment for managing customer accounts, reservation records, accommodation information, and administrative transactions.

System Response Generation

After processing user requests, the Online Booking System generates appropriate responses and sends them back to customers or administrators. Responses may include confirmation messages, error notifications, availability results, account updates, and report outputs.

The response generation process ensures that users receive immediate feedback regarding their actions and transactions within the system.

Importance of the Online Booking System as the Main Process

The Online Booking System serves as the backbone of the entire application because it coordinates all activities performed within the platform. It connects customers, administrators, and the database into a unified system capable of managing reservation operations efficiently.

Without the Online Booking System, communication between users and stored information would not be possible. The system ensures that data flows correctly between components while maintaining accuracy, consistency, and security.

The system contributes to the digital transformation of Avianna's Inland Resort by automating manual processes and providing a more convenient, accessible, and reliable reservation platform.

In summary, The Online Booking System functions as the central processing component responsible for validating user inputs, managing reservations, checking accommodation availability, storing and retrieving information, processing administrative actions, generating notifications and reports, and ensuring the overall security and efficiency of resort operations. Through these functions, the system supports effective reservation management and enhances the overall customer experience.

Database Interaction

The Database serves as the central storage repository of the proposed Avianna's Inland Resort Online Booking System. It is one of the most important components of the system because it stores, organizes, manages, and retrieves all information necessary for the daily operation of the resort. The database acts as the foundation of the entire application, ensuring that all data entered into the system is securely maintained and readily available whenever needed.

In a computerized information system, the database functions as a centralized location where all records are stored electronically. Rather than relying on paper-based documents, handwritten logs, spreadsheets, or manual filing systems, the Online Booking System utilizes a database to manage information efficiently and accurately. This centralized approach improves data accessibility, minimizes redundancy, reduces human errors, and enhances overall operational performance.

The database is designed to support the various functions of the Online Booking System, including reservation processing, customer management, accommodation monitoring, report generation, and administrative operations. Every transaction performed within the system is recorded and stored in the database to ensure data consistency and proper record keeping.

The proposed system utilizes a relational database management system (RDBMS), specifically MySQL, which allows information to be organized into related tables. This structure enables efficient storage, retrieval, updating, and management of records while maintaining data integrity and security.

Customer Information Storage

One of the primary functions of the database in the proposed Online Booking System was the storage and management of customer information. Whenever a customer registered for an account or submitted a reservation request through the website, the information provided was processed and securely stored in the database. This centralized storage mechanism allowed the resort to maintain organized, complete, and updated customer records that could be accessed whenever necessary.

Unlike traditional manual record-keeping methods, in which customer information was recorded on paper forms or reservation logbooks, the computerized database reduced the risk of misplaced documents, illegible handwriting, duplicate records, and accidental loss of important information. Each customer was assigned a unique Customer Identification Number (Customer ID), which served as the primary identifier throughout the system. This unique identifier ensured that each customer account remained distinct and helped prevent the creation of duplicate customer records.

The customer information stored in the database included the Customer Identification Number (Customer ID), full name, contact number, email address, username, encrypted password, residential address, date of registration, and account status. The account status indicated whether the customer's account was active, inactive, or suspended. These records allowed the system to properly identify customers, manage their accounts, and associate their personal information with their corresponding reservation transactions.

The centralized management of customer information improved the organization and accessibility of customer records. It also supported other system functions, such as reservation processing, customer account management, reporting, and administrative monitoring, while helping ensure that customer information remained consistent and properly maintained.

The database also stores additional information related to customer activities, including reservation history, transaction records, login activities, and account updates. Maintaining a complete reservation history enables administrators to review previous bookings, identify frequent customers, resolve customer concerns, and provide more personalized services.

Customer information is continuously updated whenever users modify their profiles or submit new reservation requests. The database automatically reflects these changes, ensuring that administrators always have access to the most accurate and current customer information.

Storing customer information in a centralized database improves communication between customers and resort management. Since customer contact details are readily available, administrators can easily send reservation confirmations, booking reminders, promotional announcements, and important notifications regarding resort services.

The system also protects customer privacy by implementing password encryption, authentication mechanisms, access restrictions, and database security protocols. Sensitive information is only accessible to authorized personnel, ensuring compliance with data privacy principles and protecting customer confidentiality.

The Customer Information Storage component provides an organized, secure, and reliable repository for managing customer records, significantly improving administrative efficiency and enhancing customer service.

Reservation Records Storage

Reservation records represented the core business transactions of the Online Booking System. Every reservation submitted through the website was processed and stored in the reservation database. This

centralized storage allowed the resort management to efficiently organize, monitor, retrieve, and analyze booking information while reducing the limitations associated with manual reservation logbooks.

Each reservation was assigned a unique Reservation Number or Booking ID, which served as its primary identifier. This unique reference number allowed customers and administrators to locate reservation records quickly and accurately whenever necessary. The reservation records stored in the database included the Reservation Number, Customer ID, reservation date, check-in date, check-out date, accommodation type, selected facility, number of guests, reservation status, payment information, booking remarks, and administrator approval details.

Every reservation submitted by a customer progressed through different stages depending on the actions taken by the administrator and the status of the booking. These stages included pending approval, confirmed reservation, cancelled reservation, completed stay, and expired booking. The reservation status was updated in the system whenever administrators performed actions related to the booking. This allowed customers and administrators to monitor the current status of each reservation accurately.

The use of electronic reservation records provided several operational advantages compared with traditional paper-based methods. Administrators were able to retrieve reservation details quickly, search for bookings using different criteria, review customer booking histories, and monitor accommodation schedules without manually checking multiple logbooks. This improved the efficiency and accuracy of reservation management.

The reservation database also served as an important source of information for generating management reports. Daily reservation reports, monthly occupancy summaries, cancellation reports, customer statistics, and accommodation utilization reports were generated using reservation information stored in the database. Since each reservation transaction was documented and retained, the database also provided historical information that supported future planning, forecasting, and business analysis.

The storage of reservation records served as a fundamental component of the Online Booking System. It ensured that reservation transactions were properly documented, securely stored, organized, and readily available for retrieval. This supported efficient reservation management and contributed to the overall reliability and effectiveness of the system.

Accommodation Information Storage

The Accommodation Information Storage module managed information regarding the facilities offered by Avianna's Inland Resort. These facilities included cottages, overnight rooms, pavilions, event areas, and other rentable spaces available for customer reservations. The module served as a centralized source of information that supported both facility management and customer access to accommodation details.

Instead of maintaining separate paper records for each accommodation, the Online Booking System stored all facility information within a centralized database. This allowed administrators to manage resort facilities more efficiently while enabling customers to access updated accommodation information through the website. The centralized storage also helped ensure that facility information remained organized, consistent, and readily available whenever needed.

The accommodation records stored in the database included the Accommodation ID, accommodation name, accommodation category, facility description, maximum capacity, rental rate, availability status, facility features, images, and maintenance status. These details provided customers with relevant information when selecting accommodations based on their preferences, group size, budget, and intended use. Information regarding capacity, amenities, rental rates, and facility features was displayed through the website to assist customers in making informed reservation decisions.

Whenever the administrator updated accommodation information, such as changing rental rates, adding new facilities, uploading photographs, or temporarily marking an accommodation as unavailable due to

maintenance, the changes were reflected in the database and made available throughout the system. This helped ensure that customers received accurate and updated information before submitting their reservation requests.

The centralized accommodation database also simplified the management of resort facilities by allowing administrators to organize available resources and maintain updated availability schedules. Through this module, the system supported efficient facility management, improved information accuracy, and provided customers with reliable accommodation information during the reservation process.

Inventory and Availability Management

Inventory and Availability Management was one of the important functions performed by the database of the Online Booking System. The module was designed to continuously monitor the occupancy and availability of the resort's facilities. It provided administrators with updated information regarding the status of cottages, rooms, pavilions, and other facilities that were available for reservation.

Whenever a reservation was approved, the system automatically updated the inventory database. The selected accommodation was marked as unavailable for the dates covered by the reservation. When the reservation period ended or a reservation was cancelled, the system restored the accommodation's availability based on the updated booking status. This process helped maintain accurate availability records and reduced the possibility of overlapping reservations.

The inventory management database stored information regarding available cottages, occupied cottages, available rooms, occupied rooms, pavilion reservations, event schedules, facility maintenance records, and the availability calendar. These records allowed administrators to monitor the current status of resort facilities and determine which accommodations were available, occupied, reserved, or temporarily unavailable.

The inventory module enabled administrators to monitor accommodation utilization through an organized scheduling system. The real-time updating of inventory records helped prevent multiple customers from reserving the same accommodation for overlapping dates. This functionality also allowed resort personnel to monitor facility usage and make appropriate adjustments based on reservation activities.

The availability calendar further assisted resort personnel in planning maintenance schedules, facility renovations, and other operational activities without unnecessarily disrupting existing customer reservations. Through effective inventory and availability management, the system helped maximize the utilization of resort facilities, improve operational planning, minimize scheduling conflicts, and provide customers with accurate and updated availability information.

Administrative Records Storage

The Administrative Records Storage component managed information related to system administrators and resort personnel who were authorized to access the administrative dashboard. Each administrator had a secure user account containing identification information and assigned system permissions. These records allowed the system to properly identify authorized personnel and control their access to administrative functions.

The administrative records stored in the database included the Administrator ID, full name, username, encrypted password, position, user role, access level, account status, login history, and administrative activities. The Administrator ID served as the unique identifier for each administrative account, while the assigned user role and access level determined the system features and functions that could be accessed by each administrator.

Different administrators were assigned different levels of authority based on their responsibilities within the resort. Managers were provided with broader administrative privileges, while staff members were given access to functions related to their assigned tasks, such as reservation and customer record management. The role-based access control mechanism ensured that administrators could only access system features appropriate

to their respective responsibilities.

The administrative records also supported accountability by documenting activities performed within the system. Whenever administrators approved reservations, modified accommodation information, generated reports, or updated customer records, these activities were recorded in the database. Login history and administrative activities provided additional information that could be reviewed when monitoring system usage and administrative transactions.

Maintaining comprehensive administrative records promoted transparency and improved operational control within the Online Booking System. These records also facilitated system auditing by providing information about user access and administrative activities. Through proper management of administrative records, the system supported secure access, accountability, and efficient administration of resort operations.

System Logs and Activity Monitoring

The System Logs and Activity Monitoring component provided continuous documentation of important activities performed within the Online Booking System. Every significant user interaction generated an activity record that was automatically stored in the database. These records provided a chronological history of system activities and allowed authorized personnel to monitor user interactions and administrative transactions.

The activity logs recorded various events, including user login and logout activities, account registration, reservation submissions, reservation approvals, reservation cancellations, profile updates, password changes, administrative actions, database modifications, and system errors. Each recorded activity provided information that could assist administrators in monitoring system usage and reviewing transactions performed by users and authorized personnel.

The activity logs enabled administrators to monitor system usage, investigate operational issues, identify potentially unauthorized activities, and troubleshoot technical problems. They also provided useful information for evaluating overall system performance and identifying areas that required improvement. By reviewing recorded activities and system errors, administrators were able to determine possible causes of problems and take appropriate corrective actions.

Comprehensive activity monitoring enhanced accountability by maintaining a record of important actions performed within the system. It also strengthened database security and operational transparency by allowing authorized personnel to review system activities when necessary. Through this functionality, the Online Booking System maintained a more organized and traceable record of user and administrative interactions.

Besides storing information, the database played an essential role in retrieving and processing data whenever users interacted with the Online Booking System. Whenever customers browsed accommodations, searched for available dates, logged into their accounts, reviewed reservation histories, or submitted booking requests, the application communicated with the database to retrieve the necessary information. The retrieved data was then processed and displayed through the system interface based on the user's request.

Similarly, administrators continuously retrieved information while performing various management tasks. These activities included viewing reservations, monitoring facility schedules, searching customer records, updating accommodation information, generating reports, and monitoring dashboard statistics. The system retrieved the required records from the database and presented the information in an organized manner to support administrative decision-making and daily resort operations.

Efficient database retrieval improved the overall responsiveness of the Online Booking System. Optimized database queries reduced waiting times and allowed customers and administrators to perform transactions more quickly and conveniently. The efficient retrieval and processing of information also reduced the need for manual searching and improved access to important records.

Accurate data retrieval supported informed decision-making by ensuring that the information displayed by the

system reflected the latest updates stored in the database. Through efficient data retrieval and processing, the system provided users with timely, accurate, and relevant information while supporting the effective management of resort reservations and operations.

Maintaining data integrity and consistency was one of the fundamental responsibilities of the database management system. Data integrity referred to the accuracy, validity, completeness, and reliability of all information stored within the database. The Online Booking System applied several database mechanisms to ensure that information remained accurate and consistent throughout the different system processes.

The system maintained data integrity through the use of primary keys, foreign keys, validation rules, required fields, referential integrity, and constraint enforcement. Primary keys uniquely identified records within database tables, while foreign keys established relationships between related records. Validation rules ensured that submitted information followed the required format and conditions, while required fields prevented essential information from being left blank. Referential integrity maintained valid relationships between related database records, and constraint enforcement prevented invalid data from being stored.

These mechanisms helped prevent duplicate customer accounts, ensured that reservation dates remained valid, and prevented customers from reserving accommodations that were unavailable. They also maintained accurate references between customers, reservations, and accommodations while ensuring that required information was properly provided. As a result, the relationships among the different database records remained consistent and reliable.

Maintaining data consistency ensured that users accessed accurate and updated information throughout the system. Customers and administrators were provided with the same current reservation, accommodation, and customer information based on the latest records stored in the database. Strong data integrity improved operational reliability, reduced data-related errors, and supported the accurate processing of reservations and other system transactions.

human errors.

Database security was one of the most critical components of the proposed Online Booking System because the application stored confidential customer information, reservation records, payment details, and administrative accounts. Protecting this information was essential for maintaining customer trust, ensuring business continuity, and supporting the proper handling of data privacy requirements.

The system incorporated multiple layers of security to safeguard the information stored within the database. These security measures included user authentication, password encryption through hashing, role-based access control, input validation, session management, database backup procedures, audit logs, and secure database connections. These mechanisms worked together to prevent unauthorized access, protect sensitive information, and maintain the reliability and integrity of stored data.

Only authorized users were permitted to access specific areas of the system according to their assigned roles and permissions. Customers were allowed to view and manage their own reservation records, while administrators were granted access to management functions required for daily resort operations. Role-based access control helped ensure that users could not access information or functions beyond the permissions assigned to their accounts.

Password encryption ensured that user credentials were not stored or displayed as readable passwords within the database. Instead, passwords were protected through hashing before being stored. Input validation was also applied to user-submitted information to reduce the risk of malicious input and unauthorized data manipulation. Session management further strengthened account security by controlling active user sessions and terminating inactive sessions when necessary, thereby reducing the possibility of unauthorized access to accounts.

The system also performed regular database backup procedures to protect against accidental data loss, hardware failures, software corruption, and unexpected system errors. Backup files allowed administrators to restore important business information when necessary, helping maintain business continuity and minimize potential operational disruptions.

Audit logs were also maintained to record important user and administrative activities within the system. These records allowed authorized personnel to monitor system activities, investigate unusual transactions, and support accountability. Secure database connections were used to protect information transmitted between the application and the database and to reduce the risk of unauthorized interception.

The database security mechanisms provided comprehensive protection for the information stored within the Online Booking System. By combining authentication, authorization, password protection, input validation, session management, activity monitoring, secure connections, and backup procedures, the system maintained a more secure and reliable database environment. These measures supported the protection of customer and business information and contributed to the long-term operational reliability of Avianna's Inland Resort.

Importance of Database Interaction

Database interaction is one of the most essential components of the proposed Online Booking System because it enables the application to efficiently store, retrieve, update, and manage all information necessary for the daily operations of Avianna's Inland Resort. A database serves as the central repository of information where all business-related data is securely organized and maintained. Without an efficient database management system, the Online Booking System would be unable to perform its primary functions such as processing online reservations, maintaining customer records, monitoring accommodation availability, generating management reports, and supporting administrative decision-making.

In traditional manual reservation systems, customer information and booking records are commonly written in logbooks, notebooks, or paper forms. Although these methods may be adequate for small-scale operations, they become increasingly inefficient as the number of customers and reservation transactions grows. Manual record-keeping is susceptible to numerous problems including duplicate entries, inaccurate information, misplaced documents, delayed updates, inconsistent records, and accidental data loss. Retrieving information from paper records also consumes significant time and effort, making it difficult for resort personnel to respond quickly to customer inquiries or generate operational reports.

The implementation of a centralized database eliminates many of these challenges by providing a structured and computerized environment for storing and managing information. Instead of relying on manual documentation, all customer details, reservation records, accommodation schedules, payment information, inventory data, and administrative activities are stored electronically in interconnected database tables. This organized approach enables the system to retrieve information rapidly, process transactions efficiently, and maintain accurate records throughout the entire reservation process.

One of the primary benefits of database interaction is its ability to support real-time reservation management. Whenever a customer submits an online reservation, the system immediately communicates with the database to verify whether the selected accommodation is available on the requested dates. If the accommodation is available, the reservation is recorded instantly and the availability schedule is automatically updated. This real-time synchronization minimizes scheduling conflicts and significantly reduces the possibility of double bookings. Unlike manual systems where staff must manually update reservation logs after each transaction, the automated database ensures that all users access the most current and accurate reservation information at any given time.

Database interaction also plays a critical role in customer information management. Every registered customer has a unique profile containing personal information such as name, email address, contact number, residential address, booking history, and account credentials. By storing these details in a centralized database, the system enables customers to access their accounts conveniently without repeatedly entering the same information for every reservation. At the same time, administrators can efficiently retrieve customer records whenever necessary, improving communication, customer support, and record management.

Another important function of database interaction is the management of accommodation information. The database stores detailed information regarding all cottages, overnight rooms, pavilions, and other resort facilities available for reservation. This includes facility names, descriptions, capacities, pricing, availability status, and related images. Whenever administrators update accommodation details, such as changing rental rates, adjusting capacities, or marking facilities as unavailable due to maintenance, the changes are immediately reflected throughout the entire system. Consequently, customers always receive updated and accurate information when browsing the resort website.

Database interaction further supports the automation of reservation processing. Every booking transaction involves several interconnected processes, including validating customer information, checking accommodation availability, recording reservation details, updating booking schedules, and notifying administrators. These processes depend heavily on the continuous exchange of information between the application and the database. Because these interactions occur automatically, the reservation process becomes significantly faster, more accurate, and more reliable than manual methods. The automation of these tasks reduces the administrative workload while improving overall operational efficiency.

Another significant contribution of database interaction is its support for report generation and decision-making. Resort administrators require accurate reports to monitor business performance, evaluate occupancy rates, analyze reservation trends, and make informed operational decisions. The centralized database enables the system to automatically generate reports containing daily, weekly, monthly, and annual reservation summaries, customer statistics, facility utilization reports, and other management information. These reports eliminate the need for manual computation and provide administrators with timely, organized, and reliable information that supports strategic planning and business development.

Database interaction is equally important in maintaining data consistency and integrity. Data integrity refers to the accuracy, completeness, and reliability of information stored within the database. The Online Booking System employs validation rules, primary keys, foreign keys, and relational constraints to ensure that only valid and consistent information is stored. For example, the system prevents duplicate customer accounts, invalid reservation dates, incomplete booking records, and inconsistent facility assignments. These built-in validation mechanisms reduce human error and ensure that all information remains accurate throughout the system's operation.

Security is another major advantage of database interaction. Since the system manages confidential customer information and business records, appropriate security measures are necessary to protect sensitive data. The Online Booking System restricts direct database access to authorized system administrators only. Customers and ordinary users cannot directly access the database; instead, all interactions occur through secure application interfaces. User authentication, encrypted passwords, session management, role-based access control, and input validation collectively protect the database against unauthorized access, accidental modification, and malicious attacks such as SQL injection. These security mechanisms help preserve customer privacy and maintain the integrity of business information.

In addition to supporting current operations, database interaction contributes to the long-term scalability and maintainability of the system. As Avianna's Inland Resort continues to expand its services by adding new accommodations, facilities, promotional packages, or online payment features, the centralized database can easily accommodate additional records without requiring major structural changes to the application. This scalability ensures that the Online Booking System remains flexible and capable of supporting future business growth and technological advancements.

Database interaction promotes better collaboration among resort personnel. Since all authorized administrators access the same centralized database, reservation information remains synchronized regardless of who processes customer requests. This shared access minimizes communication problems, reduces duplicate work, and ensures that all staff members operate using the same accurate and updated information. Such coordination contributes to improved operational efficiency and enhanced customer service.

Although customers and administrators interact with the system through web pages and graphical interfaces, every action they perform ultimately involves communication with the database. When customers register, log in, browse accommodations, submit reservations, or review booking histories, the application continuously exchanges

information with the database behind the scenes. Likewise, when administrators approve reservations, update accommodation information, generate reports, or manage customer accounts, the system retrieves and modifies data stored within the database. This indirect interaction ensures that users enjoy a simple and user-friendly experience while the application securely handles all underlying database operations.

The database serves as the backbone of the proposed Online Booking System. It provides the foundation upon which all major system functions operate, including reservation management, customer account management, accommodation monitoring, report generation, inventory management, and administrative control. Its ability to organize, secure, process, and retrieve information efficiently contributes significantly to the reliability, accuracy, and overall performance of the system.

In conclusion, database interaction is indispensable to the successful implementation of the Online Booking System for Avianna's Inland Resort. It enables the automation of manual reservation processes, improves operational efficiency, enhances customer service, strengthens data security, supports informed management decisions, and ensures the integrity and reliability of all stored information. By serving as the central repository of business data and facilitating seamless communication between users and the application, the database plays a vital role in achieving the objectives of the study and supporting the long-term digital transformation and sustainable growth of Avianna's Inland Resort.

CHAPTER 4

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.2 Overview of the Developed System

The Online Booking System of Avianna's Inland Resort was a web-based and mobile-responsive application developed to facilitate and manage the resort's reservation and accommodation operations. The system was designed to provide customers with convenient access to resort information and online booking services while enabling administrators to efficiently manage reservations, customer records, accommodation information, availability schedules, and other operational activities.

Unlike traditional manual reservation methods that relied on paper forms, logbooks, and direct communication with resort personnel, the developed system centralized reservation and customer information within a computerized database. The application allowed customers to register accounts, browse available accommodations, view resort facilities and rates, check accommodation availability, and submit reservation requests online. Administrators, on the other hand, were provided with management functions that allowed them to review and process reservations, update accommodation information, manage customer records, monitor availability, and generate reports.

The system was developed to improve the efficiency, accuracy, and accessibility of the resort's reservation process. By automating important reservation activities, the system reduced the reliance on manual record-keeping and minimized the possibility of errors such as duplicate bookings, inaccurate availability information, and misplaced customer records. The centralized database also allowed authorized users to retrieve and update information more efficiently.

The major objectives of the developed system were to:

Automate the recording and management of customer reservations.

Provide customers with convenient online access to resort information and booking services.

Manage customer accounts and reservation records through a centralized database.

Monitor the availability of cottages, rooms, pavilions, and other rentable facilities.

Prevent double bookings and scheduling conflicts.

Provide administrators with tools for reservation and accommodation management.

Generate reports to support resort monitoring and decision-making.

Improve the accuracy, accessibility, and security of resort records.

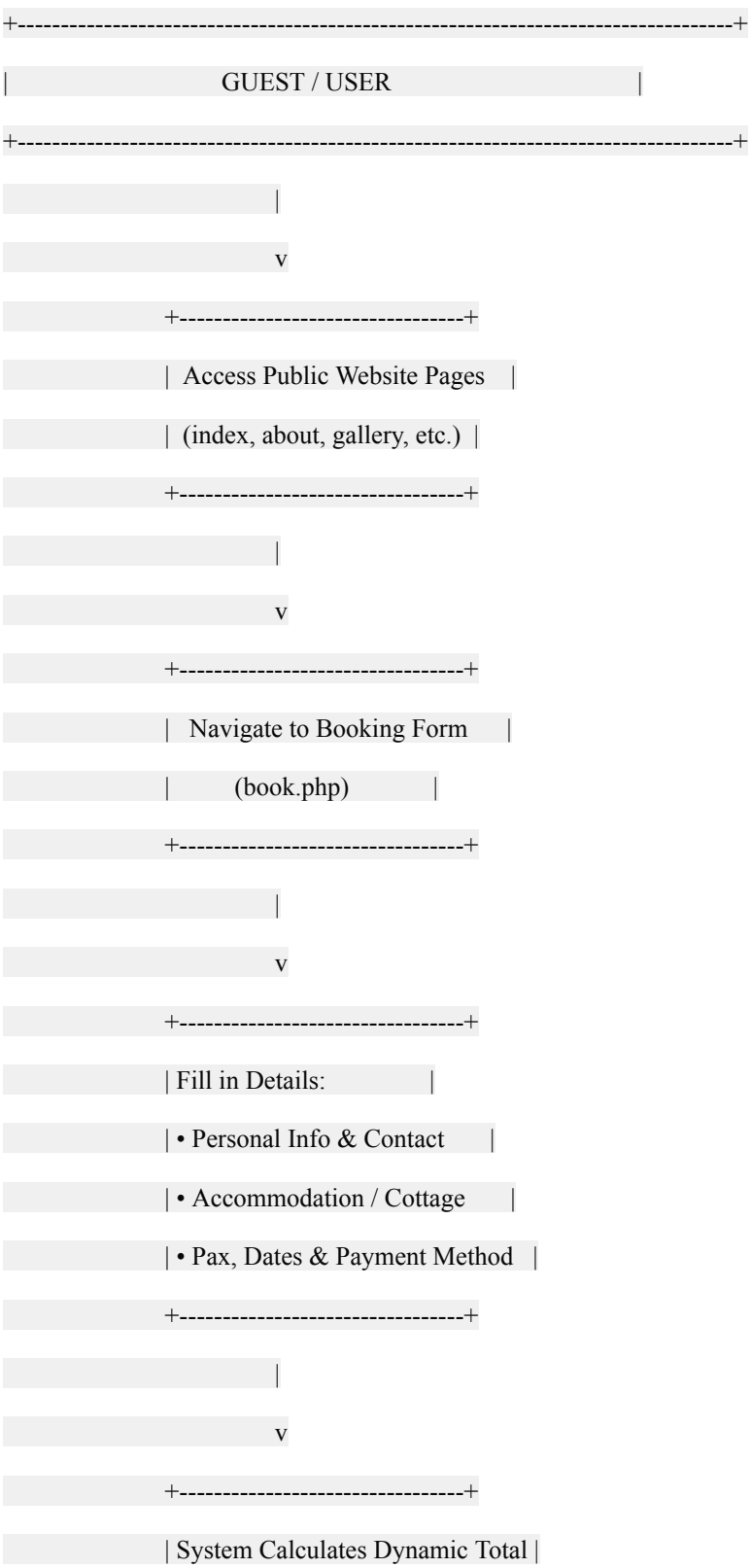
Enhance the overall efficiency of the resort's reservation and administrative operations.

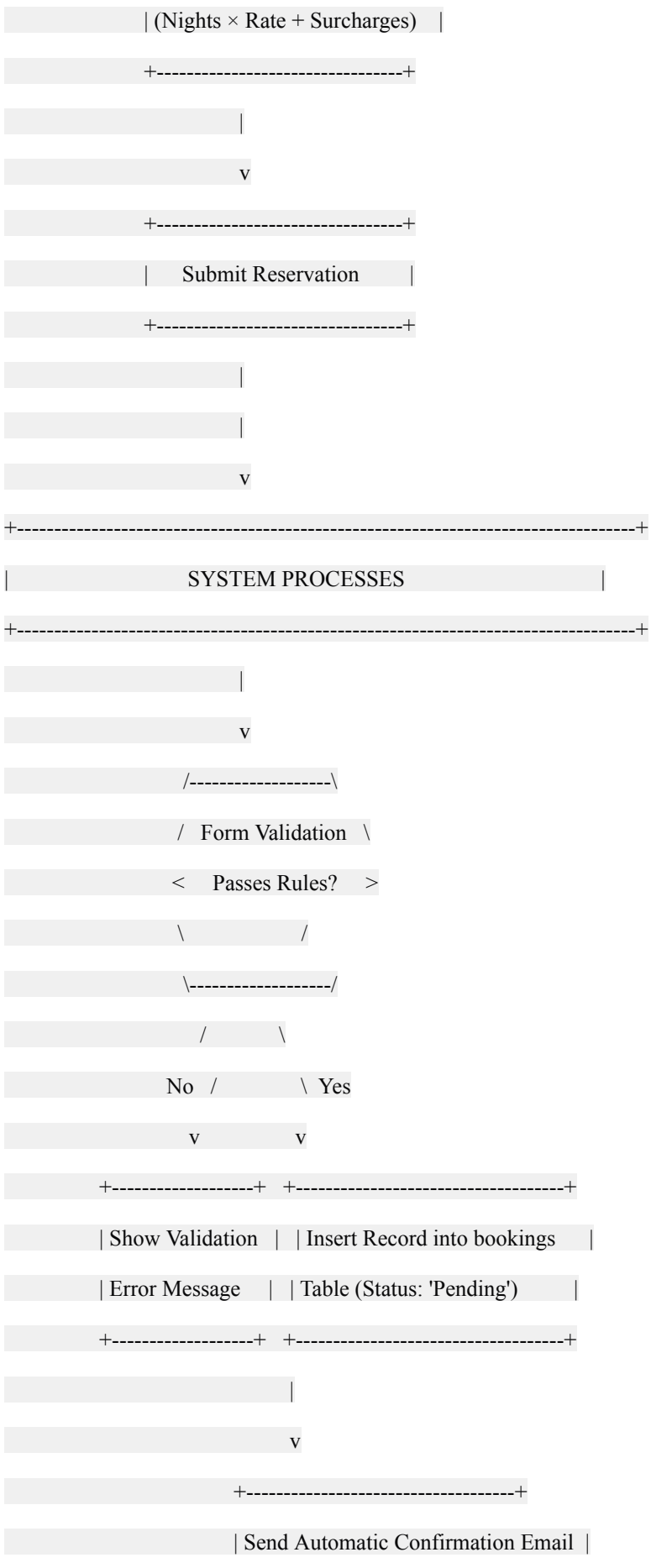
The development of the Online Booking System incorporated responsive interface design, user authentication, role-based access control, database management, input validation, reservation processing, availability monitoring, notification functions, and reporting capabilities. These components were integrated to provide customers and administrators with appropriate features based on their respective roles and responsibilities.

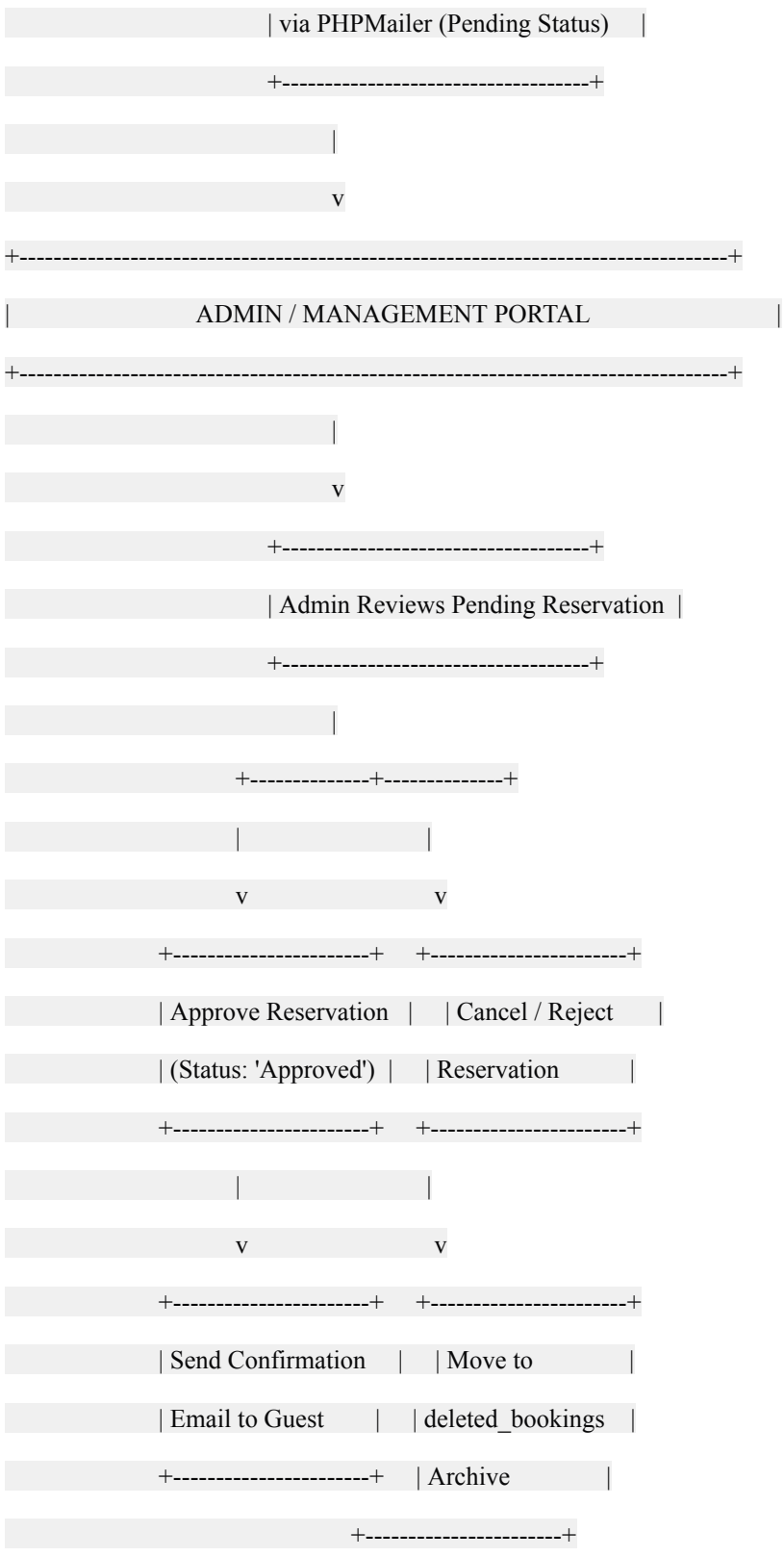
The completed system consisted of several interconnected modules that worked together to support customer

transactions and resort management activities. These modules included customer registration and authentication, accommodation information management, reservation processing, availability monitoring, customer record management, administrative management, reporting, notifications, and database management.

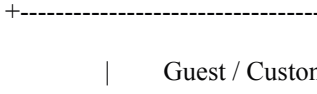
Figure 4.1 illustrates the overall workflow of the developed Online Booking System.

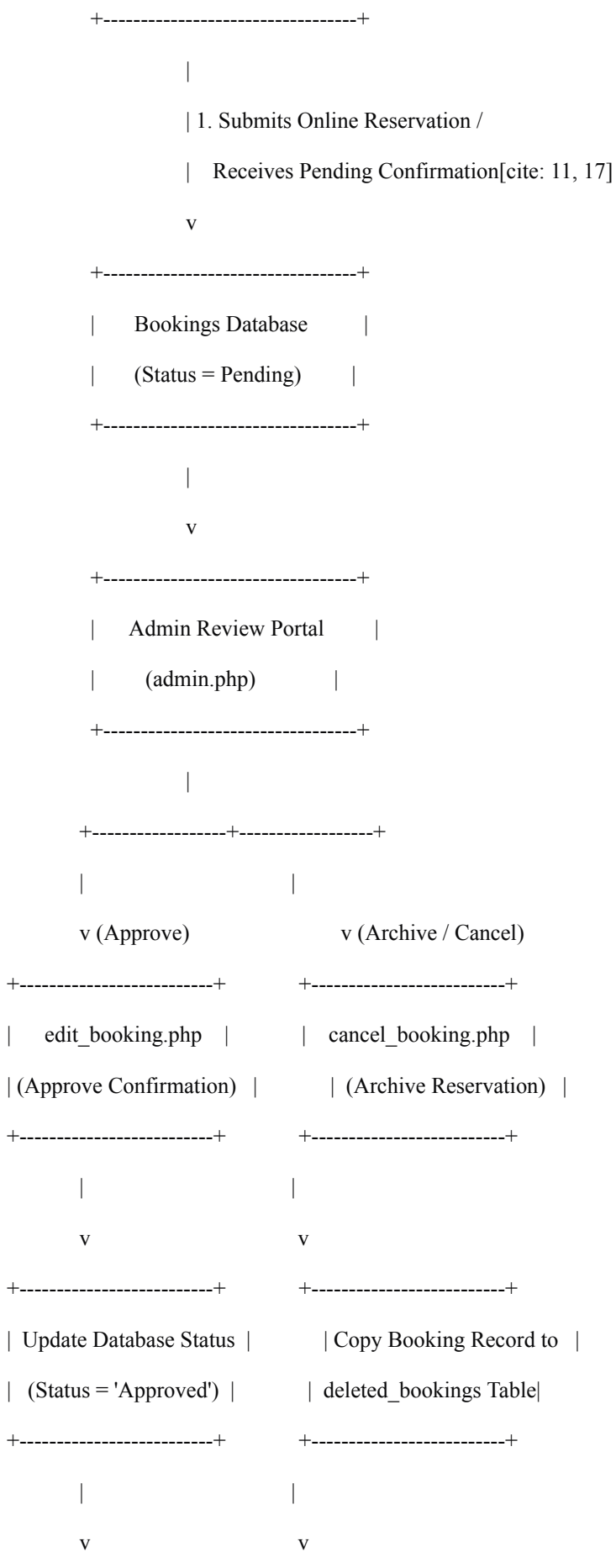



| (Nights × Rate + Surcharges) |
Submit Reservation
SYSTEM PROCESSES
Form Validation
Passes Rules?
No
Yes
Show Validation
Error Message
Insert Record into bookings
Table (Status: 'Pending')
Send Automatic Confirmation Email


| via PHPMailer (Pending Status) |
ADMIN / MANAGEMENT PORTAL
| Admin Reviews Pending Reservation |
Approve Reservation	(Status: 'Approved')
Send Confirmation	Email to Guest
Cancel / Reject	Reservation
Move to	deleted_bookings
Archive

ADMIN


Guest / Customer


1. Submits Online Reservation /
Receives Pending Confirmation[cite: 11, 17]
Bookings Database
(Status = Pending)
Admin Review Portal
(admin.php)
v (Approve)
v (Archive / Cancel)
edit_booking.php
(Approve Confirmation)
cancel_booking.php
(Archive Reservation)
Update Database Status
(Status = 'Approved')
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deleted_bookings Table

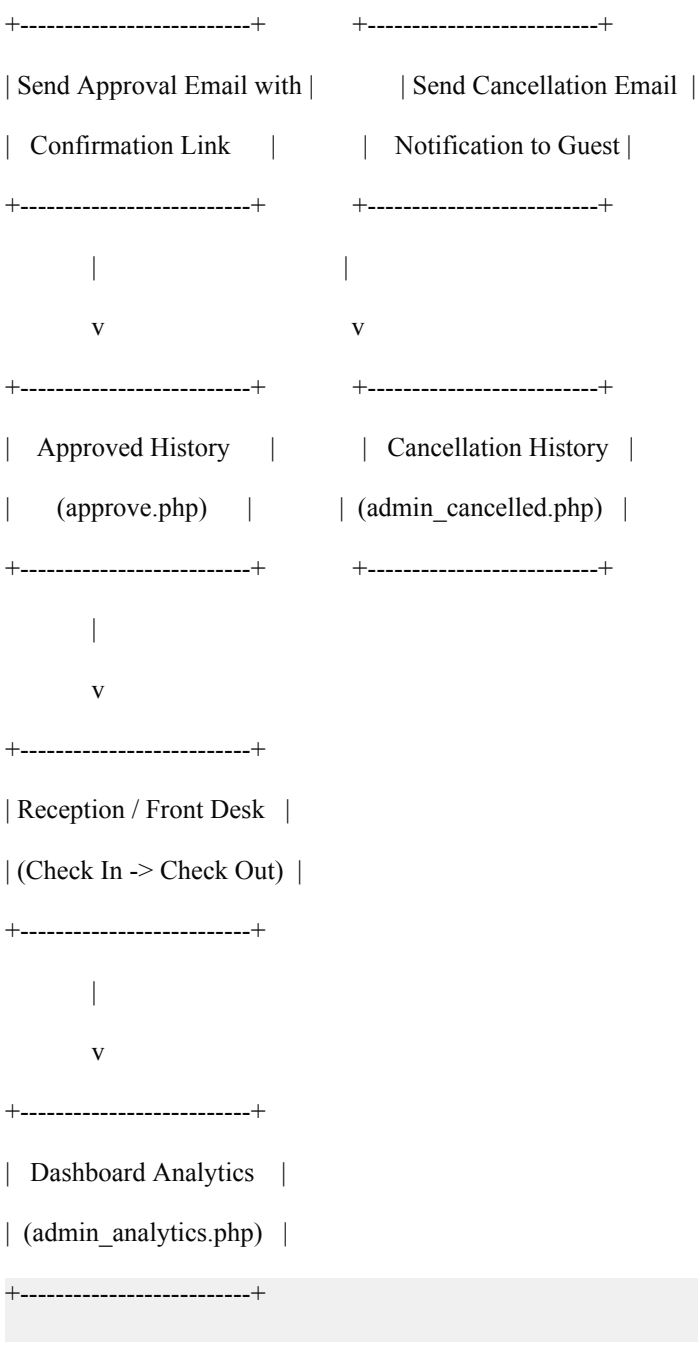
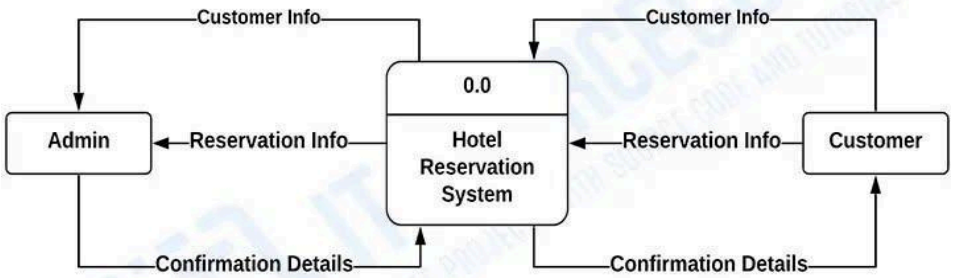


Figure 4.1. Overall Workflow of the Developed Online Booking System

(Insert System Workflow Diagram Here)

HOTEL RESERVATION SYSTEM



DATA FLOW DIAGRAM LEVEL 0

The workflow began when customers accessed the system and registered or logged into their accounts. After authentication, customers were able to browse resort facilities, view accommodation information, check availability, and submit reservation requests. The system processed the submitted information, verified the availability of the selected accommodation, and recorded the reservation details in the database.

The reservation request was then made available to the administrator for review and appropriate action. The administrator accessed the administrative dashboard to review reservation details, manage booking schedules, update accommodation information, monitor customer records, and perform other administrative functions. Once the reservation had been processed, the system updated the corresponding records and provided the appropriate confirmation or status notification.

The information generated and stored by the system was also used for administrative monitoring and report generation. Through the centralized database, administrators were able to retrieve reservation records, customer information, accommodation schedules, and other operational data. These functions supported more efficient resort management and provided accurate information for monitoring and decision-making.

The developed Online Booking System provided a centralized platform for managing the reservation and operational activities of Avianna's Inland Resort. The integration of customer-facing and administrative functions improved the accessibility of booking services, streamlined reservation management, and supported the resort's transition from manual processes to a more organized and computerized system.

4.3 System Development Methodology

The development of the Online Booking System of Avianna's Inland Resort adopted the Agile Software Development Methodology, an iterative approach that emphasized continuous collaboration, incremental development, and regular evaluation throughout the software development life cycle. Agile was selected because it allowed the researchers to accommodate changes in system requirements and incorporate feedback from the intended users. This approach helped ensure that the developed system addressed the operational needs of the resort and its stakeholders.

The Agile methodology allowed the researchers to divide the development process into manageable iterations, commonly referred to as sprints. At the end of each iteration, a functional component or feature of the system was developed, tested, and evaluated before proceeding to the succeeding iteration. This iterative process allowed the researchers to identify errors, functional problems, and usability issues at an early stage and make the necessary improvements throughout the development process.

The development process began with **requirements planning**, wherein the researchers identified and analyzed the operational needs of Avianna's Inland Resort. Information regarding the existing reservation procedures, customer requirements, accommodation information, booking processes, and administrative activities was gathered and analyzed. The problems encountered in the existing manual reservation process were also considered in determining the requirements of the proposed system. Based on the identified requirements, the researchers determined the necessary system features and functions that would support both customer and administrative activities.

After the requirements had been identified, the researchers proceeded with the **system design** phase. During this phase, the overall structure, database organization, user interface, navigation flow, and major modules of the Online Booking System were planned. The design included customer-oriented functions such as viewing resort information, checking accommodation details, and submitting online reservations. It also included administrative functions for managing reservations, customer records, accommodation information, availability, notifications, and reports. The planned design served as the basis for the succeeding development activities.

The researchers then proceeded with **iterative development**, wherein the system was developed incrementally through a series of manageable iterations. Each iteration focused on the development of specific system features or modules. Functions such as user authentication, accommodation management, reservation processing, availability monitoring, customer record management, notification functions, and administrative management were developed and integrated progressively. After each feature had been developed, it was reviewed and tested before additional functions were incorporated into the system. This process allowed the researchers to improve the system continuously as development progressed.

Throughout the development process, **testing and evaluation** were conducted to determine whether the developed features performed according to the identified requirements. The researchers tested the individual functions and integrated modules to identify errors, inconsistencies, and unexpected system behavior. Issues discovered during testing were documented and addressed in succeeding iterations. User feedback was also considered in evaluating the usability and functionality of the system. The results of the testing and evaluation activities were used as a basis for making necessary modifications and improvements.

After the major functions had been developed and tested, the researchers proceeded with the **implementation** of the Online Booking System. The developed components were integrated into a functional application and configured to support the actual reservation and administrative processes of the resort. The interaction between the customer interface, administrative portal, database, booking functions, and notification services was verified during this phase. The implemented system was then made available for further evaluation and actual use by the intended users.

Following implementation, **maintenance and continuous improvement** were carried out to ensure the continued functionality and reliability of the system. Issues encountered during actual use and evaluation were reviewed and addressed by the researchers. Necessary improvements were made to system functions, interface components, database operations, and other areas that required modification. Bug fixing and system updates were also performed when necessary to maintain the effectiveness and reliability of the Online Booking System.

The Agile Software Development Methodology provided the researchers with a flexible and iterative approach to the development of the Online Booking System. The system was developed progressively, with each component being planned, developed, tested, evaluated, and improved before or during the succeeding iterations. Through this approach, the researchers were able to identify problems at an early stage and incorporate necessary adjustments based on evaluation results and user feedback. The methodology therefore supported the development of a functional and responsive Online Booking System that addressed the identified reservation and administrative needs of Avianna's Inland Resort.

Requirements Gathering

The first phase of the development process involved gathering detailed information regarding the existing reservation and resort management processes of Avianna's Inland Resort. This phase was conducted to obtain a clear understanding of how reservations, customer inquiries, accommodation information, and other resort-related activities were currently handled. The researchers examined the existing procedures and practices used by the resort in order to identify the difficulties, limitations, and requirements that needed to be addressed through the development of the Online Booking System. The information gathered during this phase provided an important foundation for determining the appropriate functions and features that would be incorporated into the proposed system.

To obtain the necessary information, the researchers used interviews, observations, and analysis of existing records and procedures. Interviews were conducted with individuals involved in the operation and management of the resort to obtain information about the existing reservation process, the types of information required from customers, the management of accommodations, and the administrative activities performed by the resort. Through these interviews, the researchers were able to identify the specific activities that needed to be supported by the proposed system. The researchers also considered the concerns and expectations of potential customers, particularly regarding the accessibility of resort information, convenience of making reservations, and availability of accommodations.

Observation was also conducted to understand the actual flow of the resort's reservation and management activities. The researchers observed how reservation information was received, recorded, reviewed, and managed. The observation also helped the researchers identify activities that required repetitive manual work and areas where information could become difficult to organize or retrieve. Existing records and procedures were further analyzed to determine the types of information that needed to be stored and managed by the system. These included customer information, reservation details, accommodation information, booking dates, availability, and other records associated with resort operations.

The resort owner, administrators, staff, and potential customers were considered during the requirements-gathering process because each group had different interactions with the reservation process. The resort owner and administrators provided information regarding the management and monitoring of reservations and resort operations. The staff provided information regarding the actual handling of customer transactions and reservation-related activities. Potential customers provided insights into the information and services they would need when accessing the resort's online booking platform. Considering these different users allowed the researchers to identify requirements that addressed both the customer-facing and administrative aspects of the system.

Based on the information collected, the researchers identified several problems associated with the existing reservation and resort management procedures. The existing processes required considerable manual handling of reservation information and depended on direct communication and manual record keeping. These procedures could make reservation information more difficult to organize, retrieve, and monitor, particularly when several reservation requests were being handled. The researchers therefore determined that the proposed system needed to provide a centralized platform where reservation and customer information could be recorded, stored, updated, and retrieved more efficiently.

The information gathered during this phase served as the basis for identifying the functional and non-functional requirements of the Online Booking System. The functional requirements described the activities and services that the system needed to perform, while the non-functional requirements considered the expected qualities of the system, such as usability, reliability, efficiency, accessibility, and security. The researchers analyzed the collected information and translated the identified operational needs into specific system requirements that could be

incorporated into the development process.

Among the functional requirements identified were customer registration and login, accommodation browsing, availability checking, reservation submission, reservation management, customer record management, and administrative functions. The system was also designed to provide appropriate functions for administrators to review and manage reservation requests, maintain accommodation information, monitor booking records, and access reports. Notification functions were also considered to provide customers with information regarding the status of their reservation requests.

The requirements gathered during this phase were subsequently used as the reference for the system design and development activities. The researchers used the identified requirements to determine the structure of the system, the information that needed to be stored in the database, the functions that needed to be included in the user interface, and the processes that needed to be supported by the administrative portal. The requirements also served as a guide in determining the sequence in which system features would be developed and evaluated during the succeeding iterations.

The requirements-gathering phase provided the researchers with a clearer understanding of the existing reservation and resort management processes of Avianna's Inland Resort. The information obtained from the resort owner, administrators, staff, potential customers, observations, and existing records helped establish the necessary requirements of the proposed Online Booking System. Through this phase, the researchers were able to identify the essential functions and system characteristics needed to improve the organization, accessibility, and management of reservation information. The results of the requirements-gathering activities were then carried forward to the succeeding phases of the Agile development process.

System Planning

After the requirements had been identified and analyzed, the researchers proceeded with the system planning phase. This phase focused on establishing the overall direction of the Online Booking System and determining how the identified requirements would be translated into a workable development plan. The researchers defined the scope and objectives of the system based on the operational needs of Avianna's Inland Resort. The scope was established to determine the functions and activities that would be supported by the system, while the objectives served as a guide in ensuring that the proposed system addressed the problems identified during the requirements-gathering phase.

During this phase, the researchers also established the system specifications and identified the major components that needed to be developed. The planned system included customer-oriented functions such as registration and login, viewing resort and accommodation information, checking availability, and submitting reservation requests. Administrative functions were also considered, including reservation management, customer record management, accommodation management, monitoring of booking information, notifications, and report generation. The researchers organized these functions according to their importance and relationship to the overall reservation process to provide a clear direction for the succeeding development activities.

The development schedule was also established during the planning phase. The researchers identified the activities that needed to be completed during each stage of development and organized them according to the planned iterations of the Agile methodology. The development tasks were divided into manageable activities so that individual system components could be developed, tested, reviewed, and improved progressively. This arrangement allowed the researchers to monitor the progress of the project and determine whether the planned activities were being completed within the expected period.

The researchers also identified the resources required for the development of the Online Booking System. These resources included the necessary hardware, software, development tools, database resources, internet connectivity, and other materials needed during the development and testing activities. The availability of these resources was considered to ensure that the researchers had sufficient support for designing, programming, testing, and implementing the system. The researchers also considered the skills and responsibilities of the development team in assigning tasks related to system design, database development, programming, testing, documentation, and evaluation.

The appropriate technologies and development tools were also determined during this phase. The researchers considered the requirements of the proposed system when selecting the technologies to be used for the user interface, application development, database management, and other system components. The selected technologies were evaluated based on their suitability for developing a web-based and mobile-responsive Online Booking System. The researchers also considered the compatibility of the technologies with the planned system architecture and the available development resources.

The planning phase also involved considering the technical feasibility of the proposed system. The researchers evaluated whether the required technologies, hardware, software, database resources, and development skills were available and sufficient to support the project. The technical requirements of the system were reviewed to determine whether the planned features could be implemented using the available resources. This assessment helped the researchers identify potential technical limitations and make appropriate adjustments to the development plan before proceeding to the succeeding phases.

Operational feasibility was also considered to determine whether the proposed Online Booking System could be incorporated into the existing operations of Avianna's Inland Resort. The researchers examined how the system would support the activities of the resort owner, administrators, staff, and customers. Particular consideration was given to the usability and accessibility of the system, especially in relation to online reservation processing, accommodation management, customer records, and administrative monitoring. The planned system was structured to support the existing reservation activities while reducing dependence on manual record keeping and improving the organization and accessibility of reservation information.

Economic feasibility was likewise considered during the planning phase. The researchers reviewed the resources and development requirements of the project to determine whether the proposed system could be developed within the available resources and financial limitations. The expected costs associated with development tools, hardware, software, internet connectivity, maintenance, and other necessary resources were considered. The researchers also considered the potential benefits of the system, particularly its ability to improve reservation management, reduce manual work, organize customer information, and provide more convenient access to booking services.

The researchers also identified possible risks and challenges that could affect the development and implementation of the system. These included technical issues, changes in system requirements, errors discovered during development, limited resources, and possible difficulties in adapting users to the new computerized reservation process. These considerations were included in the planning process so that appropriate adjustments could be made when necessary. Since the project followed an Agile methodology, the development plan remained flexible enough to accommodate changes and improvements based on testing results and user feedback.

The planned features, resources, schedule, and feasibility considerations were evaluated to ensure that the Online Booking System could be developed and implemented within the available time, resources, and operational conditions of the resort. The researchers used the results of the planning activities as a guide for organizing the succeeding system design and development activities. The system plan also provided a basis for determining which features would be prioritized during the initial iterations and which functions could be refined during later iterations.

The system planning phase provided the researchers with a structured direction for the development of the Online Booking System of Avianna's Inland Resort. Through this phase, the researchers established the system scope, objectives, specifications, development schedule, required resources, technologies, and feasibility considerations. The planning activities helped ensure that the proposed system remained aligned with the identified needs of the resort and that the development process was organized according to the available resources and project requirements. The results of this phase served as the foundation for the succeeding system design and iterative development activities.

System Design

After the system requirements and development plans had been established, the researchers proceeded with the system design phase. During this phase, the overall structure and design of the Online Booking System of Avianna's Inland Resort were prepared based on the requirements identified during the preceding phases. The researchers translated the identified functional and non-functional requirements into specific system components, processes,

database structures, and interface designs. The purpose of this phase was to establish a clear blueprint that would guide the researchers during the actual development and integration of the system.

The researchers first designed the overall structure of the system by identifying the major components and functions that needed to interact with one another. The design considered both the customer-facing portion of the system and the administrative management portal. The customer-facing portion was designed to provide users with access to resort information, accommodation details, availability information, and online reservation services. The administrative portion was designed to provide authorized users with functions for reviewing and managing reservations, customer information, accommodation records, availability, notifications, and reports. These components were organized to ensure that information could be processed and transferred appropriately between the customer interface, system processes, database, and administrative portal.

The database structure was also designed during this phase. The researchers identified the information that needed to be stored and determined the relationships among the different data entities required by the system. The database design considered reservation records, customer information, accommodation information, booking details, and other information required for the operation of the system. The Entity Relationship Diagram (ERD) was prepared to illustrate the relationships among the entities and to serve as a reference during database implementation. The design also considered the use of appropriate primary and foreign keys to establish relationships between related records and maintain the organization and integrity of stored information.

The researchers also designed the system processes and reservation workflow. The process design described how information would move from the customer when a reservation was submitted, through system validation and database processing, and eventually to the administrator for review. The workflow included the submission of reservation information, validation of entered details, recording of the reservation with a Pending status, administrative review, approval or cancellation, and corresponding notification to the customer. The process design also considered the subsequent handling of approved reservations and the monitoring of reservation information through the administrative functions.

Various process flow diagrams were prepared to illustrate the sequence of activities performed by the system. These diagrams helped the researchers visualize the movement of information and the interaction between users and system components. The diagrams also provided a basis for identifying possible errors, unnecessary steps, and areas that required additional validation. By establishing the process flow before implementation, the researchers were able to develop the system according to a clearly defined sequence of operations.

Unified Modeling Language (UML) diagrams were also prepared to represent the functional and structural aspects of the system. These diagrams were used to describe the interactions between users and the system and to provide a visual representation of the system's functions and relationships. The researchers considered the different roles and responsibilities of customers and administrators when defining the system interactions. The customer-related functions focused primarily on accessing resort information and submitting reservations, while administrative functions focused on managing bookings, customer records, accommodation information, and other operational data.

The user interface was also designed during this phase with emphasis on accessibility, clarity, and ease of navigation. The researchers planned the layout of the different pages and system components to provide customers with a straightforward process for accessing resort information and making reservations. The booking interface was designed to organize the required customer information, accommodation selection, booking dates, number of guests, and payment information in an understandable manner. The interface design also considered the presentation of booking details and system notifications so that customers could easily understand the status of their reservation requests.

For the administrative users, an organized dashboard and management interface were designed to support the monitoring and processing of reservation information. The administrative interface included areas for reviewing pending reservations, managing booking records, maintaining customer information, monitoring accommodation availability, and accessing relevant reports and analytics. The design aimed to present important information in an organized manner so that authorized users could perform administrative activities efficiently.

The researchers also designed the navigation flow of the system to ensure that users could move between related pages and functions without unnecessary steps. The customer navigation was structured around the process of accessing resort information, viewing accommodations, checking availability, completing the booking form, and submitting a reservation. The administrative navigation was structured around reservation management, customer records, accommodation management, monitoring, and reporting. The navigation design was reviewed to ensure that the major functions were accessible and logically arranged.

Security and access control considerations were also incorporated into the system design. The researchers considered user authentication and role-based access to ensure that administrative functions were available only to authorized users. Input validation was also considered in the design of forms and reservation processes to reduce the possibility of incomplete or invalid information being submitted. These design considerations were intended to support the protection and proper management of customer and reservation information.

The design also incorporated the notification process associated with reservation transactions. The researchers planned the system so that customers would receive appropriate notifications regarding their reservation status. A pending reservation notification was considered after a reservation request had been successfully recorded, while an approval or cancellation notification was provided after the administrator had taken action on the reservation. These notification functions were incorporated into the workflow to improve communication between the resort and its customers.

The different design documents and models prepared during this phase served as references during the succeeding development activities. The process flow diagrams guided the implementation of system processes, the ERD guided the construction of the database, the UML diagrams supported the development of system functions and interactions, and the user interface layouts guided the development of the customer and administrative pages. The researchers also used these design outputs to identify inconsistencies and make necessary adjustments before and during implementation.

The system design phase provided the structural and visual foundation for the development of the Online Booking System of Avianna's Inland Resort. Through this phase, the researchers designed the database, system processes, user interfaces, navigation flow, system modules, and user interactions based on the requirements identified during the earlier phases. The resulting design provided a clear guide for the succeeding iterative development, testing, and evaluation activities and helped ensure that the developed system remained aligned with the operational requirements of the resort.

Software Development

After the system design had been completed, the researchers proceeded with the software development phase, which involved the actual coding, implementation, integration, and refinement of the designed Online Booking System of Avianna's Inland Resort. The development activities were guided by the system requirements, design documents, process flow diagrams, database structure, user interface layouts, and other specifications prepared during the previous phases. The researchers followed the planned system structure and developed the application incrementally to ensure that each component was properly implemented and could function according to its intended purpose.

The development process began with the implementation of the basic system structure and database components. The researchers developed the necessary database tables and relationships to store and organize customer information, reservation records, accommodation information, and other data required by the system. The database served as the central repository of information used by both the customer-facing and administrative components of the application. Proper relationships among the database records were implemented to allow the system to retrieve, update, and manage related information during reservation and administrative transactions.

The researchers then developed the user registration and authentication functions. These functions allowed users to create accounts, provide their required information, and securely access the appropriate features of the system. Authentication was also implemented to distinguish authorized administrative users from regular customers. The access structure was developed to ensure that users could only access functions appropriate to their assigned roles and responsibilities within the system.

The accommodation information management functions were subsequently developed to allow the system to display relevant information regarding the resort's available accommodations and facilities. The researchers implemented functions that allowed accommodation details, rates, and other related information to be stored and managed through the system. These functions were connected to the reservation process so that customers could view the available accommodation options before submitting their booking requests.

The availability checking and reservation processing functions were also implemented during the development phase. The researchers developed the necessary processes for receiving customer booking information, checking the required reservation details, calculating the applicable booking amount, and recording the reservation in the database. The reservation process was designed to prevent incomplete or invalid information from being submitted by applying appropriate validation procedures before the booking record was stored.

The reservation management functions were then integrated into the administrative portal. These functions allowed administrators to review submitted reservations, examine booking details, and take appropriate action on pending requests. Reservations that successfully passed the required validation were recorded with a Pending status and made available for administrative review. The administrator was then able to approve or cancel the reservation based on the information provided and the operational requirements of the resort. The developed workflow followed the reservation process described in the system design, where approved and cancelled transactions were handled through their corresponding procedures.
Customer record management was also incorporated into the system to allow authorized administrators to access and manage customer-related information associated with reservations. The researchers developed functions that supported the organization and retrieval of customer records from the centralized database. This reduced the dependence on separate manual records and allowed relevant customer and reservation information to be accessed through the administrative interface.

Notification functions were implemented to improve communication between the system, the resort administrators, and customers. The system was configured to send an automatic notification after a reservation had been successfully submitted and recorded as Pending. Additional notifications were implemented to inform customers when their reservations had been approved or cancelled. These functions were integrated into the reservation workflow so that customers could receive appropriate information regarding the status of their booking requests.
The administrative management functions were also developed to provide authorized users with an organized environment for managing resort reservation activities. The administrative interface included functions for reviewing reservations, managing booking information, monitoring accommodation-related information, and accessing customer records. The system also provided dashboard and analytics functions that allowed administrators to view accumulated reservation information for monitoring and decision-making purposes. These functions were integrated with the centralized database so that the information presented to administrators reflected the records stored by the system.

Reporting and analytics functions were developed as part of the administrative features of the system. The researchers implemented functions that allowed stored reservation information to be accessed and presented through the administrative dashboard. The accumulated information could then be used to support monitoring of reservations and other operational activities. The dashboard and analytics component formed part of the final administrative workflow of the developed system.

Each module was developed incrementally and evaluated after implementation. Rather than developing all system components simultaneously, the researchers worked on individual functions and integrated them progressively. After a module had been developed, its functions were checked to determine whether they operated according to the established requirements and design specifications. The researchers then proceeded with the development of succeeding components while continuously reviewing the interaction between previously developed modules.

During the development process, errors, inconsistencies, and usability concerns were identified and addressed. Necessary modifications were made to improve the functionality and interaction of the system. Changes were incorporated into the succeeding development iterations whenever problems were discovered during testing or evaluation. This allowed the researchers to refine the system continuously instead of waiting until the completion of the entire application before addressing identified issues.

The integration of the individual modules was also performed progressively. The customer-facing functions were connected to the database and reservation-processing components, while the administrative functions were connected to the same centralized database to allow authorized users to review and manage submitted reservations. The notification functions were integrated into the reservation status workflow, ensuring that appropriate messages could be sent when reservation transactions changed status. This integration allowed the different components of the system to work together as a unified application.

The researchers also ensured that the developed system followed the planned user navigation and process flow established during the system design phase. The customer was able to access the resort website, complete the booking form, submit reservation information, and receive booking notifications. The administrator was able to access pending reservations, review booking information, approve or cancel requests, and monitor reservation records through the management portal. This implementation followed the overall workflow established for the Online Booking System.

The software development phase transformed the system requirements and design specifications into a functional Online Booking System for Avianna's Inland Resort. Through incremental coding, module development, testing, integration, and continuous refinement, the researchers developed the major components required to support customer reservations and administrative management. The iterative nature of the development process allowed identified problems to be addressed throughout development and enabled the researchers to progressively integrate the individual components into a complete and functional system.

Testing

After the software development activities had been completed for each iteration, the researchers proceeded with the testing phase to determine whether the developed Online Booking System of Avianna's Inland Resort performed according to the identified requirements and design specifications. Testing was conducted throughout the development process rather than only after the entire system had been completed. This approach allowed the researchers to identify errors and functional problems at an early stage and address them before the succeeding components were integrated into the system.

The testing process initially involved **unit testing**, wherein each completed module or individual system component was tested separately. The researchers examined the functions of each module to determine whether they produced the expected results based on the specified requirements. Functions such as user registration and authentication, accommodation information management, reservation processing, availability checking, customer record management, and administrative functions were examined individually. Errors or unexpected results identified during this stage were documented and corrected before the affected components were integrated with other modules.

After the individual components had been tested, **integration testing** was conducted to determine whether the different modules worked properly when connected. The researchers examined the interaction between the customer interface, reservation processing functions, database, administrative portal, and notification functions. Particular attention was given to the transfer and retrieval of information between connected components. The researchers verified whether reservation information submitted by customers was properly stored in the database and whether the corresponding information could be retrieved and processed through the administrative portal.

The researchers also tested the reservation workflow to ensure that the system followed the intended sequence of activities. A reservation request was submitted through the customer interface and processed by the system after the required information had been validated. The researchers verified that successfully submitted reservations were recorded with the appropriate Pending status and became available for administrative review. The approval and cancellation processes were also tested to determine whether the corresponding booking status was updated correctly and whether the appropriate notification was sent to the customer. This testing was important in verifying the proper interaction between reservation processing, database management, administrative actions, and notification functions.

The database operations were also examined during testing. The researchers verified whether customer information, accommodation details, and reservation records were properly stored, retrieved, updated, and managed. The relationship between related records was checked to ensure that the system could retrieve the appropriate

information during reservation processing and administrative management. The researchers also examined whether changes made through the system were properly reflected in the corresponding database records.

Following unit and integration testing, **system testing** was conducted to evaluate the Online Booking System as a complete application. This stage focused on determining whether the integrated system performed its intended functions when all major components operated together. The researchers evaluated the overall functionality of the application, including customer access, reservation submission, availability-related processes, administrative management, notifications, record management, and reporting or analytics functions.

System testing also considered the **performance** of the application. The researchers observed how the system responded when users performed common activities such as accessing pages, entering booking information, submitting reservations, and retrieving records. The system's response and processing behavior were examined to identify delays or problems that could affect its operation. Any identified issues were reviewed and addressed as part of the iterative improvement process.

The **usability** of the system was also evaluated during system testing. The researchers examined whether the interface, navigation flow, forms, buttons, and system functions were understandable and accessible to the intended users. Particular attention was given to the customer booking process because customers needed to complete their reservations without unnecessary difficulty. The administrative interface was also evaluated to determine whether authorized users could efficiently access and manage reservation information.

The **reliability** of the system was evaluated by repeatedly performing important system functions and observing whether the expected results were consistently produced. The researchers checked whether reservation records were properly processed, database information remained consistent, and administrative actions produced the appropriate results. These activities helped determine whether the developed system could reliably support the reservation and management activities of Avianna's Inland Resort.

After the system-level testing had been conducted, the researchers performed **User Acceptance Testing (UAT)** with selected participants representing the primary stakeholders of the Online Booking System. The participants included the resort owner, administrators, resort staff, and potential customers. These participants were selected because they represented the different groups that would interact with or benefit from the developed system. Their participation provided practical feedback regarding the functionality, usability, accessibility, and suitability of the application.

During the User Acceptance Testing, the participants were allowed to interact with the developed system and perform functions relevant to their respective roles. The resort owner and administrative participants evaluated functions related to reservation management, customer records, accommodation information, and administrative monitoring. Resort staff evaluated the system based on its usefulness in supporting reservation-related activities and managing customer transactions. Potential customers evaluated the customer-facing functions, particularly the accessibility of resort information, accommodation viewing, booking form, reservation submission, and notification processes.

The participants' feedback and evaluation results were collected and reviewed by the researchers to determine whether the system had satisfied the intended requirements. Comments, observations, and identified concerns were considered in evaluating the strengths and limitations of the developed application. Issues identified by the participants were reviewed and, when necessary, incorporated into the succeeding improvement activities.

The testing results also served as a basis for determining whether additional modifications were necessary before the system was considered ready for implementation. When errors, inconsistencies, or usability concerns were identified, the researchers made the necessary corrections and conducted additional testing to verify whether the problems had been resolved. This iterative testing and improvement process allowed the researchers to refine the system progressively and ensure that the final version was more stable and aligned with the identified requirements.

The testing phase provided a systematic means of verifying the functionality, integration, performance, usability, and reliability of the Online Booking System of Avianna's Inland Resort. Through unit testing, integration testing, system testing, and User Acceptance Testing, the researchers were able to identify and address technical and usability issues throughout the development process. The results of these testing activities provided evidence

regarding the system's ability to support online reservation and administrative management activities and served as a basis for the succeeding evaluation and implementation activities.

Deployment

After the Online Booking System had successfully undergone the required testing procedures and the necessary improvements had been implemented, the researchers proceeded with the deployment phase. This phase involved preparing the developed system for operation in the target environment of Avianna's Inland Resort. The deployment activities were conducted to ensure that the completed application, database, and supporting components were properly configured and ready to be accessed by the intended users.

The researchers first prepared the deployment environment by configuring the necessary application and database components. The system files were organized and installed in the appropriate hosting environment, while the database was configured to support the storage and retrieval of customer, accommodation, and reservation information. The researchers also established the required database connection settings to ensure that the application could communicate properly with the database after deployment.

The hosting environment was also prepared to make the Online Booking System accessible through the appropriate web interface. The required application components and supporting resources were installed and configured based on the specifications established during the development process. The researchers verified that the deployed files and system components were correctly placed and that the application could be accessed through the intended environment.

After the system had been deployed, the researchers conducted verification activities to determine whether the application continued to function properly in the deployment environment. The system was accessed and tested using the deployed version rather than only the development environment. This allowed the researchers to determine whether differences between the development and deployment environments affected any of the implemented functions.

The database connection was also verified after deployment. The researchers checked whether the application could successfully connect to the configured database and perform the required database operations. Customer information, accommodation information, and reservation records were examined to ensure that the system could properly store, retrieve, and update information within the deployed environment.

User authentication was likewise verified to ensure that users could access the system according to their assigned roles. The researchers tested the login and authentication functions and confirmed that authorized users could access the appropriate system features. Administrative access was also checked to ensure that the management functions remained restricted to authorized users after deployment.

The reservation processing functions were subsequently verified in the deployed system. The researchers performed reservation-related activities to determine whether customers could successfully access the booking form, enter the required information, submit reservation requests, and have the resulting records stored in the database. The system's reservation workflow was also checked to ensure that submitted requests could be processed correctly through the administrative portal.

The administrative functions were also tested after deployment. The researchers verified whether administrators could access the management portal, review pending reservations, update reservation statuses, manage customer and accommodation information, and access the available monitoring and reporting functions. These verification activities ensured that the administrative components continued to operate as intended in the target environment.

The notification functions were also checked to determine whether the system could properly perform the required communication activities after deployment. The researchers verified the reservation notification process and examined whether the appropriate messages were generated following reservation submission and administrative actions. This verification helped ensure that the notification features remained connected to the reservation workflow after the system had been transferred to the deployment environment.

The deployed system was then made available for evaluation and actual use by the intended users. The resort owner,

administrators, staff, and selected users were given access to the system to perform functions relevant to their respective roles. Their use of the deployed application provided an opportunity to observe the system under actual operating conditions and to identify any issues that might not have been detected during development and testing.

During the initial deployment and actual use, the researchers monitored the operation of the system and reviewed any issues reported by the users. Concerns related to functionality, accessibility, navigation, database operations, or other system behavior were documented and evaluated. When necessary, appropriate adjustments were made to improve the system's performance and usability.

The deployment phase therefore served as the transition of the Online Booking System from the development and testing environment into its intended operational environment. Through proper configuration, installation, verification, and initial user access, the researchers ensured that the major components of the system were functional and interconnected after deployment. The activities conducted during this phase also provided an opportunity to confirm that the developed system could support the reservation and administrative activities of Avianna's Inland Resort under the intended operating conditions.

The deployment phase successfully prepared the Online Booking System for use by the intended users. The researchers configured the hosting environment and database, installed the required application components, verified the major system functions, and made the system accessible for evaluation and actual use. The successful deployment provided the foundation for the continued operation, maintenance, and improvement of the system.

Maintenance

After the Online Booking System had been deployed and made available for actual use, the researchers proceeded with the maintenance phase. This phase focused on monitoring the operation of the system and addressing issues that were identified during actual use, evaluation, and user interaction. The researchers continuously reviewed the behavior of the deployed system to determine whether its functions continued to operate according to the intended requirements and whether any problems affected the reservation and administrative activities of Avianna's Inland Resort.

During this phase, the researchers reviewed feedback provided by the resort owner, administrators, staff, and users who interacted with the system. User observations and reported concerns were considered in identifying areas that required correction or improvement. The researchers also examined system behavior and performance to identify errors, inconsistencies, or unexpected results that may have occurred after deployment. The information obtained from these activities served as a basis for determining the appropriate maintenance actions.

One of the major maintenance activities involved **correcting identified errors and system issues**. When errors were encountered during actual use, the researchers examined the affected system functions to determine their possible causes. Appropriate corrections were then implemented to restore or improve the affected functionality. After modifications had been made, the researchers verified the affected functions again to ensure that the corrections did not introduce additional problems into other parts of the system.

The researchers also performed **system information updates** when necessary. Information stored and displayed by the system was reviewed to ensure that it remained appropriate for the operational requirements of the resort. Updates were made to relevant accommodation information, rates, availability-related information, and other system content whenever changes were required. These updates helped ensure that customers and administrators could access information that was consistent with the current operations of the resort.

Maintenance activities also included **improving existing system functionalities**. Based on observations and user feedback, the researchers identified functions that could be improved to make the system more efficient and easier to use. Improvements were made to selected processes, interfaces, and system functions when necessary. These modifications were intended to improve the overall user experience while maintaining the existing functionality of the Online Booking System.

The researchers also monitored the **usability and performance** of the deployed system. The system was observed while users performed common activities such as accessing resort information, submitting reservations, reviewing

booking records, and managing administrative information. Issues that affected navigation, accessibility, response, or ease of use were documented and considered for improvement. This monitoring helped the researchers determine whether the system continued to meet the practical needs of its intended users.

Database-related maintenance was also considered to ensure that reservation and customer information remained properly organized and accessible. The researchers reviewed the operation of the database and its interaction with the application to identify possible inconsistencies or problems in data storage and retrieval. Proper database operation was important because the Online Booking System depended on centralized reservation and customer information for both customer-facing and administrative functions.

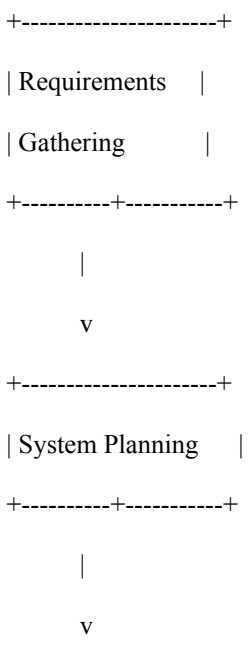
Security and access-related concerns were also monitored during maintenance. The researchers reviewed the authentication and access functions to ensure that the system continued to provide appropriate access to authorized users. Any identified concerns related to user access or system protection were considered during the maintenance process. These activities supported the continued protection and proper management of customer and reservation information.

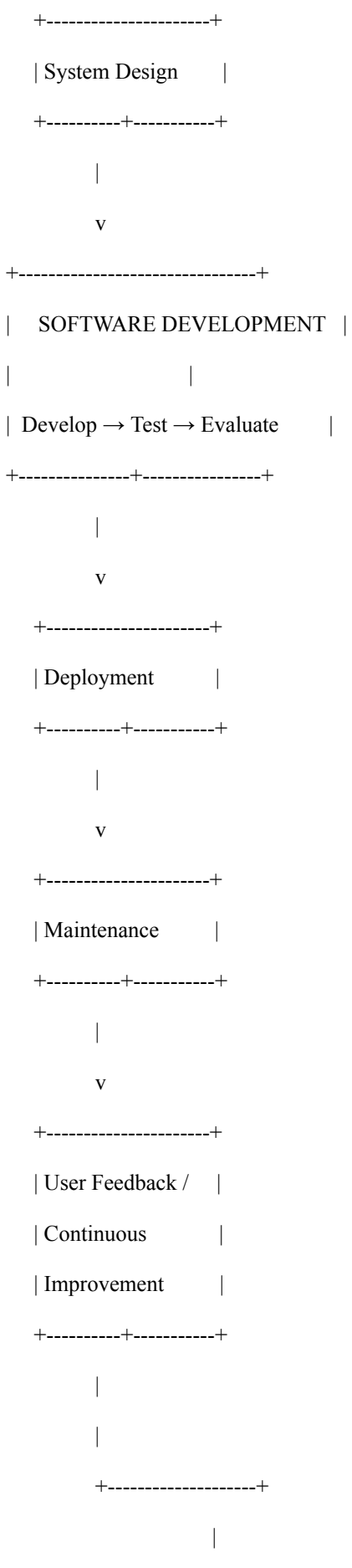
The maintenance process followed the iterative nature of the Agile methodology because improvements were introduced whenever new requirements, problems, or opportunities for enhancement were identified. Instead of treating development as completely finished after deployment, the researchers continued to evaluate the system and make necessary adjustments based on actual operating conditions. This allowed the system to adapt to the changing needs of the resort and its users.

After each significant correction or improvement, the researchers performed appropriate verification activities to determine whether the modified function operated correctly. The affected components were checked to ensure that the changes addressed the identified issue and that existing system functions remained operational. This process helped maintain the stability and reliability of the deployed application.

The maintenance phase helped ensure the continued effectiveness of the Online Booking System of Avianna's Inland Resort after deployment. Through continuous monitoring, user feedback, error correction, information updates, functionality improvements, and performance evaluation, the researchers were able to maintain the system according to the operational needs of the resort. The maintenance activities contributed to keeping the system reliable, functional, accessible, and responsive to the needs of its intended users.

AGILE SOFTWARE DEVELOPMENT METHODOLOGY




System Design
SOFTWARE DEVELOPMENT
Develop → Test → Evaluate
Deployment
Maintenance
User Feedback /
Continuous
Improvement

v

System Planning

/ Development

4.4 System Architecture

The Online Booking System of Avianna's Inland Resort utilized a **Three-Tier Client-Server Architecture**, which separated the system into three major layers: the presentation layer, application layer, and data layer. This architectural approach was used to organize the different components of the system according to their specific responsibilities. The separation of these layers allowed the researchers to manage the user interface, system processes, and database operations independently while maintaining communication among the different components.

The **presentation layer** served as the interface between the users and the Online Booking System. It included the web pages and interface components that were accessed by customers and authorized administrative users. Customers used this layer to access resort information, view accommodation details, check available booking information, complete the online booking form, and submit reservation requests. Administrators accessed the administrative interface through this layer to review reservations, manage customer and accommodation information, monitor booking records, and access available reports and analytics. The presentation layer was designed to provide an accessible and understandable interface for the different types of users.

The **application layer** served as the central processing component of the system. It handled the business logic and processes required by the Online Booking System. The application layer processed customer registration and authentication, reservation validation, booking calculations, availability-related processes, reservation status updates, administrative actions, and notification functions. When a customer submitted a reservation, the application layer processed the submitted information, validated the required data, calculated the applicable booking amount, and forwarded the appropriate information to the data layer for storage. It also processed administrative actions such as approving or cancelling reservations and updating their corresponding statuses.

The **data layer** served as the storage component of the system. It contained the database used to store and manage the information required by the Online Booking System. The stored information included customer records, accommodation information, reservation details, booking dates, and other records required for resort operations. The data layer allowed the application layer to create, retrieve, update, and manage records as users performed different activities within the system. By centralizing the information in a database, the system provided a structured method for managing reservation and customer records.

Communication among the three layers allowed the system to perform the complete reservation process. When a customer entered information through the presentation layer, the request was passed to the application layer for processing. The application layer validated and processed the information and communicated with the data layer when records needed to be stored or retrieved. The processed information was then returned to the presentation layer and displayed to the appropriate user. The same architecture supported administrative activities by allowing authorized users to access reservation and customer information through the administrative interface while the application layer handled the corresponding processes and database operations.

The architecture also supported the security and maintainability of the Online Booking System. User authentication and access control were handled through the system processes, while the database remained responsible for storing the necessary information. Separating the system into distinct layers made it easier for the researchers to identify and modify individual components without requiring extensive changes to unrelated parts of the application. This structure also supported future improvements and additional functionality that could be incorporated into the system as the operational requirements of the resort changed.

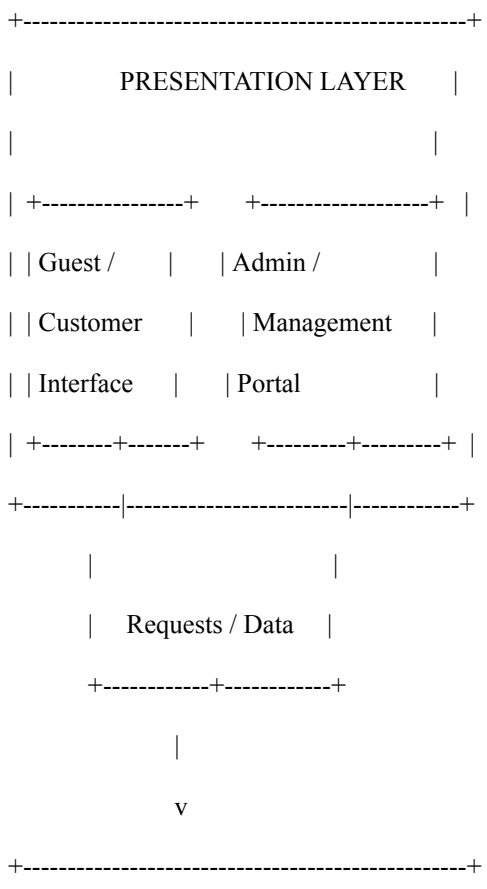
The Three-Tier Client-Server Architecture provided the structural foundation for the Online Booking System of

Avianna's Inland Resort. The presentation layer handled user interaction, the application layer processed system operations and business rules, and the data layer managed the storage and retrieval of system information. The separation of these responsibilities allowed the developed system to provide an organized approach to online reservation processing, administrative management, customer information management, and database operations.

Figure 4.3 illustrates the three-tier architecture of the developed Online Booking System and shows the relationship and flow of information among the customer and administrator interfaces, application processing, and centralized database.

FIGURE 4.3

THREE-TIER CLIENT-SERVER ARCHITECTURE



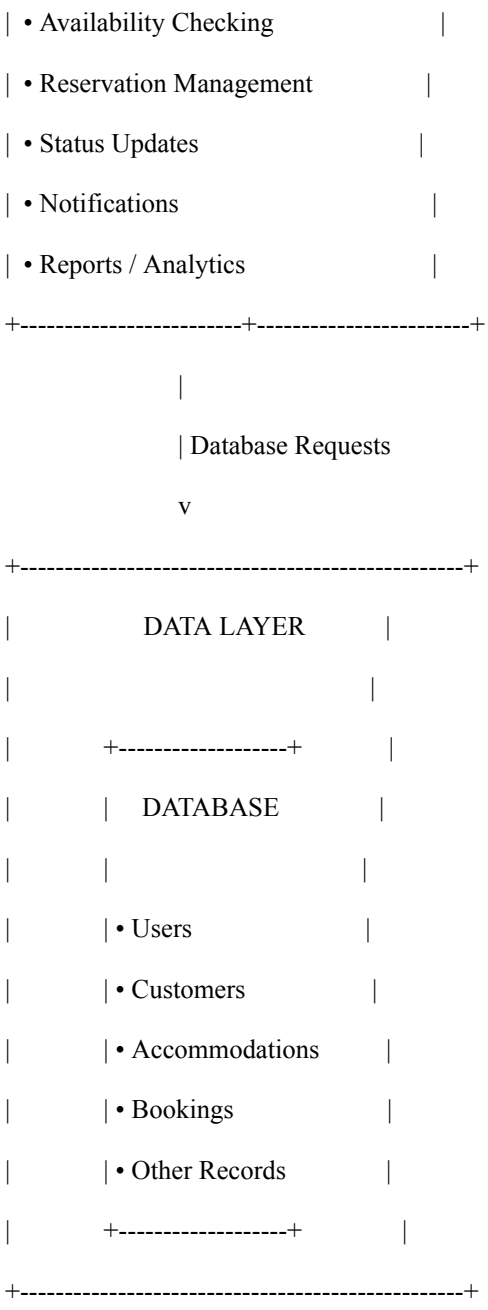
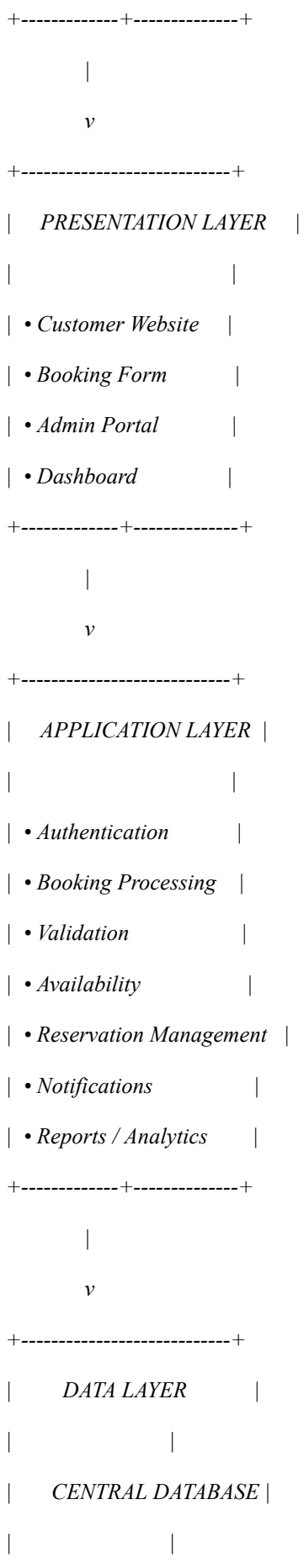


Figure 4.3 illustrates the architecture of the developed system.

(Insert System Architecture Diagram Here)




PRESENTATION LAYER
• Customer Website
• Booking Form
• Admin Portal
• Dashboard
APPLICATION LAYER
• Authentication
• Booking Processing
• Validation
• Availability
• Reservation Management
• Notifications
• Reports / Analytics
DATA LAYER
CENTRAL DATABASE

| • *Users / Customers* |

| • *Accommodations* |

| • *Bookings* |

| • *Reservation Records* |

| • *Other System Data* |

+----------+

The architecture consists of the following components:

The architecture of the Online Booking System of Avianna's Inland Resort consisted of three major components: the Presentation Layer, Application Layer, and Data Layer. Each layer performed a specific function within the system while communicating with the other layers to support the complete online reservation process. The architecture was designed to organize the user interface, system processing, and database management into separate but interconnected components.

Presentation Layer

The Presentation Layer served as the interface between the users and the Online Booking System. It was responsible for displaying resort information, receiving user input, and presenting the results of system processes. Customers accessed the system through a web browser using desktop computers, laptops, tablets, or smartphones. The customer-facing interface provided access to the resort's public information, accommodation details, booking form, and reservation functions. Through the interface, customers were able to view available information, enter their personal and contact details, select an accommodation or cottage, specify the number of guests and booking dates, provide the required payment information, and submit their reservation requests.

The Presentation Layer also included the administrative interface used by authorized administrators and resort staff. The administrative portal provided access to functions for reviewing pending reservations, managing customer records, managing accommodation information, monitoring booking records, and accessing dashboard and reporting functions. The interface was organized to provide users with clear navigation and appropriate access to the functions associated with their roles. The major interface components included the public website pages, booking form, customer access functions, administrative login, reservation management pages, customer management pages, accommodation management pages, and administrative dashboard.

Application Layer

The Application Layer contained the business logic and processing functions of the Online Booking System. It served as the intermediary between the Presentation Layer and the Data Layer by receiving user requests, processing submitted information, applying system rules, and communicating with the database. This layer handled the major processes required for online reservation and administrative management.

The application layer was responsible for user authentication, reservation processing, input validation, booking calculations, availability-related processing, reservation status management, customer record processing, administrative actions, notification functions, and reporting. When a customer submitted a reservation, the application layer received the submitted information and validated the required details before processing the request. The system then calculated the applicable booking amount and recorded the reservation in the database when the required conditions were satisfied.

The application layer also processed administrative actions performed through the management portal. When an administrator reviewed a pending reservation, the application processed the selected action and updated the corresponding reservation status. Approved reservations were processed as approved transactions, while cancelled or

rejected reservations were handled according to the cancellation procedure of the system. The application layer also supported the generation of notifications and the presentation of reservation information through the administrative dashboard.

Data Layer

The Data Layer served as the storage component of the Online Booking System. It contained the centralized relational database used to store and manage information generated and used by the system. The data layer was responsible for storing customer records, user account information, accommodation information, reservation records, booking details, and other information required for the operation and management of the resort's online reservation process.

The database stored information submitted by customers during the reservation process as well as information created or updated by authorized administrators. Reservation records included important details such as the customer associated with the reservation, selected accommodation, booking dates, number of guests, booking status, and other relevant reservation information. The database also maintained accommodation and customer information required by the administrative functions of the system.

The Data Layer supported the application layer by allowing information to be stored, retrieved, updated, and managed as users performed different system activities. Appropriate relationships among related records were maintained to support organized data management and reduce inconsistencies in stored information. The centralized database also allowed customer and reservation information to be accessed through the appropriate system functions without requiring separate manual records.

The architecture provided a structured method of communication among the three layers. Customer and administrative requests entered the system through the Presentation Layer and were forwarded to the Application Layer for processing. The Application Layer communicated with the Data Layer whenever information needed to be stored, retrieved, or updated. After processing was completed, the resulting information was returned to the Presentation Layer and presented to the appropriate user.

The three-layer architecture also supported the security and maintainability of the Online Booking System. User authentication and access control were applied to restrict administrative functions to authorized users, while the separation of application processing from database storage helped organize the system's responsibilities. This structure also made it easier for the researchers to identify, maintain, and modify individual components without significantly affecting unrelated parts of the system.

The Three-Tier Client-Server Architecture provided the structural foundation for the Online Booking System of Avianna's Inland Resort. The Presentation Layer handled user interaction, the Application Layer processed system functions and business rules, and the Data Layer stored and managed the information required by the system. The interaction among these three layers enabled the system to support online reservations, administrative management, customer record management, accommodation information management, notifications, and reporting functions.

4.5 Hardware and Software Requirements

The successful development, deployment, and operation of the Online Booking System of Avianna's Inland Resort required appropriate hardware and software resources. These resources provided the necessary environment for developing, testing, deploying, and accessing the system. The hardware requirements supported the development and operation of the application, while the software requirements provided the tools and platforms needed for programming, database management, testing, deployment, and system access.

The hardware resources were considered based on the requirements of the researchers during development and the intended users during system operation. Computers or laptops with sufficient processing capability, memory, and storage were used for system development, database management, testing, and administrative activities. Internet connectivity was also required because the system was designed as a web-based Online Booking System that

allowed customers and administrators to access its functions through a web browser.

For customers, the system could be accessed using desktop computers, laptops, tablets, or smartphones with an available internet connection and a modern web browser. This allowed users to access the resort website, view accommodation information, complete the booking form, and submit reservation requests without requiring specialized hardware.

The software requirements included the development environment, programming technologies, database management system, web server or hosting environment, and web browsers required to operate the application. The selected software tools were used to develop the customer interface, implement the application logic, manage the database, test the system, and deploy the completed application. The software environment was configured to ensure that the different components of the system could communicate properly and perform their intended functions.

The hardware and software requirements were identified and evaluated during the development process to ensure that they were sufficient for the successful implementation of the Online Booking System. The selected resources supported the development, testing, deployment, and actual use of the system and helped ensure that the application could operate according to the identified requirements of Avianna's Inland Resort.

4.5.1 Hardware Requirements

Table 4.1 presents the minimum and recommended hardware specifications required for the development, deployment, and use of the Online Booking System of Avianna's Inland Resort. The identified specifications were considered to provide sufficient resources for system development, database management, reservation processing, administrative activities, and customer access. The requirements also considered the different devices that could be used by customers and administrators when accessing the web-based system.

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3	Intel Core i5 or higher
Memory (RAM)	4 GB	8 GB or higher
Storage	256 GB SSD	512 GB SSD
Monitor	1366 × 768 Resolution	Full HD Display
Internet Connection	10 Mbps	50 Mbps
Mobile Device	Android 10	Android 13 or higher

The **processor** requirement was considered to ensure that the computer could adequately handle the development tools, web browser, database applications, and other software required for the system. An Intel Core i3 processor was considered sufficient for basic system access and development activities, while an Intel Core i5 or higher processor was recommended for smoother operation when performing multiple development, testing, and administrative tasks simultaneously.

The **memory (RAM)** requirement was established to support the operation of the web browser, development environment, database tools, and other applications used during system development and administration. A minimum of 4 GB of RAM was considered sufficient for basic system operation, while 8 GB or higher was recommended to provide better performance when several applications or system components were being accessed simultaneously.

The **storage** requirement was also considered to provide sufficient space for the operating system, development tools, project files, database files, system backups, and other resources required during development and operation. A 256 GB SSD was identified as the minimum storage requirement, while a 512 GB SSD was recommended to provide additional space for system-related files and future maintenance activities.

The **monitor** requirement was considered to ensure that users could properly view the system interface and its different pages and functions. A resolution of 1366 × 768 was considered the minimum requirement for accessing the system, while a Full HD display was recommended to provide a clearer and more comfortable viewing experience, particularly when administrators accessed the management interface and dashboard.

An **internet connection** was required because the Online Booking System was developed as a web-based application. A minimum connection speed of 10 Mbps was considered sufficient for basic access to the system, including viewing resort information, completing booking forms, and submitting reservation requests. A 50 Mbps connection was recommended to provide more stable and responsive access, particularly when several users or devices were accessing the system simultaneously.

A **mobile device** was also considered because customers could access the Online Booking System using smartphones. Android 10 was identified as the minimum mobile operating system requirement, while Android 13 or higher was recommended to provide improved compatibility, security, and performance when accessing the web-based application through a modern mobile browser.

The hardware specifications presented in Table 4.1 provided the recommended resources for supporting the major activities associated with the Online Booking System. The configuration was considered sufficient for system development, reservation processing, database operations, administrative management, and customer access. The recommended specifications also provided additional processing capacity and storage to support the continued use and maintenance of the system as the resort's reservation records and system requirements increased over time.

4.5.2 Software Requirements

Table 4.2 presents the software requirements used for the development, testing, deployment, and operation of the Online Booking System of Avianna's Inland Resort. The identified software tools provided the necessary environment for developing the system, managing the database, writing and maintaining source code, testing system functions, and accessing the completed web application.

Software	Purpose
Windows 11 / Ubuntu Linux	Operating System
Visual Studio Code	Source Code Editor
MySQL	Database Management System

PHP / Node.js	Backend Development
ReactJS	Frontend Development
Google Chrome	Web Browser
Git	Version Control
XAMPP	Local Development Server

The **Windows 11 and Ubuntu Linux operating systems** provided the working environment used during the development and testing of the Online Booking System. These operating systems supported the installation and operation of the development tools, database software, local server environment, and web browsers required throughout the project.

Visual Studio Code was used as the primary source code editor for writing, organizing, and modifying the source files of the system. It provided an environment where the researchers could manage the different components of the application and make necessary modifications during the iterative development process. The editor also supported the organization and maintenance of the project files throughout development.

MySQL was used as the database management system for storing and managing the information required by the Online Booking System. The database contained records associated with users, customers, accommodations, reservations, booking information, and other system-related data. MySQL supported the creation, retrieval, updating, and management of records required during reservation processing and administrative activities.

PHP and Node.js were identified as backend development technologies for implementing the server-side functions of the system. The backend environment was responsible for processing requests, validating submitted information, communicating with the database, and executing the system processes required for reservation and administrative management. These technologies supported the implementation of the application logic connecting the user interface with the database.

ReactJS was used for frontend development to create and organize the user interface of the web application. It supported the development of interactive pages and interface components used by customers and administrators. The frontend provided users with access to functions such as viewing resort information, completing the booking process, and accessing administrative management features.

Google Chrome was used as a web browser for accessing and testing the Online Booking System. The researchers used the browser to verify the appearance, navigation, responsiveness, and functionality of the developed web pages. It also provided an environment for testing the customer-facing and administrative interfaces during development and deployment.

Git was used for version control and source code management. It allowed the researchers to track changes made to the project files throughout the development process. The use of version control supported the organization of development activities and provided a means of managing modifications made during different iterations of the system.

XAMPP was used as the local development server environment during system development and testing. It provided the necessary local server and database environment for running and testing the web application before deployment.

The researchers used the local environment to verify the interaction between the application and database and to identify and correct errors before the system was transferred to the target deployment environment.

The software tools presented in Table 4.2 were selected based on their suitability for the development requirements of the Online Booking System. Their use provided the researchers with the necessary environment for source code development, database management, frontend and backend implementation, local testing, browser-based access, and version control. These software resources supported the development process and contributed to the implementation and maintenance of the Online Booking System of Avianna's Inland Resort.

4.6 Database Design

The database served as one of the major foundations of the Online Booking System of Avianna's Inland Resort because it was responsible for organizing, storing, and managing the information required by the system. The database provided a centralized location for maintaining customer records, user account information, accommodation details, reservation records, booking information, and other data associated with the operation of the resort's online reservation process. A properly structured database was necessary to ensure that information could be stored systematically and retrieved efficiently whenever it was required by customers or authorized administrators.

The researchers employed a **relational database model** using MySQL for the development of the Online Booking System. The relational model allowed the researchers to organize information into related tables, with each table representing a specific type of information used by the system. Relationships were established among the tables through primary keys and foreign keys to ensure that related records could be properly connected. This structure supported the reservation process by allowing customer information, accommodation information, and booking records to be associated with one another.

The database was designed based on the information requirements identified during the requirements-gathering and system design phases. The researchers determined the types of information that needed to be stored and organized them according to their respective entities. The database design considered information related to users, customers, accommodations or facilities, bookings, and other records required for reservation and administrative management. The structure was designed to support the different functions of the system while maintaining an organized and consistent collection of records.

The **Booking** entity served as one of the central components of the database because it stored the reservation transactions submitted by customers. Each booking record contained information necessary to identify and manage a reservation, including the customer associated with the booking, the selected accommodation or facility, booking date, check-in date, check-out date, number of guests, and reservation status. The Booking entity also served as a connection between customer records and accommodation or facility records, allowing the system to determine which customer had submitted a particular reservation and which facility had been requested.

The **User** and customer-related information was also stored in the database to support account authentication and reservation management. User records allowed the system to identify individuals accessing the application and determine the appropriate functions available to them. Customer information associated with reservations was maintained so that administrators could retrieve the necessary details when reviewing and managing booking requests.

The **Accommodation or Facility** entity was designed to store information regarding the resort's available accommodations and facilities. This information supported the customer-facing portion of the system by allowing customers to view available options before submitting their reservations. It also supported administrative functions by allowing authorized users to manage accommodation-related information and monitor the facilities associated with reservation transactions.

The researchers also applied **database normalization techniques** during the database design process to minimize unnecessary duplication of information and maintain consistency among related records. Information was organized into appropriate tables rather than being repeatedly stored in a single table. This approach reduced data redundancy

and made it easier to update specific information without unnecessarily modifying unrelated records. The relationships established among the tables also helped maintain the integrity of the information stored in the database.

Primary keys were used to uniquely identify individual records within the different entities, while foreign keys were used to establish relationships between related entities. For example, the Booking entity used identifiers that connected a reservation to the corresponding customer or user and selected accommodation or facility. These relationships allowed the system to retrieve related information when processing reservations and when administrators reviewed booking records.

The database also supported the reservation status management process of the system. When a customer successfully submitted a reservation, the booking information was stored in the database with a **Pending** status. The administrator could then review the reservation through the management portal and update its status based on the action taken. Approved reservations were recorded with an **Approved** status, while cancelled or rejected transactions were handled according to the system's cancellation and archiving process. This allowed the database to maintain the current state of each reservation and support the proper processing of booking transactions.

The database was also designed to support the retrieval of information required by the administrative dashboard and reporting functions. Reservation records stored in the database could be accessed and processed to provide administrators with information regarding bookings and other resort-related activities. By maintaining centralized records, the system provided a consistent source of information for reservation monitoring and administrative decision-making.

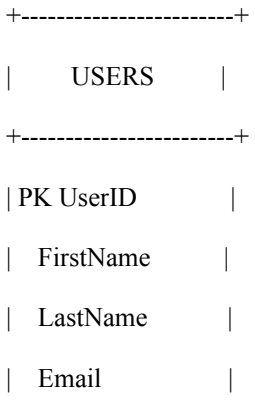
Database security and data integrity were also considered during the design process. Access to database information was controlled through the application's authentication and authorization mechanisms so that sensitive customer and reservation information could only be accessed through appropriate system functions. Validation procedures were also incorporated into the application to reduce the possibility of incomplete or invalid information being stored in the database.

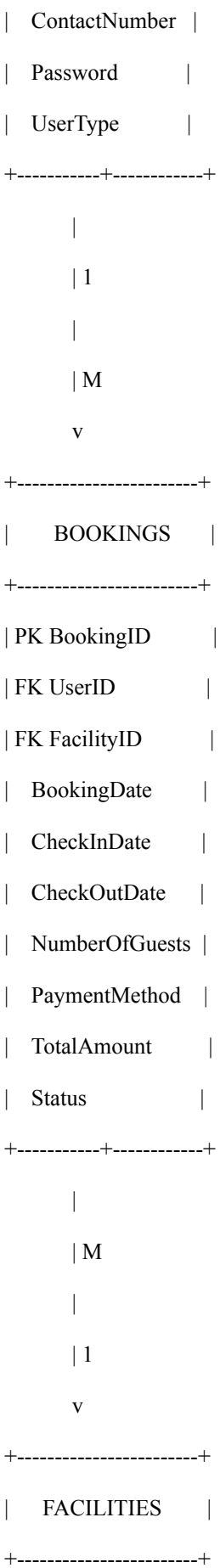
The database design provided the structural foundation for storing and managing the information required by the Online Booking System of Avianna's Inland Resort. The use of a relational database model with MySQL allowed the researchers to organize information into related entities, minimize redundancy, maintain data integrity, and support efficient data retrieval. The database design also provided the necessary support for customer management, accommodation management, reservation processing, administrative monitoring, and reporting functions.

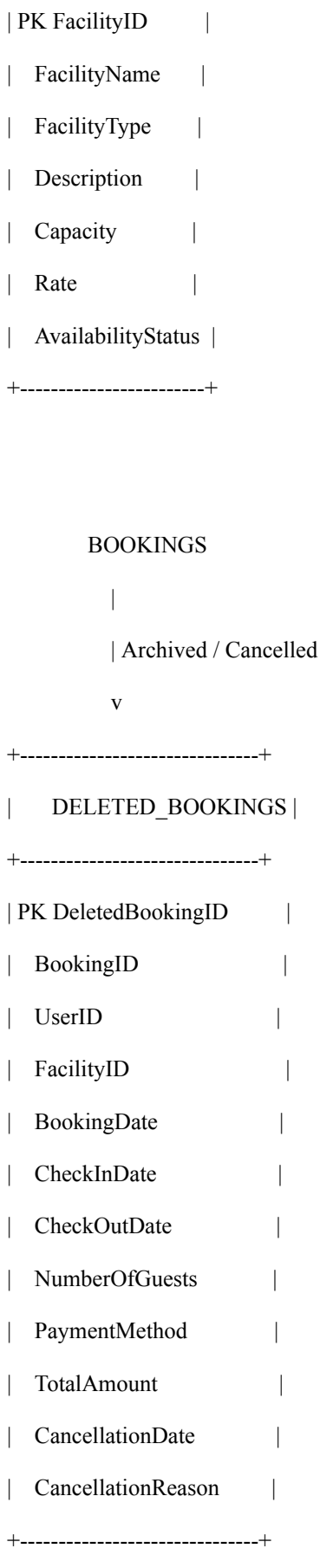
Figure 4.4 Presents the Entity Relationship Diagram (ERD) of the

system.

(Insert ER Diagram Here)




| ContactNumber |
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| UserType |
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| M
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| BOOKINGS |
+----------+
| PK BookingID |
| FK UserID |
| FK FacilityID |
| BookingDate |
| CheckInDate |
| CheckOutDate |
| NumberOfGuests |
| PaymentMethod |
| TotalAmount |
| Status |
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|
| M
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| 1
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| FACILITIES |
+----------+



The primary entities of the database include:

The primary entities of the Online Booking System of Avianna's Inland Resort were designed to organize and manage the information required for online reservation and resort administration. Each entity represented a specific type of information used by the system, while relationships among the entities allowed the application to connect customers, facilities, and reservation records. The major entities included the Users, Customers, Facilities, Bookings, and Deleted Bookings tables.

Users Table

The Users table stored the account information required for accessing the Online Booking System. It was primarily used for user authentication and access control. Each user account was assigned a unique identifier that served as the primary key. The table contained information needed to identify the user and determine the appropriate level of access within the system.

Field	Description
user_id	Primary Key
fullname	Complete Name
username	Login Username
password	User Password
role	User Access Level

The Users table allowed the system to distinguish between authorized administrative users and other users who accessed the system. The role field was used to determine the functions and areas of the system that could be accessed by the user.

Customers Table

The Customers table contained the personal and contact information of customers who submitted reservations through the Online Booking System. The information stored in this table supported the reservation process and allowed administrators to retrieve customer information associated with booking records.

Field	Description
customer_id	Primary Key
fullname	Customer Name

email	Customer Email Address
contact_number	Customer Contact Number
address	Customer Address

The customer information stored in this table was associated with reservation records so that the system could identify the customer who submitted a particular booking. This relationship also allowed administrators to manage and retrieve customer information through the administrative portal.

Facilities Table

The Facilities table stored information about the accommodations, cottages, and other reservable facilities available at Avianna's Inland Resort. The table provided the information needed by the customer-facing booking interface and the administrative management functions.

Field	Description
facility_id	Primary Key
facility_name	Facility or Accommodation Name
facility_type	Type of Facility
description	Facility Description
capacity	Maximum Number of Guests
rate	Facility Rate
availability_status	Current Availability Status

The Facilities table allowed the system to organize the different accommodation and facility options available for reservation. The information stored in this table could be displayed to customers when selecting a facility and could also be managed by authorized administrators.

Bookings Table

The Bookings table served as one of the central entities of the Online Booking System because it stored the reservation transactions submitted by customers. Each booking record represented a reservation request made for a particular facility. The table contained information necessary for identifying the customer, selected facility, reservation dates, number of guests, payment method, total amount, and current reservation status.

Field	Description
booking_id	Primary Key
user_id	Foreign Key
facility_id	Foreign Key
booking_date	Date Reservation Was Created
check_in_date	Scheduled Check-In Date
check_out_date	Scheduled Check-Out Date
number_of_guests	Number of Guests
payment_method	Selected Payment Method
total_amount	Total Reservation Amount
status	Current Booking Status

The Bookings table was connected to the Users and Facilities tables through foreign keys. These relationships allowed the system to determine which user or customer was associated with a reservation and which facility had been selected. The status field was used to monitor the progress of each reservation. Newly submitted reservations were recorded with a Pending status and were subsequently reviewed by an administrator. Approved reservations were updated to an Approved status, while cancelled reservations were processed according to the system's cancellation and archival procedure.

Deleted Bookings Table

The Deleted Bookings table was used to store records of cancelled or rejected reservations that were removed from the active booking records. Instead of permanently losing the information associated with a cancelled reservation, the system transferred or copied the relevant booking information into this table for archival and historical purposes.

Field	Description
deleted_booking_id	Primary Key
booking_id	Original Booking Identifier
user_id	Associated User
facility_id	Associated Facility
booking_date	Original Booking Date
check_in_date	Scheduled Check-In Date
check_out_date	Scheduled Check-Out Date
number_of_guests	Number of Guests
payment_method	Payment Method
total_amount	Total Reservation Amount
cancellation_date	Date of Cancellation
cancellation_reason	Reason for Cancellation

The Deleted Bookings table allowed the system to maintain a historical record of cancelled or rejected reservations. This supported administrative monitoring and prevented cancelled transactions from being permanently lost. The archived information could also be used when reviewing previous reservation activities.

Relationships Among the Database Entities

The relationships among the database entities allowed the Online Booking System to properly connect users, customers, facilities, and reservation records. A user could be associated with multiple booking records, while each booking was associated with the appropriate user or customer. Similarly, a facility could be associated with multiple booking records over different dates, while each booking referred to a specific facility. These relationships allowed the system to retrieve complete reservation information by connecting the customer details, selected facility, and booking information.

Primary keys were used to uniquely identify records within each table, while foreign keys were used to establish relationships between related tables. The use of foreign key constraints helped maintain referential integrity and prevented invalid references between database records. Proper indexing was also considered to support efficient retrieval of frequently accessed information, particularly during reservation processing, administrative review, and record management.

Overall, the database entities and their relationships provided the foundation for the efficient management of reservation information within the Online Booking System of Avianna's Inland Resort. The database structure allowed the system to record customer information, manage facilities, process reservations, monitor booking statuses, and maintain records of cancelled transactions. The organization of the database also supported the administrative functions of the system and contributed to the reliability, consistency, and accessibility of reservation information.

CHAPTER 5

5.1 CONCLUSION AND RECOMMENDATIONS

This chapter presented the conclusion and recommendations derived from the development, implementation, testing, and evaluation of the Online Booking System of Avianna's Inland Resort. It provided a synthesis of the major findings obtained throughout the study and discussed how the developed system addressed the problems and operational needs identified during the initial stages of the research. The chapter also reviewed the extent to which the objectives of the study were achieved through the implementation of the developed system and the evaluation conducted with the intended users.

The development of the Online Booking System was undertaken to provide Avianna's Inland Resort with a more organized and accessible method of handling reservation activities. The system was developed to support customers in accessing resort information, viewing available accommodations or facilities, submitting reservation requests, and receiving notifications regarding their bookings. At the same time, the system provided administrators and resort staff with functions for reviewing and managing reservations, maintaining customer and accommodation information, monitoring booking records, and accessing relevant administrative information.

The results obtained from the development and testing activities were used as the basis for determining the effectiveness and suitability of the developed system. The researchers considered the functionality, usability, reliability, performance, and overall operation of the application based on the identified requirements and feedback from the intended users. The testing and evaluation activities also provided opportunities to identify errors, usability concerns, and areas that required improvement before and after deployment.

This chapter therefore consolidated the findings of the study and provided an overall assessment of the developed Online Booking System. The conclusion discussed the accomplishments of the project and the contribution of the system to the reservation and administrative processes of Avianna's Inland Resort. The recommendations identified possible improvements and additional features that could be considered in future versions of the system to further enhance its functionality, security, accessibility, and efficiency.

The recommendations were also intended to support the continued maintenance and improvement of the system after implementation. As the operational requirements of the resort and the expectations of customers may change

over time, continuous evaluation and updating were considered important to maintain the effectiveness of the Online Booking System. Through appropriate maintenance, future enhancements, and continued user feedback, the system could remain responsive to the needs of the resort and provide sustained support for its online reservation operations.

5.2 SUMMARY OF FINDINGS

The development and evaluation of the Online Booking System of Avianna's Inland Resort yielded significant findings that were aligned with the objectives of the study. The results demonstrated that the developed system provided a centralized and organized platform for managing online reservations and supporting the resort's administrative activities. The findings were based on the system development, testing, implementation, and evaluation conducted throughout the study.

The developed system successfully provided customers with an online platform for accessing resort information and submitting reservation requests. Customers were able to provide the necessary personal and contact information, select their preferred accommodation or facility, specify the booking dates and number of guests, select a payment method, and submit their reservation requests through the online booking form. The system processed the submitted information and recorded valid reservation requests in the database, reducing the resort's dependence on manual reservation recording and direct processing.

The system also successfully implemented reservation validation and status management. Submitted reservation requests were initially recorded with a Pending status and were made available to authorized administrators for review. Administrators were able to examine the details of pending reservations and take appropriate actions by approving or cancelling the requests. Approved reservations were updated accordingly, while cancelled or rejected reservations were processed through the system's archival procedure. This provided the resort with a more organized method of monitoring the status of reservation transactions.

The developed database successfully stored and managed the information required for reservation processing and administrative management. Customer records, user accounts, facility information, and booking records were organized in a centralized relational database. The relationships among the database entities allowed the system to associate customers with their reservations and selected facilities. This organization improved the accessibility and consistency of reservation information and provided administrators with a centralized source of booking records.

The system also provided notification functions to improve communication between the resort and its customers. Customers received appropriate notifications regarding their reservation requests and subsequent booking status. The implementation of these functions helped provide customers with timely information about their reservations and reduced the need for repeated direct inquiries regarding the status of submitted booking requests.

The administrative functions of the system also provided authorized users with tools for managing reservation and customer information. Administrators were able to review pending bookings, manage reservation records, access customer information, and manage accommodation or facility information through the administrative portal. The system also provided dashboard and reporting functions that supported the monitoring of accumulated reservation information and assisted administrators in reviewing the resort's booking activities.

The testing activities demonstrated that the major components of the system were able to operate according to their intended functions. Unit testing was used to examine individual modules, while integration testing was conducted to verify the interaction among the different system components. System testing was subsequently performed to evaluate the overall functionality, usability, performance, and reliability of the integrated application. Issues identified during these testing activities were addressed through corrections and improvements as part of the iterative development process.

User Acceptance Testing was also conducted with selected participants representing the intended stakeholders of the system. The participants included the resort owner, administrators, resort staff, and potential customers. Their evaluation and feedback provided information regarding the usability, functionality, accessibility, and suitability of

the developed system. The feedback obtained from the participants was considered by the researchers in identifying areas that required further improvement and in determining whether the system adequately addressed the identified operational requirements.

The findings indicated that the developed Online Booking System provided Avianna's Inland Resort with a functional web-based platform for supporting reservation processing and administrative management. The system addressed the identified need for an online reservation facility by providing customers with a more accessible booking process and administrators with a centralized method of managing reservation information. The results of the development, testing, and evaluation activities demonstrated that the system provided the essential functions required to support the resort's online booking operations while also providing opportunities for future enhancement and continuous improvement.

5.3 CONCLUSION

The Online Booking System of Avianna's Inland Resort was successfully developed as a web-based solution for improving the resort's reservation and booking management processes. The system provided customers with a more convenient way to access resort information, select available facilities, submit reservation requests, and receive booking notifications. It also provided administrators and resort staff with a centralized platform for reviewing reservations, managing customer and facility information, and monitoring booking records.

The developed system addressed the identified need for an online booking platform by reducing the reliance on manual reservation procedures and providing a more organized method of recording and managing booking information. The use of a centralized relational database allowed customer, facility, and reservation records to be stored systematically and retrieved when needed. The implementation of reservation validation and status management also helped ensure that booking requests were properly processed and monitored.

The results of the testing and evaluation indicated that the major functions of the system operated according to the intended requirements. The system supported the essential processes involved in online reservation, including customer information submission, booking processing, reservation status management, administrative review, and notification. User feedback obtained during the evaluation also provided useful information for improving the system's functionality, usability, and reliability.

The study concluded that the developed Online Booking System provided Avianna's Inland Resort with a functional and accessible platform for supporting its reservation operations. The system helped improve the organization and management of booking information while providing customers with a more convenient reservation process. Although further enhancements and continuous maintenance may be implemented in the future, the developed system successfully addressed the identified operational needs and provided a suitable foundation for improving the resort's online booking and administrative activities.

5.4 Recommendations

To ensure the continued effectiveness, reliability, and relevance of the Online Booking System of Avianna's Inland Resort, the following recommendations are proposed. These recommendations are intended to support the continuous improvement of the system and address possible needs that may arise during its continued use.

System Improvements

The resort may enhance the notification function by implementing automated alerts for important reservation events, such as newly submitted bookings, approved reservations, cancelled reservations, and upcoming check-in dates. This improvement could help administrators monitor booking activities more efficiently and provide customers with timely information regarding their reservations.

The security of the system may also be strengthened by implementing additional authentication and access-control

mechanisms. Features such as two-factor authentication, stronger password policies, secure password storage, and improved user access restrictions may be considered to provide better protection for customer, user, and reservation information.

The system may also be enhanced by providing additional reporting and export functions. Reports containing reservation records, customer information, booking history, and other relevant administrative data could be made available in formats such as PDF, CSV, or Excel. This would allow administrators to easily save, print, share, and review reservation information for management and documentation purposes.

Future developers may further improve the availability and reservation management functions by implementing more advanced real-time availability checking. This could help prevent overlapping reservations and provide customers with clearer information regarding available accommodations and facilities before submitting a booking request.

The system may also be continuously evaluated based on feedback from the resort owner, administrators, staff, and customers. Regular maintenance, security updates, database backups, bug fixing, and usability improvements should be conducted to ensure that the Online Booking System remains functional, secure, and responsive to the changing operational requirements of Avianna's Inland Resort.

Future Research Directions

Future researchers may explore the integration of additional features that could further improve the functionality and usefulness of the Online Booking System. One possible direction would be the development of more advanced analytics and reporting features that could help the resort analyze booking patterns, customer preferences, peak reservation periods, and other relevant reservation trends. These features could provide the resort management with additional information for planning and decision-making.

Future studies may also conduct a longitudinal evaluation of the system to determine its long-term effects on the resort's reservation processes, customer service, and overall operational efficiency. Such a study could compare the resort's operations before and after the continued implementation of the system and identify improvements or additional challenges that may arise over time.

Another possible direction would be the development of a mobile application or enhanced mobile version of the system. This could provide customers and administrators with more convenient access to booking and management functions through mobile devices. Future researchers may also investigate additional features that could improve accessibility and usability when internet connectivity is limited.

Implementation Strategies

The resort management and staff should be provided with appropriate training to ensure that they can properly use the Online Booking System and its administrative functions. Training should cover reservation management, customer record handling, facility information management, notification functions, and other relevant system features. Proper training would help promote consistent system use and reduce errors during daily operations.

Regular database backups and system maintenance should also be conducted to protect reservation and customer information and reduce the risk of data loss. The system should be regularly monitored for errors, security issues, and performance concerns. Software updates, bug fixes, and necessary improvements should be implemented as needed to maintain the reliability and security of the system.

The resort is also encouraged to continuously collect feedback from customers, administrators, and staff regarding their experience with the system. The feedback may be used to identify usability issues, determine additional requirements, and guide future system improvements. Through continuous monitoring, maintenance, training, and evaluation, the Online Booking System can remain useful and responsive to the operational needs of Avianna's Inland Resort.

5.5 Impact of the Study

The development of the Online Booking System of Avianna's Inland Resort contributed to the practical application of information technology in improving reservation and resort management processes. The system provided a digital platform that helped reduce dependence on manual booking procedures and provided customers with a more convenient way to submit reservation requests. It also provided administrators with a centralized system for managing customer information, facility details, and reservation records.

For Avianna's Inland Resort, the system provided an organized approach to reservation management and improved the accessibility of booking information. The administrative functions allowed authorized personnel to review, approve, cancel, and monitor reservations through a centralized platform. This could help reduce errors associated with manual record keeping and improve the efficiency of reservation-related activities.

For customers, the Online Booking System provided greater convenience by allowing them to access resort information and submit reservation requests through a web-based platform. Customers could provide their booking details without having to rely entirely on in-person or manual reservation procedures. The notification functions also supported communication between the resort and its customers regarding reservation status.

The study also contributed to the field of Information Technology by demonstrating the practical use of web-based systems, database management, and online reservation technologies in a resort environment. It showed how a digital system could be developed to address specific operational needs of a local resort while providing an accessible platform for both customers and administrators.

Despite its benefits, the system still required continuous maintenance, monitoring, and improvement to address future operational requirements. The effectiveness of some system functions also depended on reliable internet connectivity, accurate information entered by users and administrators, and proper maintenance of the database and application. Nevertheless, the developed Online Booking System demonstrated the potential of information technology to improve the organization, accessibility, and efficiency of reservation operations at Avianna's Inland Resort.

5.6 Summary

In summary, the development of the Online Booking System of Avianna's Inland Resort successfully applied software development principles and web-based technologies to address the resort's reservation and management needs. The system provided customers with a convenient platform for submitting reservation requests and provided administrators with organized tools for managing bookings, customer information, facility information, and reservation records.

The developed system demonstrated the practical application of information technology in improving the organization and efficiency of resort reservation processes. Through its implementation, testing, and evaluation, the system addressed the identified operational needs and provided a foundation for more accessible and organized online booking services at Avianna's Inland Resort.

The study also provided a basis for future improvements and enhancements of the system. Continuous maintenance, user feedback, security updates, and additional features could further improve the system's functionality and usefulness as the operational requirements of the resort developed over time.

End of Proposal

Note to the instructors handling this subject-

CHED BSIT ALIGNMENT STATEMENT

This capstone project complies with CHED BSIT standards by demonstrating competencies in application development, database management, systems analysis and design, and emerging technologies such as machine learning. The project follows outcomes-based education principles and integrates ethical considerations, user safety, and data privacy.